OSDP 3.0 Proposed PIV Enhancements

Security Industry Association

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Changelog

2026-11-11

- Added TWIC Mode 2 Active Card Authentication as a PIV Auto profile, with a capability flag, certificate-derived FASC-N, and a proposed 127-byte challenge and RSA/ECC prehash construction. Added NexGen V10 AID recognition and clarified unknown minor-version handling. Updated the Appendix C ACA example and documented Mode 4 composition using commanded fingerprint verification.

- Specified maximum card response-length requests as `0xFF` (short) and `0xFFFF` (extended), with shorter nonzero lengths permitted. Documented TSA's interpretation of ISO response-length encoding, its conflict with TWIC Appendix E, and potential compatibility issues with other federally issued credentials.

- Renamed the OSDP channel requirement to Secure Channel 3.0. Required protected TPK loading, biometric-object reads, biometric matching, and match replies with deployment-specific keys, per-fragment checks, and cleanup on channel loss or replacement to protect the ACU–PD link against line tapping.

- Required biometric signature validation and identity binding for PIV and TWIC. Allowed maintained enrollment-time validation, with the reply hash identifying the verified reference record.

- Made PIVMODE flag `0x02` require SM for PUT DATA and kept SELECT ordinary. Added protected and ordinary provisioning examples, with completion tied to the card's result.

- Enabled contact and contactless NEXGEN TWIC free read/write E-Stickers in the selected TWIC application. Defined native TLV encoding, accepted `0x6282` for complete reads, and added write/read examples and a captured write/read/erase sequence.

- Distinguished TWIC fingerprint verification, Reference Picture Verification, and PACS-held alternate-biometric matching, with FASC-N and transaction binding under 33 CFR 101.105 and 101.520(c).

- Specified TWIC AES-128-ECB decryption, padding and reference checks. Defined Result `0x07` for TPK/decryption failure, `0x0A` for an absent reference, and `0x0B` for an unusable reference.

- Required multiple-card rejection, PIVERROR `0x102A`, and a one-second RF removal interval between TWIC presentations.

- Required CS2 and CS7 for credential SM preparation and defined biometric working-data and TPK zeroization. Specified the SHA-256-derived card identifier and coordinate-only KDF and key-confirmation inputs, with the uncompressed-point prefix retained in the card command.

- Defined Function Code 15 capability bits for legacy SPE, explicit PIV SPE, and ACU-armed Auto Execute. Simplified LOADTPK to a 16-byte key for one biometric operation in the current presentation, with unconditional key cleanup.

- Aligned PIVMODE's Bluetooth interface bit with Function Code 13 at `0x10`, and reserved `0x08` in both global and per-entry masks.

- Updated TWIC Part 4 to October 2026, corrected registration and operational citations, and added the 2026 EC384 technical advisories. Referenced 33 CFR 101.520 for authentication and validity checks. Corrected digest byte-size notation in the autonomous-PIV table.

2026-10-14

- Added per-profile key references and certificate-based algorithm discovery so the same PIV Auto configuration supports RSA and ECC cards.
- Added a resume option after pairing or PIN entry, defined counter handling and the challenge format, and updated the appendix with PIV and Legacy/NextGen TWIC examples.

2026-10-07

- Added the Invalid TPK result for TWIC Privacy Key failures and set the TPK field length to 16 bytes.
- Clarified PIV/TWIC application selection and added a Data Model byte to PIVSELECT.
- Updated the Legacy/NextGen fingerprint and facial workflows and the BIOMATCH hash description.
- Clarified CRAUTH cleanup so each new request starts a fresh transaction while the previous reply remains available for retransmission.

2026-09-16

- Renamed `osdp_PIVDATA`/`osdp_PIVDATAR` to `osdp_PIVGETDATA`/`osdp_PIVGETDATAR`, retaining command code `0xA3`. Added identifier and tag lengths, two-byte offsets and requested lengths, and whole-object or unique direct-child selection with the original tag and length bytes preserved. Defined OSDP 3.0 multi-part delivery and an all-zero first reply for empty success. Command data length distinguishes the new request from the fixed-length OSDP 2.2 form.
- Added Format Marker `0xFF` to the reassembled `osdp_CRAUTH` request, retaining command code `0xA5`. Multi-part lengths and offsets cover the marker, algorithm, key reference, and challenge. OSDP 3.0 ACUs use the new CRAUTH and PIVGETDATA forms with supporting PDs; PDs may additionally support the OSDP 2.2 forms, with format-specific response handling.
- Renamed `osdp_PDERROR` to `osdp_PIVERROR` and added GET DATA selection and transfer errors. The new CRAUTH and PIVGETDATA forms return one terminal PIVERROR on failure, replacing CRAUTH's empty failure reply and optional follow-up error. Renumbered tentative PIV-specific errors consecutively from `0x1020` through `0x1029`; credential removal is now `0x1026`.
- Separated credential SM setup, VCI pairing, and PIN verification. Clarified automatic preparation over contact and contactless, PIV Auto failure reporting, and ACU-directed recovery. Authenticated PIN and pairing rejections preserve SM; SM processing and integrity failures end the session.

- Added osdp_CARDSTATUS Type 0x02 diagnostics containing the SM outcome and credential CVC hash. Every CARDSTATUSR now includes Status Type after reply code 0x85: 0x01 for standard status and pairing completion, 0x02 before the multi-part fields in each diagnostic fragment. Extended osdp_PIVSTATUS with SW1/SW2 when a verified credential failure status is available; the status word follows the signed response in the reassembled payload.
- Defined cached-PIN reuse without extending the original deadline, silent idle expiration, and completion of requests affected by expiration. Bounded PD timeout defaults to 1–255 seconds. Reassigned SPE flags 0x01 for retention across removal and 0x02 for Auto Execute; added result 0x0B for credential communication failure or timeout, requiring preparation to restart after an unconfirmed PIN VERIFY. Applied the same restart rule to pairing VERIFY timeouts, reported through existing PIVERROR 0x0002 (smart-card timeout).
- Added workflow diagrams and worked packet examples for selection, SM/VCI recovery, PIN-cache timing, and multi-part replies. Standardized table sizes and protocol values on hexadecimal notation.

2026-09-02

- Made proposal builds self-contained with the SIL-licensed Carlito typeface in regular, bold, italic, and bold-italic faces. Documented the XeLaTeX and latexmk requirements.
- Completed the Enhanced PIV message-name cleanup by applying the compact OSDP naming convention consistently to osdp_PIVLOADTPK, osdp_PIVXMITPAIRING, osdp_PIVVCILOADTA, osdp_PIVVCITASTATUS, and osdp_PIVVCITASTATUSR; command and reply codes, payloads, and behavior are unchanged.
- Standardized multi-part messages on the generic OSDP *MpSizeTotal*, *MpOffset*, and *MpFragmentSize* field names and moved the multi-part fields ahead of the reassembled osdp_PIVMODE and osdp_PIVVCILOADTA data.
- The current osdp_PIVVCILOADTA definition supersedes the anchor-only multi-part sizing described in the 2026-05-30 entry: its total now includes the 12-byte metadata block.
- Updated the PIV Auto simulator, generated fixtures, proposal PDF, and redline PDF to exercise and publish the same framing and naming.

2026-08-12

- Proposed clear, mode-specific SPE behavior for status, PIN acquisition and verification, cached-PIN verification, cache/abort, and attempts-remaining queries. The proposal also defines finite default timeouts, cache and PIN-verified lifecycles, and ACK/poll completion behavior.
- Proposed a one-byte osdp_PIVSPER reply whose value is interpreted by the requested mode. The OSDP 2.2 osdp_PR01SPER reply remains unchanged.
- Standardized Enhanced PIV names, including renaming the autonomous PIV Auto reply to osdp_PIVSTATUS. Corrected tentative code conflicts by assigning osdp_PIVSELECT to 0xB0 and osdp_PDERROR to 0x8A.

2026-06-18

- Added `osdp_PIVSELECT` as an OSDP-native direct AID selection command using tentative command code `0xA7`; clarified its ISO/IEC 7816 `SELECT by DF name` mapping, selected-AID status update, and application-state clearing rules.
- Removed stale `osdp_CRAUTHR` `osdp_PDERROR` wording so codes `0x0006`/`0x0007` retain their common meanings for secure-channel and multi-part errors.
- Clarified that VCI pairing codes are transmitted from the ACU to the PD only over OSDP Secure Channel 3.0 and are used by the PD to establish VCI secure messaging with the credential.
- Tightened and clarified verbiage by replacing imprecise "legacy" terminology throughout normative text with neutral wording such as unchanged, OSDP 2.2-format, and poll-based; retained "legacy" only for withdrawn algorithms such as RSA-1024 test material.

2026-05-30

- Added NIST-derived VCI and PIV APDU fixture coverage, GSA-backed demonstration keys, and reproducible worked examples for PIV Auto and VCI validation. RSA-1024 material is retained only as informative legacy/non-current test-vector material.
- Clarified PIV Auto cryptographic inputs and response handling, including KMAC-derived challenge construction, RSA and ECC `GENERAL AUTHENTICATE` templates, nonzero UUID requirements, and complete returned `0x7C` Dynamic Authentication Templates in `osdp_PIVSTATUS`.
- Aligned related command semantics for capability advertising, `osdp_CARDSTATUS` errors, selective `osdp_PIVDATA` replies, `osdp_PIVSPE` status/abort modes, and VCI trust-anchor inventory.
- Reworked Appendix C and the references section for standards review: examples are explicitly informative, vector provenance is documented, source fixtures are framed as public test material, and normative references now include the needed APDU, KMAC, X.509, and requirement-language specifications.
- Added supporting scripts and tests for deterministic simulation, fixture replay, VCI import/validation, Appendix C generation, YubiKey material loading, live APDU capture, and reproducible PDF/table builds.

2026-05-27

- Version renamed from OSDP 2.3 to OSDP 3.0 throughout.
- Removed `osdp_PIVSPE` flag `0x01` (retry continuation); each `osdp_PIVSPE` command now initiates exactly one PIN-entry session and `osdp_PIVSPER` terminates the session unconditionally, including on PIN mismatch. ACUs wishing to allow additional attempts **MAY** re-issue `osdp_PIVSPE`, using the *Attempts Remaining* count from the preceding `osdp_PIVSPER` to guide retry policy. Bit `0x01` is now reserved.

- Aligned osdp_PIVSPER and osdp_PIVBIOMATCHR result codes for credential removal, changed osdp_PIVBIOMATCHR 0x03 to PIN verification required, reserved osdp_PIVBIOMATCHR 0x07, clarified biometric assurance limits, and specified that only off-card biometric match success returns the 32-byte SHA-256 hash of tag 0xBC.

- Clarified the Function Code 14 biometric-support reference for osdp_PIVLOADTPK and added CAC as PIV mode data model 0x03.

- Added PIV Auto KMAC challenge derivation, required series and sequence counters, fixed-width supplemental entropy, and the osdp_PIVSTATUS poll reply for reporting credential identity and signed challenge responses. PIV Auto is configured through osdp_PIVMODE and is restricted to Secure Channel 3.0 sessions that are not using factory keys.

- Replaced the SP 800-73 Intermediate CVC anchor model with an OSDP-defined VCI trust anchor record (tag 0x7F50) carrying a Profile Identifier (0x80 = 0x01), an eight-byte Issuer Identifier Number (0x42), and a DER-encoded SubjectPublicKeyInfo (0x7F49). Maximum conforming record size is 576 bytes (0x0240), derived from the RSA-4096 key case.

- Added normative IIN derivation rule: the IIN in osdp_PIVVCILOADTA and in the trust anchor record **SHALL** be the leftmost eight bytes of the subjectKeyIdentifier from the content-signing X.509 certificate.

- Tightened osdp_PIVVCILOADTA multi-part encoding: Anchor Length and MpSizeTotal ranges corrected to 0x0000-0x0240; added normative constraint that MpOffset + MpFragmentSize **SHALL NOT** exceed Anchor Length; PD **SHALL** verify the reassembled byte string is exactly *Anchor Length* bytes and a single 0x7F50 record before committing.

- Documented variable-length PD storage accounting: PDs **SHALL** account for anchors by their actual *Anchor Length*, not a fixed maximum; replacement is atomic (prior anchor remains if replacement fails); delete releases actual stored bytes.

- Specified the two-tier PD verification path during VCI establishment: direct (trust anchor IIN matches secure messaging CVC IIN, anchor SPKI verifies CVC signature) and intermediate (no direct match triggers retrieval of PIV data object 0x1017/5FC122; Intermediate CVC Subject Identifier must equal CVC IIN, Role 0x5F4C must be 0x12, anchor IIN must match Intermediate CVC IIN, anchor SPKI verifies Intermediate CVC, Intermediate CVC verifies secure messaging CVC). PDs **SHALL NOT** perform X.509 path validation or revocation for VCI trust anchors.

- Added normative ACU post-VCI validation requirements: after osdp_CARDSTATUSR reports VCI established, the ACU **SHALL** obtain the content-signing certificate (from cache or via osdp_PIVDATA for 0x1017/inner tag 0x70), verify SKI matches *VCI Anchor IIN*, verify SPKI matches the stored anchor record, verify PIV or PIV-I content-signing EKU OID, verify validity period (expired certificate **SHALL** cause session rejection and anchor unload), and perform X.509 path validation and revocation per deployment policy.

- Restored *VCI Trust Anchor ID* to osdp_CARDSTATUSR with normative language; added *VCI CVC IIN* and *VCI Anchor IIN* fields (each 8 bytes) so the ACU receives the IIN pair needed to distinguish direct and intermediate verification paths.

- Updated osdp_PIVVCITASTATUSR: clarified that *Available Bytes* reflects variable-length accounting and that the per-record IIN is the anchor selector from the stored trust anchor record, not necessarily the secure messaging CVC IIN reported in osdp_CARDSTATUSR.
- Added VCI security preconditions: all VCI-related commands **SHALL** be rejected unless OSDP Secure Channel 3.0 is active and not using factory/default keys; failures reported via osdp_PDERROR 0x0006.

2026-01-21

- Error-handling cleanup and code assignments: osdp_PIVPUTDATA now **SHALL** report failures via osdp_PDERROR (status word optional) so writes can't silently succeed.
- osdp_PIVSPE adds result code 0x09 in osdp_PIVSPER for "VCI not established."
- PDERROR is credential/global only; PD capability gaps stay osdp_NAK.
- OSDP 2.2-format replies (e.g., osdp_CRAUTHR, osdp_PIVDATAR) remain unchanged, with optional follow-up osdp_PDERROR detail if the PD has it.
- Documented tentative command/reply codes for the new PIV set: commands 0xA6 (VCI trust-anchor status) through 0xAF (VCI trust-anchor load), and replies 0x85-0x89 for the new responses.

2026-01-07

- Added a PIV Mode interface-mask table aligned to Function Code 13 bits.
- Clarified CRAUTHR's full GENERAL AUTHENTICATE response payload, with status words carried through PDERROR.
- Clarified CARDSTATUS and CARDSTATUSR behavior: solicited status after PD preparation, with an error on credential removal.
- Remapped PIV error codes to the 0x10xx block and added PIV APDU references.
- Aligned PIVDATA selection of Discovery (0x7E) and BIT (0x7F61) subtags with SP 800-73-4.
- Clarified PDERROR's single-error reply and PIVXMITPAIRING submission before VCI establishment when the credential requires pairing.

2025-12-10

- Consolidated application-level errors on osdp_PDERROR (two-byte codes, optional SW1|SW2 for APDU failures, multi-error support)
- moved the PIV error set to 0x0100-0x01FF and updated PIV commands to use osdp_PDERROR (transport faults remain osdp_NAK)

- made the Global Interface Mask in `osdp_PIVMODE` a required byte (no TLV tag) with an updated field layout
- clarified that `osdp_CARDSTATUSR` is solicited-only after PD preparation with poll-based `osdp_RAW` as the aggregated card-present indication
- and noted the deprecation of `osdp_GENAUTH` with a recommendation to migrate to `osdp_CRAUTH`.

2025-12-04

- Clarified that `osdp_CRAUTHR` returns the full `GENERAL AUTHENTICATE` data including the Dynamic Authentication Template (excluding only the ISO/IEC 7816 status word).

2025-11-12

- Added PIV Biometric Match request/reply and tables (one-bit type values only)
- moved Enhanced PIV NAKs to a 0x20+ base (added 0x27 Credential Not Present/Removed, 0x28 Biometric Type Not Supported; clarified 0x23 Credential Security Status Not Satisfied)
- aligned CRAUTHR error handling to reference the enhanced NAK set
- relocated Enhanced PIV (0x04) and PIV Auto (0x08) indicators to Function Code 12 and cleaned Function Code 13 to interface-only bits (removed non-interface flags; removed PIV Auto from Function Code 14)
- updated CARDSTATUS gating to Function Code 12 bit 0x04
- added a PIV Mode Flags table and clarified PIV Mode AID encoding
- enabled PDF TOC hyperlinks and improved table/caption pagination.

2025-10-30

- Clarified CRAUTH command/reply behaviour (including RSA, AES, and ECC support), overhauled PIV Mode configuration with TLV-based application and interface controls (including PIV Auto), added card status request/reply messages, expanded PIV data retrieval semantics, introduced PIV Put Data, PIV Secure PIN Entry, TWIC privacy-key load, VCI pairing transmit, and VCI trust-anchor load/status commands, and documented the enhanced error-code set (including VCI/PIV Auto conditions).

2025-09-17

- Major PIV Mode and SPE revisions.

2025-06-25

- Added multi-AID usage and VCI support notes.

2025-05-27

- New function code definitions for 3.0 biometric handling.

DRAFT

1 Introduction

This document captures proposed enhancements to the SIA Open Supervised Device Protocol (OSDP) needed to carry FICAM-compliant Personal Identity Verification (PIV) workflows. The material consolidates draft content, reviewer comments, and interoperability feedback collected during the 2025 working sessions and re-expresses it as normative guidance for command authors and device implementers.

The chapters that follow highlight changes required for existing OSDP 2.2 commands, introduce new transactions where no equivalent existed, and provide supplemental examples that reflect current federal credentialing practices. Wherever the proposal diverges from the baseline specification, the text records both the behavioral requirements and the error handling expectations so that access control units (ACUs) and peripheral devices (PDs) can adopt the profile without relying on vendor extensions.

The key words **SHALL**, **SHALL NOT**, **SHOULD**, **SHOULD NOT**, and **MAY** are interpreted as described in RFC 2119 and RFC 8174 when they appear in capitals.

Tables express protocol sizes, lengths, offsets, and numeric values in hexadecimal with the prefix `0x`. Packet and APDU dumps use hexadecimal byte pairs without repeated prefixes. Standard names, bit positions, and elapsed-time annotations retain their usual notation.

OSDP 3.0 introduces a new PIVGETDATA request format at command code `0xA3`. PDs support the new format and may also accept the OSDP 2.2 format; the command data length distinguishes the two. OSDP 3.0 ACUs use the new format with PDs that support it. The revised CRAUTH and PIVGETDATA transactions return data on success and one terminal PIVERROR on failure.

2 Changes to OSDP 2.2

This section documents the normative deltas that apply when an OSDP 2.2 implementation adopts the Enhanced PIV profile. Each subsection references the original command definition and records the precise behavioral updates, encoding clarifications, and new error handling rules introduced by this proposal.

2.1 Updated Command Definitions

2.1.1 Authentication Challenge (osdp_CRAUTH)

The `osdp_CRAUTH` command, code `0xA5`, transports a `GENERALAUTHENTICATE` operation to the credential. An OSDP 3.0 ACU **SHALL** use the new format with a PD that supports it. An OSDP 3.0 PD **SHALL** support the new format and **MAY** also support the OSDP 2.2 format.

Request format and recognition The new format uses the OSDP 3.0 multi-part fields immediately after the command code. The reassembled data consists of Format Marker `0xFF`, Algorithm, Reference Identifier, and Challenge, in that order. The three leading fields are each one byte. *MpSizeTotal* counts all of these bytes, including the marker, algorithm, and key reference.

The first data fragment **SHALL** start at offset `0x0000` and contain at least one data byte. The PD **SHALL** select the new format when that first byte is `0xFF`. Otherwise, the PD **SHALL** select the OSDP 2.2 format if supported, or reject the request with `osdp_NAK` `0x09` (unable to process command). This proposal reserves `0xFF` in that position as an OSDP format marker. A dual-format PD **SHALL NOT** interpret that byte as a legacy algorithm identifier. The marker is separate from the Algorithm field and is removed before the PD constructs the credential command.

The PD **SHALL** retain the selected format until completion or cancellation. Later fragments carry the next bytes of the same data record; the marker and other fields are not repeated. Fragment boundaries may split the leading fields or the challenge. The PD **SHALL** validate the complete record before sending the credential command. The minimum record length is `0x0003`; the selected algorithm determines the required challenge length. A malformed new-format record **SHALL NOT** be retried as an OSDP 2.2 record. Multi-part framing errors use the applicable OSDP NAK rules. Invalid reassembled parameters return PIVERROR `0x0005` (invalid parameter) after acceptance.

A PD accepting the OSDP 2.2 form **SHALL** use that form's request and response rules for the transaction. The rules below apply to the new form. The worked example in Appendix E shows the marker, algorithm, and key reference divided between command fragments.

- Codes `0x05`, `0x06`, and `0x07` identify RSA private-key operations. Codes `0x08`, `0x0A`, and `0x0C` identify AES encryption with 128-, 192-, and 256-bit keys respectively. Codes `0x11` and `0x14` identify ECC P-256 and P-384 mechanisms; the selected key reference determines whether the operation is ECDSA signature generation or ECDH key establishment. Algorithm selection remains subject to the applicable NIST requirements.
- The *Key* field retains its role as the card's reference identifier. PDs **SHALL NOT** reinterpret or remap the identifier.

- For RSA algorithms (0x05, 0x06, 0x07), the *Challenge* field carries the encoded datum for the RSA private-key operation. The PD forwards the value unmodified; padding and digest encoding are the responsibility of the ACU.
- For AES algorithms (0x08, 0x0A, 0x0C), the *Challenge* field contains the plaintext block(s) to be encrypted. PDs **SHALL** send the data to the credential without padding; ACUs **SHALL** provide data aligned to the block size dictated by the selected key length and card profile.
- For ECC signature operations using key references 0x9A, 0x9C, or 0x9E with algorithms 0x11 or 0x14, the *Challenge* field carries the pre-hashed input to be signed. The PD **SHALL** issue GENERALAUTHENTICATE with command data encoded as 7C Len 82 00 81 Len Challenge; the credential returns the ECDSA signature in tag 0x82. Exponentiation tag 0x85 **SHALL NOT** be used with 0x9A, 0x9C, or 0x9E.
- For ECC key-establishment operations using key reference 0x9D or retired key-management references, algorithms 0x11 and 0x14 use the *Challenge* field to convey the peer public key. The PD **SHALL** issue GENERALAUTHENTICATE with command data encoded as 7C Len 82 00 85 Len (04 || X || Y), where X and Y are the uncompressed coordinates for the selected curve and are left-padded with 0x00 as required. The credential performs elliptic-curve Diffie-Hellman in accordance with SP 800-73.

Table 1: Authentication Challenge (osdp_CRAUTH)

Size (bytes)	Name	Meaning	Value
0x01	CMND	Command identifier.	0xA5
0x02	MpSizeTotal	Length of the complete record, including Format Marker, Algorithm, Reference Identifier, and Challenge; least-significant byte first.	0x0003–0xFFFF
0x02	MpOffset	Offset of this fragment within the complete message, least-significant byte first.	0x0000–0xFFFF
0x02	MpFragmentSize	Length of the fragment payload, least-significant byte first.	0x0000–0xFFFF
0x01	Format Marker	First byte of the reassembled record. Identifies the OSDP 3.0 format; removed before constructing the credential command.	0xFF

Size (bytes)	Name	Meaning	Value
0x01	Algorithm	Cryptographic mechanism. Supported values include 0x05/0x06/0x07 (RSA private-key operation), 0x08/0x0A/0x0C (AES encryption with 128/192/256-bit keys), and 0x11/0x14 (ECC P-256/P-384; operation depends on the key reference).	0x00–0xFF
0x01	Reference Identifier	Key reference passed to GENERALAUTHENTICATE.	0x00–0xFF
0x00-n	Challenge	For RSA algorithms, the encoded datum for the private-key operation. For AES algorithms, the plaintext block(s) to be encrypted, pre-padded by the ACU as required. For ECC 0x9A/0x9C/0x9E, the pre-hashed signature input in tag 0x81; for ECC 0x9D/retired key-management references, the peer public key in tag 0x85.	0x00–0xFF

When credential SM is established, the PD **SHALL** protect the resulting credential command using that context. The credential enforces the selected key's access conditions. An available card-authentication key (0x9E) may be used without SM or PIN verification when the credential permits it. Shared card-transport rules appear in Subsection 3.1.

Deprecated osdp_GENAUTH The osdp_GENAUTH command is deprecated and expected to be removed in a future revision of this specification. Using a PIV credential to decrypt or process arbitrary data is unsafe and unnecessary for credential verification; implementations **SHOULD** use osdp_CRAUTH flows instead.

2.1.2 Response to Challenge (osdp_CRAUTHR)

OSDP 3.0 uses the same completion rule for CRAUTH and PIVGETDATA: data replies carry successful results; one terminal PIVERROR reports a failed accepted operation. This replaces CRAUTH's OSDP 2.2 failure behavior.

After accepting the complete CRAUTH request, the PD **SHALL** return the result on subsequent osdp_POLL commands. On success, the PD **SHALL** return osdp_CRAUTHR using the OSDP 3.0 multi-part format. On failure, the PD **SHALL** return one osdp_PIVERROR with the applicable error code from Subsection 2.2. That reply ends the operation. The PD **SHALL NOT** precede the error with an empty CRAUTHR. If CRAUTHR fragments have already been returned, the ACU **SHALL** discard the partial response when PIVERROR arrives.

OSDP retransmissions repeat the same reply without repeating the credential operation. Command rejection and transport faults continue to use the applicable OSDP NAK rules.

After returning the final `osdp_CRAUTHR` fragment or a terminal `osdp_PIVERROR`, the PD **SHALL** end the operation and clear its assembled request, request length, response offset, pending error, and completion state. The PD **SHALL** then accept a new `osdp_CRAUTH` beginning at offset zero. Clearing the completed operation **SHALL NOT** invalidate the cached OSDP reply required for link-layer retransmission. A retransmission replays that reply without advancing the response offset or repeating the credential operation.

Reply structure

`osdp_CRAUTHR` **SHALL** return the credential's full ISO/IEC 7816 `GENERALAUTHENTICATE` response payload, including the Dynamic Authentication Template, framed by the standard OSDP multi-part fields. PDs **SHALL NOT** include SW1|SW2 in `osdp_CRAUTHR`; any status words from the credential **SHALL** be carried only in `osdp_PIVERROR` under the status-word rules in Subsection 2.2.1. PDs **SHALL NOT** strip outer BER-TLV tags or lengths from the returned payload.

Table 2: Response to Challenge (osdp_CRAUTHR)

Size (bytes)	Name	Meaning	Value
0x01	CMND	Reply identifier.	0x82
0x02	MpSizeTotal	Length of the complete response, least-significant byte first.	0x0000–0xFFFF
0x02	MpOffset	Offset of this fragment within the complete response, least-significant byte first.	0x0000–0xFFFF
0x02	MpFragmentSize	Length of the fragment payload, least-significant byte first.	0x0000–0xFFFF
0x00-n	Response Data	Signature, encrypted block, or shared-secret material returned by the credential. The credential's BER-TLV structure is preserved exactly as received (status word excluded).	0x00–0xFF

Card response-length requests For logical card commands that request response data, the PD **SHALL** use an explicit nonzero response length appropriate to the requested data. When requesting the maximum response length, the PD **SHALL** use `0xFF` for short APDUs or `0xFFFF` for extended APDUs, requesting up to 255 or 65,535 response-data bytes, respectively. Shorter nonzero lengths **MAY** be used. This rule applies to application selection, commanded data retrieval and authentication, PIV Auto, and internal biometric-object reads. The PD **SHALL** use the APDU format supported by the credential and encode Le with one byte in short format, two bytes following extended Lc, or a leading `0x00` followed by two Le bytes in a case-2 extended command.

For response continuation with status 61XX, the PD **SHALL** issue GET RESPONSE with an explicit length bounded by 0xFF and the available count. SW2 0x00 denotes at least 256 available bytes, so the next short GET RESPONSE requests 0xFF bytes. It **SHALL** concatenate response data until the operation completes. A corrected nonzero length in 6CXX may be used within these bounds; a credential requiring the zero-encoded maximum is reported through the command's existing credential-failure path, with its status word when available. Commands completing without response data retain their absent Le. Credential SM wrapping follows SP 800-73-5 Part 2, including its outer Le and protected expected-length encoding.

For TWIC GET DATA and its GET RESPONSE continuations, status 0x6282 indicates the end of available data before the requested Le (TWIC Part 2, Appendix E). The PD **SHALL** retain the returned bytes, finish response assembly, and accept the read when the assembled bytes contain one complete requested object, including its tag and length. A truncated object **SHALL** produce PIVERROR 0x1028 (malformed credential-object encoding). Normal object validation and selection rules then apply; the PD **SHALL** remove the status word before returning object data. This rule applies to commanded reads, PIV Auto, and internal biometric-object reads. Appendix B.22 shows a captured write/read/erase sequence.

TSA interpretation and interoperability ISO/IEC 7816-4:2020, Section 5.2, defines short Le 0x00 as 256 bytes and extended all-zero Le as 65,536 bytes. TWIC Part 2, Section 5.2, note 3 (page 37), instead describes Le 0x00 yielding an empty 0x9000 response as ISO-compliant. Appendix E (page 59) of the same TWIC specification uses the ISO zero-encoded maximum and directs T=1 GET DATA to use Le 0x00. This proposal uses explicit nonzero response-length requests to accommodate TSA's interpretation. Other federally issued credentials may require the ISO zero-encoded maximum or different length handling. The PD reports a credential length rejection through the existing failure path.

2.1.3 Get PIV Data (osdp_PIVGETDATA)

OSDP 3.0 retains command code 0xA3 and introduces the PIVGETDATA request format defined here. An OSDP 3.0 ACU **SHALL** send the new format to a PD that supports it. An OSDP 3.0 PD **SHALL** support the new format and **MAY** also support the OSDP 2.2 format. The ACU **SHALL** confirm Enhanced PIV support (Function Code 12 bit 0x04) before using the new format. The reply uses the OSDP 3.0 multi-part layout. Worked exchanges appear in Appendix B.

Recognizing the request format The PD **SHALL** determine the format from the command data length, excluding the command code, OSDP header, security block, and trailer. The OSDP 2.2 format contains exactly 0x05 data bytes: a three-byte object identifier, a one-byte element identifier, and a one-byte offset. The new format contains 0x06 plus Identifier Length plus Tag Length data bytes, giving a range of 0x07 through 0x0C bytes. These lengths do not overlap.

A PD that supports both formats **SHALL** interpret exactly 0x05 data bytes using the OSDP 2.2 definition, including its selection and response rules. A new-format-only PD **SHALL** reject that request with PIVERROR 0x0005 (invalid parameter) on a later poll after ACK. For a new-format request, the PD **SHALL** validate both length fields and the exact total length before accessing the credential. Any other data length or invalid new-format field length produces the same error. The PD **SHALL NOT** reinterpret a malformed new-format request as an OSDP 2.2 request.

Table 3: Get PIV Data (osdp_PIVGETDATA)

Size (bytes)	Name	Meaning	Value
0x01	CMND	Command identifier.	0xA3
0x01	Identifier Length	Number of bytes in Object Identifier.	0x01–0x03
0x01-0x03	Object Identifier	Complete BER tag identifying the credential data object. Native byte order; no leading padding.	BER tag
0x01	Tag Length	Number of bytes in Element Tag. Zero selects the complete object and omits Element Tag.	0x00–0x03
0x00-0x03	Element Tag	Complete BER tag of a unique direct child of the object's outer container.	BER tag
0x02	Offset	Starting byte within the selected object or element. Zero identifies its first tag byte. Least-significant byte first.	0x0000–0xFFFF
0x02	Requested Length	Maximum bytes to return, least-significant byte first. Zero requests all remaining bytes.	0x0000–0xFFFF

Request encoding The ACU **SHALL** provide valid Object Identifier and Element Tag encodings. The PD **SHALL** pass Object Identifier to the card unmodified and use Element Tag unmodified to select a matching element from the response. Each length field **SHALL** equal the number of tag bytes that follow it. Identifier Length is one to three; Tag Length is zero to three. Tag Length zero omits Element Tag. The PD **SHALL** reject invalid field lengths, truncated requests, and trailing request bytes with `osdp_PIVERROR` 0x0005 (invalid parameter).

~~The PD~~ For biometric and privacy-key objects, the request and every data-reply fragment **SHALL** use Secure Channel 3.0 with deployment-specific keys as specified in paragraph 3.11. The PD checks this requirement before accessing the credential, independently of the requested element or byte range.

The PD **SHALL** issue the card's `GET DATA` operation for the requested object on the active credential interface, using the response-length and continuation rules in paragraph 2.1.2. Each request reads the credential; the PD **SHALL NOT** substitute object data retained from an earlier request. If the credential denies access through that interface, the PD **SHALL** report the failure using `osdp_PIVERROR`. When credential Secure Messaging (SM) is established, the PD **SHALL** use that SM context. The credential enforces VCI availability and the requested object's access conditions. The PD **SHALL** complete card-side response chaining and remove status words and secure-messaging envelopes before selecting response data. A requested OSDP length does not limit the card-side transfer needed to complete that operation.

Selection When Tag Length is zero, the PD **SHALL** select the complete object, including its outer tag, length, and value. The PD **SHALL** preserve the original bytes returned by the credential. PIV responses use `0x53` for ordinary objects and native `0x7E` and `0x7F61` containers for Discovery and the BIT Group. Other application profiles may define other containers; whole-object selection does not substitute or remove them.

For nonzero Tag Length, the PD **SHALL** parse the outer container and select exactly one matching direct child, including that child's tag, length, and value. The enclosing object's tag and length are not included. The PD **SHALL NOT** search recursively, decompress the value, or re-encode the selected element. The PD **SHALL** inspect the complete child sequence before accepting a unique match. No match produces `0x1024` (data object or tag not found); multiple matches produce `0x1027` (ambiguous element selection); malformed container or child encoding produces `0x1028` (malformed credential-object encoding). Element extraction requires definite BER lengths, a single complete outer container, and child boundaries within that container. To select an inner tag, the PD **SHALL** search the elements directly inside the object's outer container. In BER-TLV encoding, *primitive* means the tag marks its value as data bytes; *constructed* means the tag marks its value as a sequence of TLV elements. For PIV objects returned under tag `0x53`, the PD **SHALL** treat the contents as TLV elements even though the tag marks them as primitive. Child values are opaque regardless of their constructed bit; the PD **SHALL NOT** recursively validate them as ASN.1 structures. Whole-object selection does not require child parsing. An empty card response is valid whole-object data; a nonzero Tag Length against an empty response **SHALL** produce `0x1024` (data object or tag not found).

Offset and requested length The PD **SHALL** apply Offset and Requested Length once, before dividing the data into fragments. Offset zero identifies the first byte of the selected object's or element's tag. The PD **SHALL** start at Offset and return no more than Requested Length bytes. Requested Length zero requests all remaining bytes. If fewer bytes remain, the PD **SHALL** return those bytes. The resulting bytes are the *requested data*. Offset equal to the selected-data length produces an empty success; Offset beyond that length produces `0x0005` (invalid parameter). The requested data **SHALL NOT** exceed 65,535 bytes. An oversized through-end result produces `0x1029` (transfer length exceeded); a bounded range within a larger object is permitted. A nonzero Offset can skip tag and length bytes. The requested range may begin or end within a tag, length, or value. The PD **SHALL** return those bytes unchanged. The ACU is responsible for interpreting a partial result.

Figure 1 shows how selection and fragmentation relate.

Acceptance and completion The PD **SHALL** acknowledge an accepted operation with `osdp_ACK`. Acceptance does not indicate successful card access or parameter validation. The PD **SHALL** return the result on a subsequent `osdp_POLL`: `osdp_PIVGETDATAR` for success or `osdp_PIVERROR` for failure. While processing is incomplete, the PD **SHALL** answer polls with `osdp_ACK`. Unsupported commands and transport faults retain their OSDP `osdp_NAK` responses.

Only one operation may be outstanding at that PD during this read. Until the read completes or is aborted, the ACU **SHALL** send only `osdp_POLL` or `osdp_ABORT` to that PD. The PD **SHALL** reject a competing command with `osdp_NAK` `0x09` (unable to process command record) without replacing the active operation. Traffic to other PDs is unaffected. On `osdp_ABORT`, the PD **SHALL** discard pending data or errors, cancel further delivery, and return `osdp_ACK`. It **SHALL NOT** subsequently report completion for that operation.

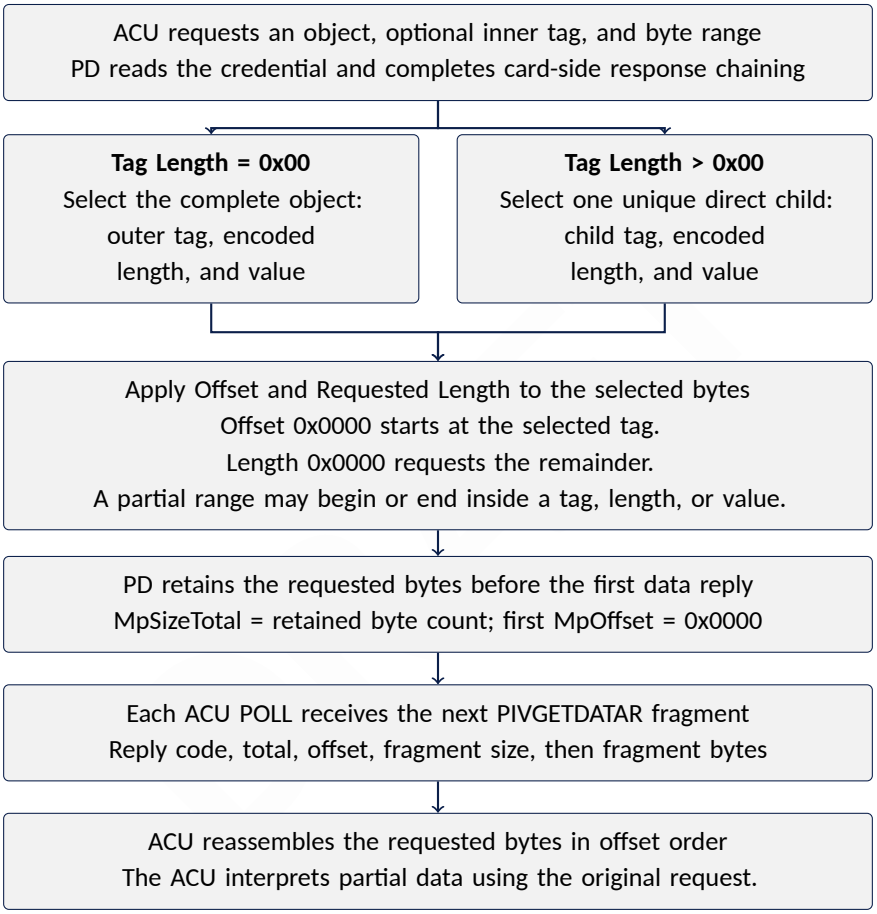


Figure 1: GET DATA selection precedes byte-range selection and OSDP fragmentation. ACK acknowledges acceptance. A failed read or selection ends with PIVERROR on a poll instead of data fragments. Child-selection errors are defined in the text.

The PD **SHALL** prepare and retain all requested data before sending the first data fragment. The data **SHALL** come from one credential operation. The PD **SHALL NOT** combine data from different credential presentations. Once delivery begins, the retained bytes remain unchanged. Card removal before all requested data is retained produces `0x1026` (credential not present or removed); removal afterward does not change the retained data. A card-operation timeout produces `0x0002` (smart-card timeout), insufficient PD resources produce `0x1021` (insufficient PD memory or storage), and unsatisfied credential security conditions produce `0x1023` (credential security status not satisfied). A failure produces one terminal `osdp_PIVERROR`, not an empty `osdp_PIVGETDATAR` followed by optional error detail. Error definitions appear in Subsection 2.2.

2.1.4 PIV Data Reply (osdp_PIVGETDATAR)

Table 4: PIV Data Reply (osdp_PIVGETDATAR)

Size (bytes)	Name	Meaning	Value
0x01	CMND	Reply identifier.	`0x80`
0x02	MpSizeTotal	Total number of requested data bytes across all fragments. Least-significant byte first.	`0x0000–0xFFFF`
0x02	MpOffset	Position of this fragment's first byte within the requested data. Least-significant byte first.	`0x0000–0xFFFF`
0x02	MpFragmentSize	Number of Card Data bytes in this reply. Least-significant byte first.	`0x0000–0xFFFF`
0x00-n	Card Data	The bytes returned in this reply, in their original order.	`0x00–0xFF`

The PD **SHALL** transfer the requested data using the OSDP 3.0 multi-part structure and usage rules. The ACU retrieves each fragment with `osdp_POLL`. Every reply contains the reply identifier, the three two-byte multi-part fields, and the next portion of the requested data. The multi-part fields use least-significant byte first, in the order shown in the table.

MpSizeTotal **SHALL** specify the number of requested data bytes, not the original object length. Its value **SHALL** remain unchanged throughout the transfer. *MpOffset* identifies the first byte of this fragment within the requested data. The first fragment **SHALL** have *MpOffset* zero, regardless of Offset in the command. Each subsequent fragment **SHALL** start immediately after the preceding fragment. *MpFragmentSize* **SHALL** equal the number of data bytes in that reply.

Fragment boundaries may occur anywhere within the requested data, including within a tag or length field. The PD **SHALL NOT** add or repeat TLV headers for individual fragments. The ACU reconstructs the requested data by joining the fragments in order. Reply identifiers, multi-part fields, and OSDP packet headers and trailers are not part of that data and are not counted in *MpSizeTotal* or *MpFragmentSize*.

For example, a request for 0x0104 bytes starting at Offset 0x0100 produces *MpSizeTotal* 0x0104. With 0x0072 data bytes per reply, the three fragments have *MpOffset* values 0x0000, 0x0072, and 0x00E4, and sizes 0x0072, 0x0072, and 0x0020.

For an empty success, the PD **SHALL** return one osdp_PIVGETDATAR with all three multi-part fields zero and no data bytes. The ACU **SHALL** interpret this first reply as successful completion with an empty result. For PIVGETDATAR, the early-termination indication applies only when *MpSizeTotal* is nonzero. For a nonempty result, delivery completes when the distinct accepted fragment lengths sum to *MpSizeTotal*. Normal OSDP retransmission rules apply: replaying a reply does not advance the data offset or execute the card operation again. The ACU **SHALL** discard partial data on cancellation or the OSDP 3.0 early-termination indication (zero fragment size and offset at or beyond total). That indication is not successful completion of a nonempty transfer.

2.2 PIV Error Codes

PD error codes are carried in osdp_PIVERROR using two-byte codes (least-significant byte first). The common PD codes remain available: 0x0001 (smart-card error), 0x0002 (smart-card timeout), 0x0004 (required TLV missing), 0x0005 (invalid parameter), 0x0006 (~~secure channel~~ Secure Channel 3.0 required), 0x0007 (invalid multi-part transfer), 0x0008 (multi-part fragment out of sequence), and 0x0009 (multi-part fragment out of bounds). The following PIV-specific codes occupy the 0x1000-0x10FF range:

Table 5: Enhanced PIV Error Codes

Code	Description
0x1020	Function not supported by credential (PD capability gaps **SHALL** use osdp_NAK function-not-supported).
0x1021	Insufficient PD memory or storage to complete the requested operation.
0x1022	Insufficient credential memory or storage space.
0x1023	Credential security status not satisfied.
0x1024	Data object or tag not found.
0x1025	VCI not established.
0x1026	Credential not present or removed.
0x1027	Ambiguous element selection: more than one direct child matches the requested tag.
0x1028	Malformed credential-object encoding ~~encountered during element extraction~~or invalid encrypted-reference length.
0x1029	Selected response exceeds the supported transfer length.
0x102A	Multiple contactless credentials detected; all presented cards rejected.

The assignments are tentative. Unlisted codes remain unassigned by this proposal.

2.2.1 PIV Error Reply (osdp_PIVERROR)

The PD **SHALL** use `osdp_PIVERROR` for errors in PIV operations that are not reported by a command-specific result. Each reply contains one two-byte error code, least-significant byte first. The originating command defines when the error is returned; PIVGETDATA errors are returned on a later poll. Transport faults continue to use `osdp_NAK`. This reply replaces the name `osdp_PDERROR`; its tentative byte value `0x8A` and payload structure are unchanged.

Table 6: PIV Error Reply (osdp_PIVERROR)

Size (bytes)	Name	Meaning	Value
0x01	CMND	Reply identifier.	`0x8A (tentative)`
0x02	Error Code	Two-byte error code, least-significant byte first.	`0x0001–0x10FF`
0x00 or 0x02	Status Word	SW1 followed by SW2 when a card command returned the failure being reported. Omitted for failures detected locally by the PD. See the status-word rules below.	–

When a credential command returns the failure being reported, the PD **SHALL** include that command's status word in SW1, SW2 order. For a valid protected response, the PD **SHALL** use the status word inside the SM response. The PD **SHALL** omit the status word for failures detected locally, including invalid OSDP parameters, missing trust anchors, timeouts, card removal, and failed verification of a protected response. A success status from an earlier credential command **SHALL NOT** be attached to a later failure.

2.3 Updated Function Codes

2.3.1 Function Code 12 – Smart Card Support

Function Code 12 advertises smart-card transport capabilities and profile support at the PD. The compliance byte uses the following bit assignments:

Table 7: Function Code 12 Compliance Flags

Mask	Description
`0x01`	PD supports transparent reader mode.
`0x02`	PD supports extended packet mode.
`0x04`	PD supports the OSDP 3.0 Enhanced PIV profile, including `osdp_PIVGETDATA` refinements and related commands.
`0x08`	PD supports OSDP 3.0 autonomous PIV operation (PIV Auto).
`0x10`	PD supports the TWIC Mode 2 Active Card Authentication profile of PIV Auto (tentative). Requires flags `0x04` and `0x08`; see Subsection 3.12.

Bits not listed are reserved and **SHALL** be transmitted as zero.

2.3.2 Function Code 13 – Reader Interfaces

OSDP 2.2 defined Function Code 13 as a reader count indicator with a compliance byte fixed at 0x00. Enhanced PIV-capable PDs **SHALL** continue to populate the reader count field, but they MAY set bits within the compliance byte to advertise the downstream credential technologies they expose. A value of 0x00 remains valid and retains the original OSDP 2.2 meaning of "interface presence unspecified."

When a PD reports specific interfaces, it **SHALL** encode the compliance byte using the flags in Table 8. The reader count byte continues to report the number of attached credential interfaces.

Table 8: Function Code 13 Compliance Flags

Mask	Description
0x01	PD presents a contact card interface (ISO 7816 or equivalent).
0x02	PD presents a high-frequency contactless credential interface (13.56 MHz NFC).
0x04	PD presents a low-frequency credential interface (approximately 125 kHz prox).
0x08	PD supports credential Secure Messaging and Virtual Contact Interface (VCI) for applicable interfaces and modes.
0x10	PD presents a Bluetooth credential interface.
0x20	PD presents a barcode credential interface.
0x40	PD presents an ultra-high-frequency credential interface (UHF/RAIN RFID).

All other bits are reserved for future use and **SHALL** be transmitted as zero.

2.3.3 Function Code 14 – Biometric Support

Function Code 14 remains backward compatible with OSDP 2.2. A compliance value of 0x00, 0x01, 0x02, or 0x03 retains its original meaning and indicates that the PD does not support the OSDP 3.0 biometric command set. Devices that support the enhanced commands **SHALL** set Bit 7 (0x80). When Bit 7 is asserted, the remaining bits of the compliance byte are interpreted as a capability bitfield. Bits not explicitly defined in Table 9 are reserved and **SHALL** be transmitted as zero.

Table 9: Function Code 14 Compliance Flags

Mask	Description
0x80	PD supports the OSDP 3.0 biometric command set.
0x01	PD performs off-card fingerprint comparison using FIPS 201 BIO ANSI INCITS 378 templates.
0x02	PD performs on-card fingerprint comparison using FIPS 201 OCC processes.
0x04	PD performs off-card iris comparison aligned with FIPS 201 BIO guidance.
0x08	PD performs off-card facial comparison using ANSI INCITS 385-2004 images stored on the credential.

2.3.4 Function Code 15 – Secure PIN Entry Support

For a PD advertising Enhanced PIV (Function Code 12 bit 0x04), Function Code 15's compliance byte is the bitmask below. Values 0x00 and 0x01 retain their OSDP 2.2 Appendix B.16 meanings: no SPE support and legacy SPE support, respectively. The ACU **SHALL** confirm Enhanced PIV support before interpreting the additional bits.

Table 10: Function Code 15 Compliance Flags

Mask	Description
0x01	PD supports OSDP 2.2 XWR Mode-01 Secure PIN Entry and PR01SPER.
0x02	PD supports explicit PIVSPE commands and PIVSPER replies.
0x04	PD supports ACU-armed PIVSPE Auto Execute when a credential is presented. Requires bit 0x02.

Bit 0x04 requires bit 0x02. Other bits are reserved and **SHALL** be transmitted as zero. Auto Execute begins only after the ACU arms it using PIVSPE, as specified in the command definition.

3 New or Enhanced PIV Commands

This section introduces commands defined exclusively for the OSDP 3.0 Enhanced PIV profile.

Unless specified otherwise, each multi-part command or reply defined in this section places *MpSizeTotal*, *MpOffset*, and *MpFragmentSize*, in that order, immediately after *CMND*. Each field is a two-byte unsigned integer encoded least-significant byte first. The bytes following *MpFragmentSize* are a fragment of the logical message data: *MpSizeTotal* is the complete logical-data length, *MpOffset* is the zero-based position of the fragment, and *MpFragmentSize* is the exact number of fragment bytes in the current message. The receiver **SHALL** apply the OSDP multi-part sequencing and bounds rules and reassemble the complete logical data before interpreting command-specific fields. Fixed fields shown after the multi-part fields in a packet table occur at their stated position in the reassembled data and are not repeated in every fragment. The separate `osdp_FILETRANSFER` framing remains unchanged.

3.1 Credential preparation and Secure Messaging

OSDP Secure Channel 3.0 protects communication between the ACU and PD. Credential Secure Messaging (SM) protects credential commands and replies between the PD and the credential. Credential SM can operate over contact or contactless. The Virtual Contact Interface (VCI) allows a contactless credential to perform operations that otherwise require contact, subject to the credential's access rules. When the credential requires pairing, the PD establishes SM before submitting the eight-digit pairing code. Pairing is the default, recommended configuration in SP 800-73-5 Part 1, Section 5.5. The pairing code and the cardholder's PIN serve separate purposes: pairing enables VCI; PIN verification authorizes PIN-protected operations.

Credential presentation The PD **SHALL** select the configured application on an interface allowed by `osdp_PIVMODE`. For a PIV application, the PD **SHALL** inspect the SELECT response's application property template (tag `0x61`). Its cryptographic algorithm template (tag `0xAC`) contains algorithm identifiers in tag `0x80`. The PD **SHALL** examine all such entries. An identifier of `0x27` or `0x2E` advertises credential SM; SP 800-73-5 Part 2, Section 3.1.1, requires this indication when the PIV application supports SM. The PD **SHALL** distinguish a malformed response from a valid response that advertises neither SM algorithm.

When a TWIC application is enabled in PIVMODE, the PD **SHALL** reject all cards if two or more contactless cards are detected in the RF field (TWIC Part 3, Section 8, item 3). It **SHALL** cancel the current credential operation, clear its credential context, and report PIVERROR `0x102A`; an active Auto attempt instead reports PIVSTATUS Result `0x05`, Detail `0x04`, and a pending SPE credential operation reports PIVSPER Result `0x08` (security condition not met). During this condition the PD reports no selected credential or AID in CARDSTATUSR and emits no credential identifier. After the transaction ends, the PD **SHALL** observe an empty RF field continuously for at least one second before reading a new contactless presentation (Section 8, item 4). ACA, data reads, and commanded biometric verification within one uninterrupted transaction share the same presentation.

Required credential SM cipher suites A PD supporting credential SM preparation through PIVMODE flag `0x02` **SHALL** implement both CS2 (`0x27`) and CS7 (`0x2E`) and use the suite advertised by the selected PIV application. SP 800-73-5 Part 2, Section 4.1.4 and Table 18, define their ECDH curves, CVC signatures, KDFs,

nonces, and AES key sizes. An application supporting VCI with a P-384 digital signature, key-management, or retired key-management key requires CS7 under that section. The PD **SHALL** validate the CVC and perform key establishment and SM with the advertised suite's parameters.

Discovery Object The Discovery Object (`0x7E`) describes the credential's PIN and VCI policy. Its PIN Usage Policy (tag `0x5F2F`) identifies the supported PIN options, the preferred PIN when both PIN types are enabled, and whether VCI requires pairing. The PD uses this information to prepare PIN and VCI operations. SP 800-73-5 Part 1, Section 3.3.2, defines the policy bytes.

The PD **SHALL** read the Discovery Object when preparing a credential-side PIN operation or attempting SM setup, unless the PD already holds the policy for the selected application on the current credential. Automatic reads **SHALL** be limited to those workflows. Collecting a PIN for later use requires no credential read. A status query reports the policy already read, or `0xFFFF` when unavailable. The ACU can also request the object explicitly with `osdp_PIVGETDATA`.

In the first policy byte, mask `0x08` indicates VCI support. When VCI is supported, mask `0x04` permits VCI without a pairing code; otherwise pairing is required. If the object is absent, VCI is unavailable. The SELECT response remains the source of SM capability information.

Automatic SM preparation After selecting a PIV application, the PD checks the requirements for SM setup. If PIVMODE flag `0x02` is set, the PD supports an SM algorithm advertised by the credential, and at least one trust anchor is loaded, the PD **SHALL** attempt SM setup. This applies to both contact and contactless credentials, in commanded mode and PIV Auto.

Selecting a different application starts this check again. Selecting the same PIV application again preserves the current SM session and PIN-verification status. Reselection alone therefore requires neither new SM setup nor repeated PIN verification. The credential's access rules still apply to each operation. These selection rules follow SP 800-73-5 Part 2, Section 3.1.1.

The PD's trust store contains the anchors loaded through `osdp_PIVVCILOADTA`. The PD checks its own store before starting SM. An empty store prevents setup. If anchors are present, the PD must still find one that can validate this credential's certificate, directly or through an intermediate certificate. Loading an anchor makes it available for the next attempt. To start a new attempt with a credential already present, the ACU reapplies ordinary PIVMODE~~with a fresh Auto Series; the~~ ; when Auto is enabled, it supplies a fresh Series. The mode-change clearing rules apply. The Resume extension in Subsection 3.12 continues only an eligible pairing/PIN prerequisite failure and does not restart blocked SM.

The credential returns its SM card-verifiable certificate (CVC) during key establishment. The PD **SHALL** validate that certificate using a loaded anchor, directly or through an intermediate CVC, and verify the authentication cryptogram as specified in SP 800-73-5 Part 2, Section 4.1. The PD **SHALL** report SM as established only after those checks succeed.

SM identifier and ephemeral-key encoding For both CS2 and CS7, the PD **SHALL** compute $ID_{sICC} = T_8(\mathrm{SHA256}(C_{ICC}))$, where T_8 selects the leftmost eight bytes and C_{ICC} is the complete

encoded SM CVC, including the outer 0x7F21 tag and length. SP 800-73-5 Part 2, Section 4.1.1, Table 15, step H6, defines this identifier. The KDF uses SHA-256 for CS2 and SHA-384 for CS7 under Section 4.1.6 and Table 18.

The PD **SHALL** encode the ephemeral point coordinates as fixed-width, big-endian $X \| Y$: 32 bytes per coordinate for CS2 and 48 for CS7. For KDF *OtherInfo*, it **SHALL** use the first 16 bytes of $X \| Y$. For key confirmation, the CMAC input **SHALL** be KC_1_V $\| ID_{sICC} \| ID_{sH} \| X \| Y$, using the full coordinates. These encodings follow Sections 4.1.6–4.1.7. The GENERAL AUTHENTICATE command carries the uncompressed point 0x04 followed by $X \| Y$, as specified in Section 4.1.8. Thus the point-format byte is present in the command encoding; the KDF and key-confirmation inputs consist of the coordinate bytes above.

Proceeding without SM When SM is disabled, unsupported, or unavailable because a trust anchor is missing, the PD **SHALL** check whether the operation requested by the ACU or configured in PIV Auto requires SM. PUT DATA requires SM whenever PIVMODE flag 0x02 is set. If SM is required, the PD **SHALL** terminate the operation without sending the operation's command to the credential and return the defined failure reply. Otherwise, the PD **SHALL** perform the requested operation under the credential's access rules. The PD **SHALL NOT** substitute another key, algorithm, interface, or operation to avoid an SM requirement.

PIV key 0x9A requires PIN verification or a supported biometric comparison within the credential, as specified in SP 800-73-5 Part 1, Section 5.1, Table 5. The PIN-based contactless PIV Auto workflow with key 0x9A therefore requires SM, VCI, and successful PIN verification. The PD **SHALL** check those requirements in that order before sending the authentication command. Missing PIN verification returns Result 0x05, Detail 0x02 (PIN verification required). If SM is unavailable, the PD reports osdp_PIVSTATUS Result 0x02 (SM failure) with the applicable detail code. PIV Auto with key 0x9E can continue without SM and reports the credential-authentication result. A commanded PIN verification over contact can also proceed without SM.

The PD may start key establishment before discovering that the required trust anchor is missing. In that case, the PD **SHALL** discard any session keys derived during setup. The PD then handles the requested operation using the rules above.

In commanded mode, the ACU obtains the preparation result with osdp_CARDSTATUS. The PD reports the established SM and VCI state in osdp_CARDSTATUSR. A missing anchor leaves permitted data reads available. For example, the ACU can use osdp_PIVGETDATA to read the Secure Messaging Certificate Signer object (0x5FC122) and inspect the signer certificate and any intermediate CVC. The credential's own SM CVC is returned during key establishment; that CVC is separate from the signer data object. The ACU can request osdp_CARDSTATUS Status Type 0x02 to obtain the setup outcome and the hash of the CVC received during that attempt, including when setup failed.

Failed security checks A missing or invalid credential CVC, invalid authentication cryptogram, or failure to verify an SM reply **SHALL** end the current operation with a failure result. The PD **SHALL** block further data access and authentication commands until the credential is removed, a different application is successfully selected, or PIVMODE is reapplied. PIVSELECT remains available to select a different application; selecting the same application preserves the blocked state. This prevents continued use of a session that failed its

security checks. The rule also applies when automatic SM setup fails before an ACU command arrives. PIV-MODE, trust-store management, and SM diagnostic requests remain available. For a standard status request (Status Type `0x01`), the PD terminates the request without sending further commands or data to the credential and returns `osdp_PIVERROR` `0x1023` (credential security conditions not satisfied). This distinguishes a blocked credential from operation without SM.

For blocked operations, the PD reports `osdp_PIVSPER` `0x08` (security condition not met) for SPE, or `osdp_PIVERROR` `0x1023` (credential security conditions not satisfied) for commands that use PIVERROR.

If an operation requires SM, VCI, or PIN verification and that requirement is unmet, the PD **SHALL** report failure in the operation's completion reply. PIV Auto uses `osdp_PIVSTATUS`; SPE uses `osdp_PIVSPER`.

Figure 2 summarizes credential preparation and the decision to proceed with the requested operation.

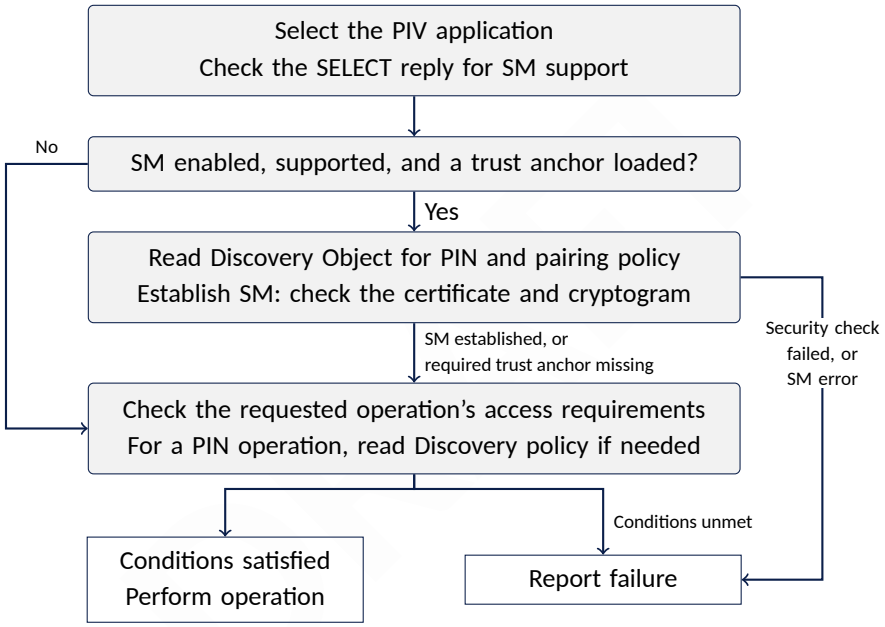


Figure 2: How the PD prepares a credential. The PD establishes SM when enabled and supported, using a loaded trust anchor. The credential's access rules determine which operations can proceed. Contactless PIN verification requires VCI.

Contactless operation When pairing is required, the ACU sends `osdp_PIVXMITPAIRING` after SM is established. The PD submits the pairing code to the credential through SM. A successful verification enables VCI. For a credential whose Discovery Object permits VCI without pairing, establishing SM also enables VCI. PIN verification over contactless requires VCI. Biometric comparison within the credential and pairing-code submission can use contactless SM without VCI, subject to the credential's access rules. SM and VCI preserve the credential's operation-specific restrictions, including the contact-interface restriction on ordinary PIV PUT DATA.

Protected credential commands When credential SM is established, the PD **SHALL** use that SM session to protect PIV commands that support SM. The PD **SHALL** verify and unwrap each protected reply before interpreting the credential data or status. These requirements apply to data access, authentication, PIN verification, and biometric operations. The PD **SHALL** follow SP 800-73-5 Part 2, Section 4.2, for encryption, MACs, counters, and credential-command and response chaining. The PD's GET RESPONSE exchanges follow that section's chaining rules.

The PD **SHALL** discard the SM session keys on credential reset, an SM error, or replacement by newly established session keys, as specified in SP 800-73-5 Part 2, Section 4.3. This requirement applies to the outer SW processing status defined in Section 4.2.7. The protected command status in tag 0x99 reports the result of the credential operation separately. When SM ends, the PD **SHALL** clear dependent VCI and PIN-verification status and apply the PIN-cache clearing rules.

A verified protected reply reporting a credential rejection **SHALL** end the requested operation with that failure while preserving SM. The PD **SHALL** update verification status according to the credential's response. An incorrect PIN (0x63Cx) clears PIN-verification status and the cached PIN; established pairing and VCI remain available. An incorrect pairing code (0x6300) clears pairing verification, making VCI unavailable when pairing is required. These credential rules are defined in SP 800-73-5 Part 2, Sections 3.2.1.1 and 3.2.1.3. The SM diagnostic outcome remains unchanged. A failed certificate, cryptogram, or protected-reply check follows the blocked-state rules above.

After an authenticated PIN rejection, the ACU can query remaining attempts with PIVSPE Mode 0x06, acquire a corrected PIN, and request verification again. After a pairing rejection, the ACU can submit a corrected pairing code. Neither retry requires a new SM session.

After an SM failure, the ACU **SHALL** reapply PIVMODE to start a new SM attempt before retrying a protected operation. For a commanded retry, the ACU disables PIV Auto in that configuration, waits for SM establishment, supplies any required pairing code, and requests PIN acquisition and verification again. The PD **SHALL NOT** retry VERIFY automatically. Figures 3 and 4 show the ACU and PD exchanges after authenticated rejection. Figure 5 shows recovery after SM failure. The PD **SHALL** keep a completed operation's reply until it is delivered or canceled.

Commanded PIN verification For a PIN entered before credential presentation, the ACU first requests Mode 0x04 (acquire and cache) and waits for completion. After credential presentation, the PD prepares the credential. The ACU checks credential status and supplies any required pairing code before requesting cached-PIN verification. Figure 6 shows a successful contactless workflow with SM enabled, a suitable trust anchor loaded, and PIV Auto disabled. The ACU obtains the pairing code during enrollment or through another deployment-defined process. On contact, pairing is omitted and Mode 0x03 uses the selected contact interface, with SM when established.

ACU recovery after a PIV Auto failure PIV Auto reports one result for the configured authentication attempt. The ACU can then request SM diagnostics, correct missing trust material, or choose another authentication workflow. Loading an anchor makes that anchor available to the PD. Reapplying ordinary PIVMODE starts a new automatic attempt and clears the previous PIN and SM state. The ACU **SHALL** finish trust recovery before pairing and PIN verification that depend on the new SM session.

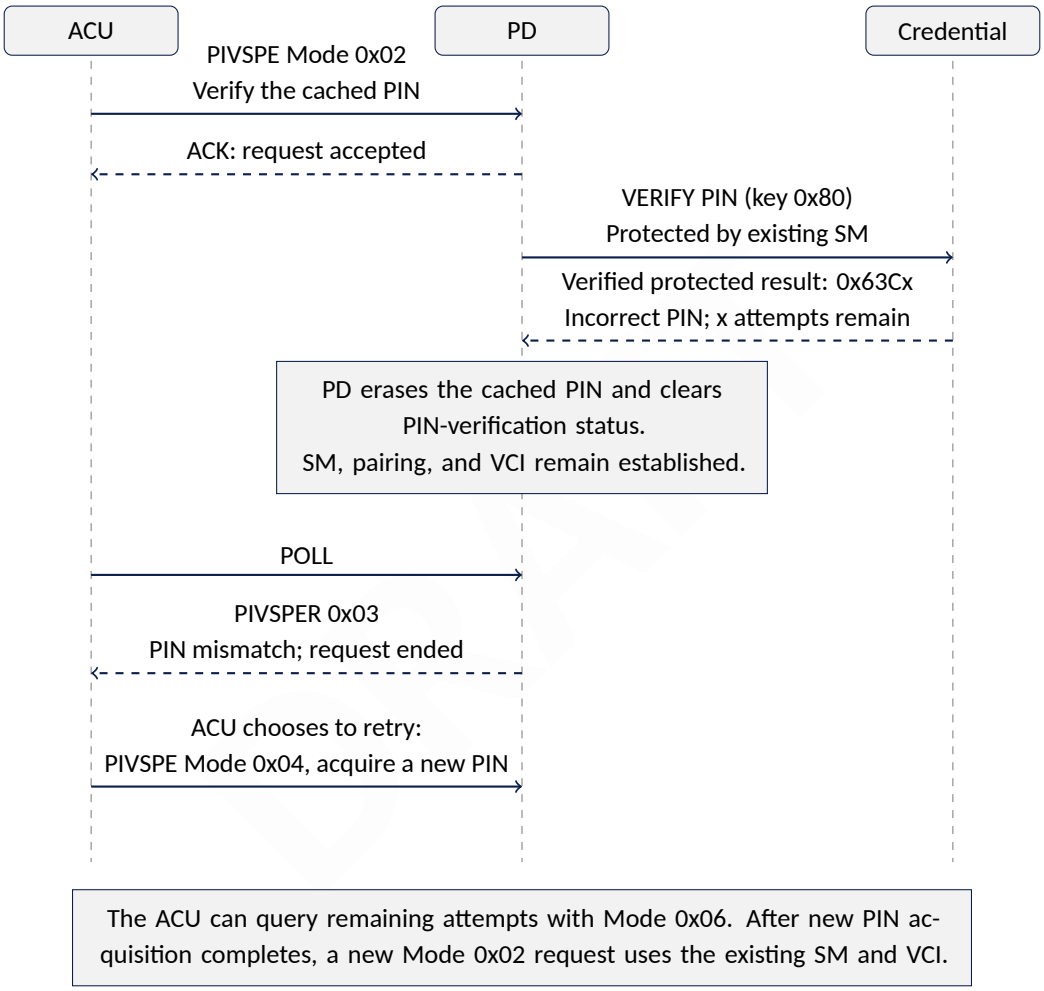


Figure 3: Incorrect PIN over established VCI. The PD verifies the protected reply before reporting the rejection. PIN entry occurs at the PD; the ACU never receives the PIN. Acquisition and its completion follow the normal ACK/POLL exchange.

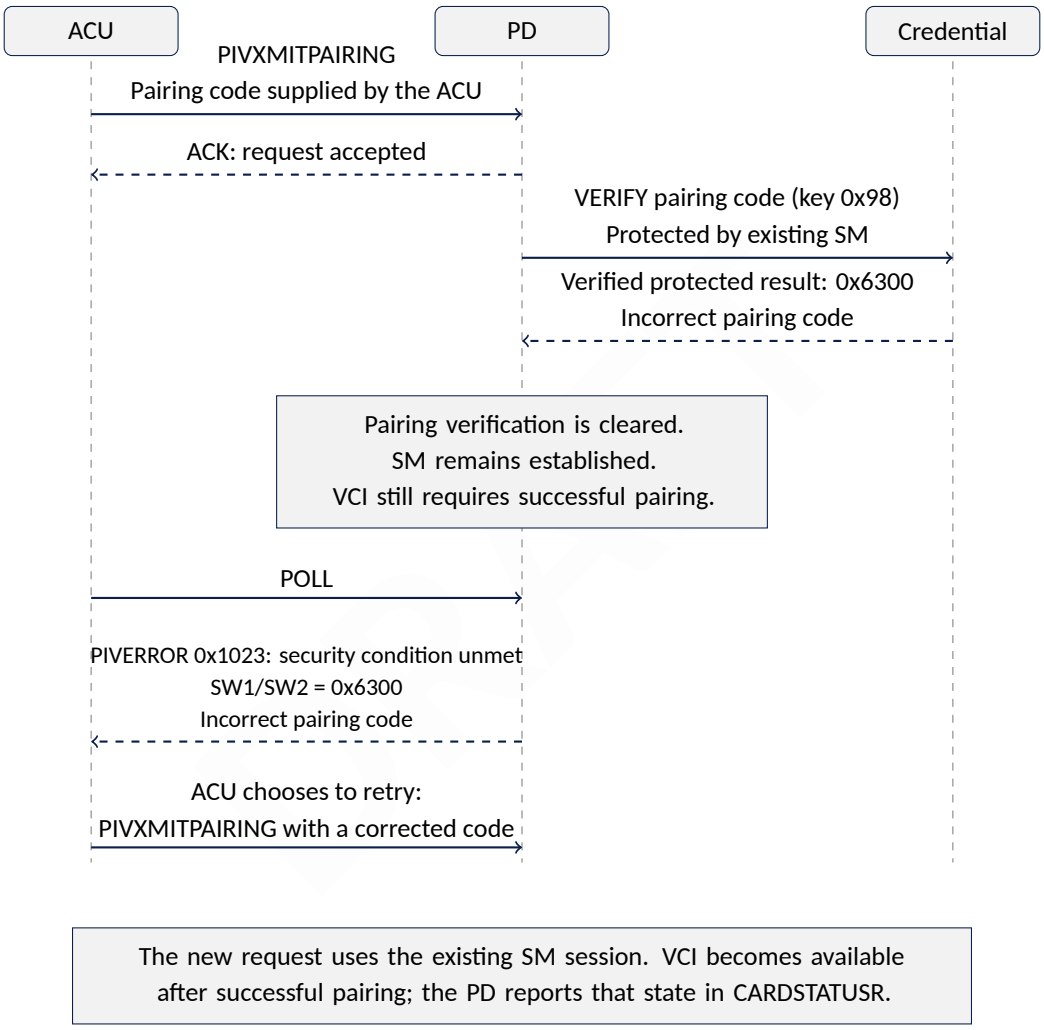


Figure 4: Incorrect pairing code on a contactless credential that requires pairing. The pairing code originates at the ACU. Each retry is a new command; the PD never repeats VERIFY automatically.

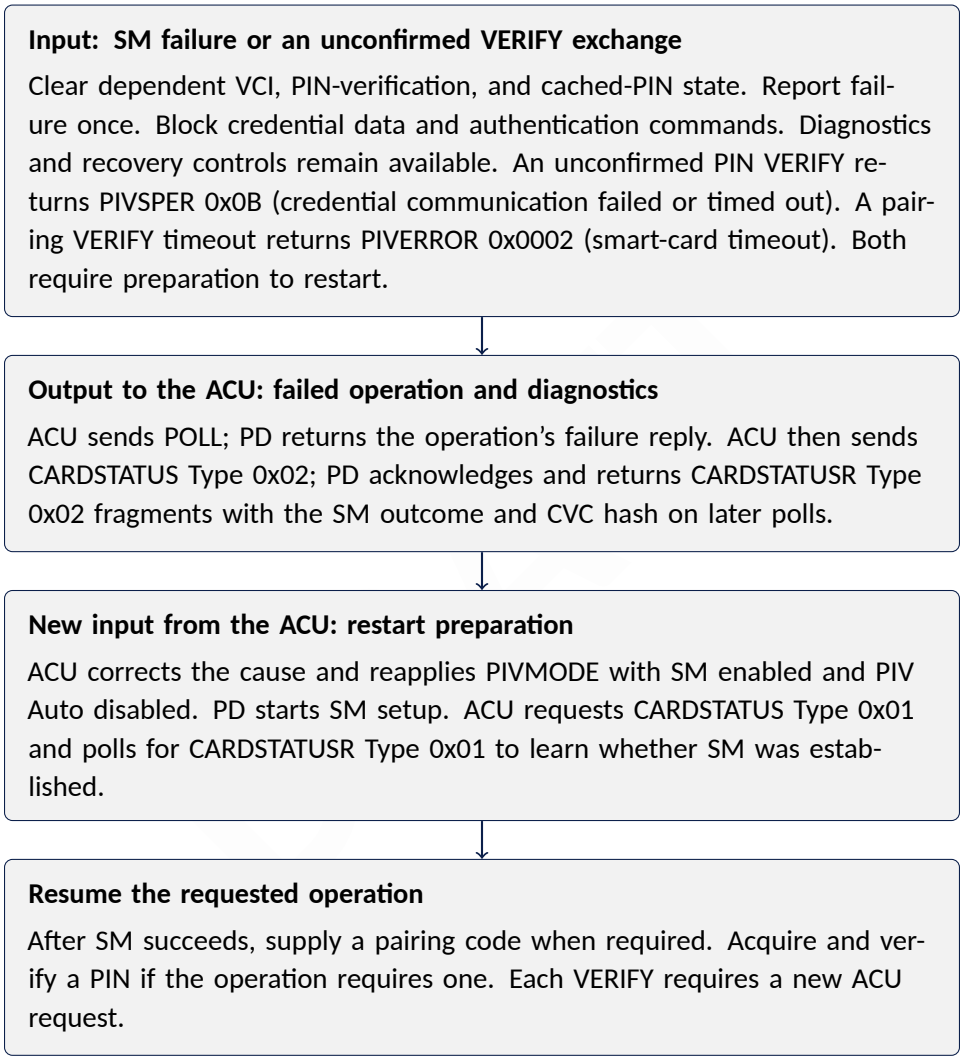


Figure 5: Recovery after SM failure or an unconfirmed VERIFY. The ACU receives the failed operation's result before requesting diagnostics and restarting preparation.

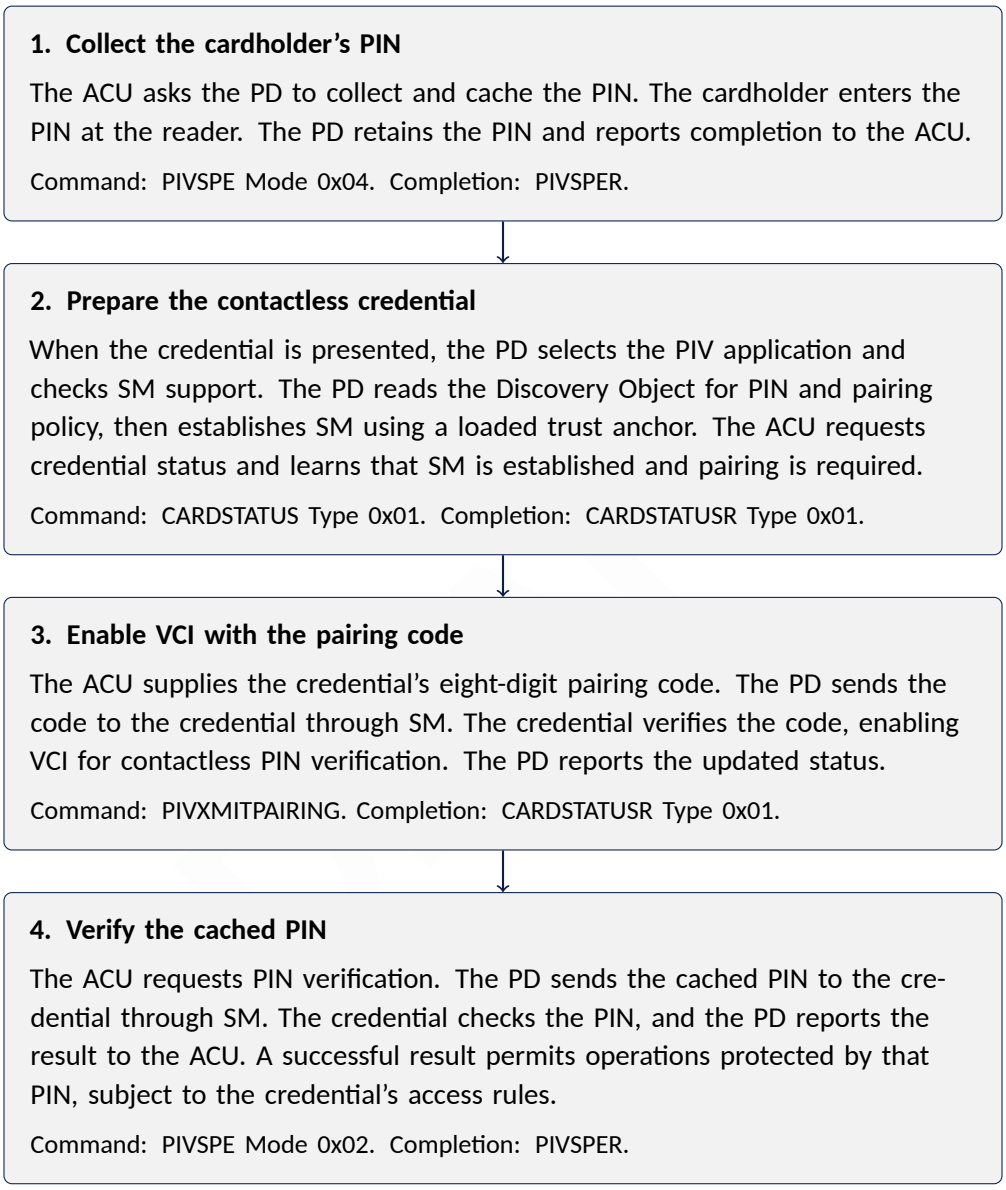


Figure 6: Contactless PIN verification with a pairing code. The ACU directs the reader (PD); the credential verifies the pairing code and PIN. Each OSDP command receives ACK when accepted and a completion reply on a later POLL. Appendix D shows the packet exchanges.

After complete delivery of an eligible PIVSTATUS 0x05/0x01 or 0x05/0x02, the ACU may complete commanded pairing and PIN verification while Auto remains enabled, then send PIVMODE Resume with that status's pair and the next unused counters. Resume preserves completed SM, VCI, pairing, and PIN state instead of ordinary mode-change clearing. It performs one additional attempt with the same entry and key, without automatic VERIFY or authentication retry. Context loss, blocked SM, and counter exhaustion follow Subsection 3.12; they do not permit implicit fallback.

When SM is established but the configured authentication requires unsupported VCI, the PD **SHALL** terminate the attempt and report osdp_PIVSTATUS Result 0x05, Detail 0x03 (required VCI unsupported). The PD sends no authentication command for that attempt. Missing SM support is reported as Result 0x02, Detail 0x05 (required SM unsupported). A trust-anchor update cannot add either capability. The ACU must select a supported workflow or reject the transaction. Appendix D.12 shows trust recovery followed by commanded pairing, PIN verification, and authentication.

3.2 PIV Put Data (osdp_PIVPUTDATA)

The osdp_PIVPUTDATA command (command code 0xA8) ~~instructs the PD to deliver an ISO 7816-4 PUT DATA operation to the credential~~delivers a complete BER-TLV write payload to the selected credential application. The ACU ~~**SHALL** construct the payload as a complete BER-TLV object , including the desired outer tag (~~supplies the object encoding required by the card: 0x7E for the Discovery Object, 0x7F61 for a BIT Group Template, or the 0x5C/0x53 ~~sequences for other data objects, and so on)~~sequence for other PIV objects. For TWIC Data Model 0x02, a NEXGEN free read/write E-Sticker uses its native 0xE1–0xEA tag, length, and value, as specified in TWIC Part 2, Sections 4.7.8 and 5.4 and Appendix F. The PD **SHALL** ~~forward the payload to the credential using CLA = 0x00,~~ issue PUT DATA to the application selected by PIVMODE or a successful PIVSELECT, using that application's Data Model and interface.

With PIVMODE flag 0x02 set, credential SM is required for PUT DATA. The PD **SHALL** complete SM establishment and validation before issuing the write, then protect the command and response using that session. Missing trust anchors, unsupported SM, or setup failure **SHALL** terminate the request with PIVERROR 0x1023 before issuing PUT DATA. An established session is reused. With flag 0x02 clear, the PD **SHALL** issue ordinary PUT DATA. Ordinary PIVMODE replacement clears the previous credential-SM state.

The card command uses INS = 0xDB, P1 = 0x3F, and ~~P2 = 0xFF. If credential Secure Messaging (SM) is established, the PD **SHALL** use the established SM context and adjust the CLA accordingly. Chained APDUs **SHALL** be generated by the PD when the payload exceeds a single-card APDU .~~ P2 = 0xFF. Ordinary PUT DATA uses CLA 0x00, or 0x10 for non-final command-chain fragments, and absent Le. Protected PUT DATA uses CLA 0x0C, or 0x1C for non-final fragments, with encryption and MAC processing from SP 800-73-5 Part 2, Section 4.2. The logical command has absent Le, so its SM encoding omits tag 0x97; the protected APDU uses the outer Le 0x00 specified by Section 4.2.4. The PD protects the complete logical command before command chaining and verifies the complete protected response before accepting the card's application status.

Table 11: PIV Put Data (osdp_PIVPUTDATA)

Size (bytes)	Name	Meaning	Value
0x01	CMND	Command identifier.	0xA8 (tentative)
0x02	MpSizeTotal	Total size of the complete PUT DATA payload, least-significant byte first.	0x0000–0xFFFF
0x02	MpOffset	Offset of this fragment within the complete payload, least-significant byte first.	0x0000–0xFFFF
0x02	MpFragmentSize	Number of payload bytes carried in this fragment, least-significant byte first.	0x0000–0xFFFF
0x00-n	Data	BER-TLV payload to be written to the credential. This field already includes the outer object tag selected by the ACU.	0x00–0xFF

Responses ~~PIV~~ For PIV Data Model 0x01, PUT DATA requires the contact interface, including when SM is established. ~~On~~ For that model on contactless, the PD **SHALL** terminate the request without sending PUT DATA to the credential and return osdp_PIVERROR 0x1023 (security conditions not satisfied). ~~A successful write is~~ For TWIC Data Model 0x02, NEXGEN free read/write E-Stickers support both contact and contactless (TWIC Part 2, Section 4.5, access table). ACKs for non-final OSDP fragments acknowledge receipt. The final successful completion **SHALL** be acknowledged with osdp_ACK ~~.~~ after the card reports application status 0x9000, verified under SM for a protected write. A failed protected write terminates the request through the existing error path. A timeout after submission reports unconfirmed completion; another write requires a new ACU request. Normal OSDP retransmissions replay the existing reply. If the PD does not implement the command, the PD **SHALL** return osdp_NAK function-not-supported. For a credential-side failure, the PD **SHALL** report osdp_PIVERROR, either immediately or on a subsequent osdp_POLL. The applicable codes are 0x1020 (function not supported by credential), 0x1021 (insufficient PD buffer space), 0x1022 (insufficient credential storage), 0x1023 (security conditions not satisfied), and 0x1026 (credential absent or removed). The status-word rules in Subsection 2.2.1 apply.

Provisioning examples Both examples use contact cards and Secure Channel 3.0 on the ACU–PD link. In each case, the card-management system establishes write authorization in the selected application after selection and preparation, before sending PIVPUTDATA.

Card A, protected PUT DATA. Card A supports protected provisioning. The ACU loads its SM trust anchor and installs commanded PIVMODE data 0201010103010100: flags 0x02, one entry, contact interface, PIV data model, and default PIV AID. The PD sends ordinary SELECT, establishes and validates SM, and reports security status 0x02 in CARDSTATUSR. The ACU sends PIVPUTDATA with the BER-TLV write payload. The PD sends

the protected card command and acknowledges completion after verifying the protected 0x9000 response. Failed SM preparation returns PIVERROR 0x1023 before the write.

Card B, ordinary PUT DATA. Card B accepts ordinary provisioning. The ACU installs commanded PIVMODE data 0001010103010100, with flag 0x02 clear. The PD sends ordinary SELECT and reports security status 0x00. The same PIVPUTDATA payload becomes `00 DB 3F FF Lc Data`, with absent Le. Card status 0x9000 completes the write; a rejected write follows the existing PIVERROR path.

TWIC NEXGEN E-Sticker example Over Secure Channel 3.0, the ACU installs commanded PIVMODE data 00 0102010C020209A00000036720000001: flags zero, one TWIC entry, contactless global and entry masks, and the common nine-byte TWIC AID prefix. The PD selects the TWIC application; the ACU confirms a NEXGEN AID in CARDSTATUSR before writing. Alternatively, PIVSELECT with Data Model 0x02 and that prefix selects the TWIC application on the current interface.

The ACU sends PIVPUTDATA with the native E-Sticker TLV E1058103414243, containing child tag 0x81 with value ABC. The PD issues 00DB3FFF07E1058103414243, with absent Le, and acknowledges completion after card status 0x9000. PIVGETDATA A301E10000000000 reads the complete object through 00CB3FFF035 C01E1FF; PIVGETDATAR returns E1058103414243. TWIC Part 2, Sections 5.2–5.4 and Appendix F, define these object transfers. A site-signed PACS authorization record uses the same transfer: the ACU supplies and verifies its signature, card-identifier binding, and validity, following the use described in TWIC Part 4, Section 3.1.

3.3 PIV Application Select (osdp_PIVSELECT)

The `osdp_PIVSELECT` command (command code 0xB0, tentative) instructs the PD to select the application identified by the supplied Application Identifier (AID) on the currently active credential. The command provides an OSDP-native selector for ISO/IEC 7816-4 `SELECT by DF name`~~; it is not a raw APDU tunnel and does not expose APDU header bytes to the ACU~~. PDs **SHALL** advertise support via Function Code 12 bit 0x04 (Enhanced PIV) before controllers attempt to use this command.

Table 12: PIV Application Select (osdp_PIVSELECT)

Size (bytes)	Name	Meaning	Value
0x01	CMND	Command identifier.	0xB0 (tentative)
0x01	Selection Flags	Selection controls. All bits are reserved for future use and **SHALL** be transmitted as zero.	0x00
0x01	Data Model	Data model for subsequent commands: 0x01 PIV, 0x02 TWIC, 0x03 CAC; 0x04-0x7F reserved; 0x80-0xFF private use. 0x00 is invalid.	0x01–0xFF
0x01	AID Length	Length of the Application Identifier.	0x05–0x10

Size (bytes)	Name	Meaning	Value
0x05-0x10	AID	Application Identifier to select by ISO/IEC 7816 DF name.	0x00–0xFF

The *Data Model* field uses the same identifiers as a osdp_PIVMODE Configuration Entry: 0x01 PIV, 0x02 TWIC, and 0x03 CAC. Values 0x04-0x7F are reserved; 0x80-0xFF are private-use models supported by agreement between the ACU and PD. 0x00 is invalid. The ACU supplies the model and AID together; a matching osdp_PIVMODE Configuration Entry is not required. The command retains the current interface and mode-wide security settings. The ~~model does not override the~~ credential's access conditions apply to the selected data model. The payload is exactly four fixed bytes, including *CMND*, followed by *AID Length* bytes, for a total of 0x09-0x14 bytes. This layout replaces the earlier draft without a Data Model byte; PDs **SHALL NOT** interpret that earlier layout as this command.

The PD **SHALL** issue SELECT with CLA = 0x00, INS = 0xA4, P1 = 0x04, P2 = 0x00, Lc = AID Length, and Data = AID~~, and Le = 0x00~~. The PD selects a nonzero Le and handles response continuation under paragraph 2.1.2. The PD **SHALL** request the normal application property template over the interface selected by the current PIV mode. ~~A selection within an established SM context follows the shared transport rules in~~ SELECT **SHALL** use its ordinary card-command form, including when a credential SM session is established. Session preservation and clearing follow Subsection 3.1. When a successful selection changes the application or Data Model, the PD **SHALL** clear the previous application's SM session, VCI status, pairing code, cached PIN, PIN-verification status, and application information. ~~This prevents the PD from carrying security state from one application into another.~~ Selecting the PIV application when a credential is first presented preserves a PIN collected before presentation. Selecting the same PIV application again with the same Data Model preserves its SM session and PIN-verification status. If the credential rejects SELECT, the PD keeps the current application and its state.

The *AID Length* field **SHALL** be in the ISO/IEC 7816 application identifier range of 5 to 16 bytes. If the payload is malformed, its size is not exactly 0x04 plus AID Length, the AID length is outside that range, the Data Model is zero, reserved, or unsupported by the PD, or any reserved selection flag is nonzero, the PD **SHALL** report osdp_PIVERROR 0x0005 (invalid parameter) without issuing SELECT to the credential. A successful selection is acknowledged with osdp_ACK. Only after success, the PD **SHALL** commit the selected Data Model and AID as one operating context. Subsequent commands use that context until credential removal, another successful selection, or a mode change. A rejected command or credential SELECT **SHALL NOT** change the previous context. The PD **SHALL** update the selected AID reported by osdp_CARDSTATUSR; if the credential returns an FCI template containing tag 0x4F, that value **SHALL** be reported, otherwise the PD **SHALL** report the requested AID. The ACK and CARDSTATUSR layouts are unchanged.

On a TWIC credential, the PIV and TWIC applications have distinct AIDs. The ACU determines the intended application through system configuration, using an osdp_PIVMODE configuration entry or osdp_PIVSELECT. For osdp_PIVSELECT, the ACU supplies Data Model 0x01 with the PIV AID, or Data Model 0x02 with the TWIC AID; 0x02 covers both Legacy and NextGen TWIC applications. The PD **SHALL** apply the data model and access conditions of the selected application to subsequent commands. Selecting the PIV application invokes normal PIV processing, including PIN, SM, and VCI requirements where applicable. Biometric data access through the TWIC application does not require the PIV PIN; deciphering its encrypted biometric data requires the appropriate TPK.

Informative selection examples The following hexadecimal command payloads omit OSDP framing and Secure Channel 3.0 for readability. The bytes are CMND, Selection Flags, Data Model, AID Length, then AID; neither example depends on a matching PIVMODE profile.

- Select the PIV application, including the PIV application on a TWIC:
  `B0 00 01 0B A0 00 00 03 08 00 00 10 00 01 00`.
- Select the TWIC application using its common nine-byte AID prefix:
  `B0 00 02 09 A0 00 00 03 67 20 00 00 01`.

After ACK, CARDSTATUSR reports the selected AID. A full TWIC AID ending in `01 01` identifies Legacy; `01 03` identifies NextGen V8, and `01 04` identifies NextGen V10. For major version `0x01`, an otherwise valid, non-reserved minor version not listed here **SHALL NOT** be rejected solely because that minor version is unknown. The PD reports the full selected AID and applies the backward-compatible TWIC data model, without assuming support for additional features from the unknown minor version alone. If only the requested prefix is available, it does not identify the generation; the ACU **SHALL NOT** infer NextGen facial support from that prefix alone. These AID distinctions are specified in *TWIC Specification*, *Part 2*, Section 4.1 and Appendix C.

If the PD does not implement the command, the PD **SHALL** return `osdp_NAK` function-not-supported. Credential-side failures **SHALL** be reported using `osdp_PIVERROR`: `0x1020` (application selection unsupported), `0x1023` (security conditions not satisfied), `0x1024` (requested AID absent), or `0x1026` (credential absent or removed). The status-word rules in Subsection 2.2.1 apply.

3.4 Card Status (osdp_CARDSTATUS)

The `osdp_CARDSTATUS` command (command code `0xA9`, tentative) requests the current credential context from the PD. PDs **SHALL** advertise support via Function Code 12 bit `0x04` (Enhanced PIV) before controllers attempt to use this command. The payload selects standard credential status or SM diagnostics.

Table 13: Card Status Request (osdp_CARDSTATUS)

Size (bytes)	Name	Meaning	Value
0x01	CMND	Command identifier.	`0xA9 (tentative)`
0x01	Status Type	`0x01`: standard PIV credential status. `0x02`: SM diagnostic record. Values `0x00` and `0x03-0x7F` are reserved and return `osdp_PIVERROR` `0x0005` (invalid parameter). Values `0x80-0xFF` are reserved for private use.	`0x00-0xFF`

PDs **SHALL** return `osdp_NAK` function-not-supported when the command itself is not implemented. If the requested status type is not supported or the payload is malformed, PDs **SHALL** report the condition via `osdp_PIVERROR` (for example `0x0005` (invalid parameter)) on a subsequent `osdp_POLL`.

3.5 Card Status Reply (osdp_CARDSTATUSR)

Every osdp_CARDSTATUSR reply **SHALL** include Status Type immediately after the reply code. Type 0x01 selects the standard credential status fields below. Type 0x02 selects the diagnostic multi-part layout in Subsection 3.5. The ACU **SHALL** select the reply layout from this byte and verify that the type matches the pending request. An unknown or mismatched type **SHALL** be treated as an invalid reply; the ACU discards the reply and any partially reassembled record. The ACU can cancel the pending operation with ABORT and issue a new request.

The PD accepts osdp_CARDSTATUS Type 0x01 with osdp_ACK and returns osdp_CARDSTATUSR on a subsequent osdp_POLL after the current credential preparation completes. The PD **SHALL** use a finite, documented preparation timeout. If SM is established and the credential requires a pairing code, the PD reports that state so the ACU can supply the code. The ACU **SHALL** interpret the following combination as pairing required: Detected Credential Type 0x02, Credential Security Status 0x02, and an available PIN Usage Policy whose first byte has mask 0x08 set and mask 0x04 clear. Security Status 0x02 alone indicates SM without VCI; contact SM requires no pairing. While status is pending, the ACU sends only osdp_POLL or osdp_ABORT.

osdp_CARDSTATUSR also completes an accepted osdp_PIVXMITPAIRING command, using Status Type 0x01. Controllers continue to use osdp_RAW for ordinary card-data indications. Whenever credential SM is established, the status reply reports the CVC IIN and the loaded anchor IIN, including for contact SM. The values match for direct verification and differ when an Intermediate CVC is used. The *SM Trust Anchor ID* identifies the corresponding osdp_PIVVCITASTATUSR inventory entry, whose IIN matches *SM Anchor IIN*.

Table 14: Card Status Reply (osdp_CARDSTATUSR)

Size (bytes)	Name	Meaning	Value
0x01	CMND	Reply identifier.	0x85 (tentative)
0x01	Status Type	Standard credential status, including successful pairing completion.	0x01
0x01	Detected Credential Type	Current card interface: 0x00 none, 0x01 ISO 7816 contact, 0x02 ISO 14443 contactless, 0x03 other ISO 7816 credential (for example DESFire, Seos), 0x04 low-frequency credential, 0x05 unknown type, 0x06-0x7F reserved, 0x80-0xFF private use.	0x00–0xFF
0x01	Credential Security Status	0x00 SM absent; 0x01 SM established and VCI available; 0x02 SM established without VCI. Contact SM uses 0x02. Values 0x03-0x7F reserved, 0x80-0xFF private use.	0x00–0xFF

Size (bytes)	Name	Meaning	Value
0x01	SM Trust Anchor ID	Identifier assigned in osdp_PIVVCILOADTA for the anchor used to validate credential SM. The anchor's inventory IIN matches *SM Anchor IIN*. Set to 0x00 when SM is absent.	0x00–0xFF
0x08	SM CVC IIN	Issuer Identification Number from the credential SM CVC tag 0x42. Set to all zero when SM is absent.	0x00–0xFF
0x08	SM Anchor IIN	Issuer Identification Number of the loaded trust anchor used to validate credential SM. Set to all zero when SM is absent.	0x00–0xFF
0x02	PIN Usage Policy	PIN Usage Policy bytes (tag 0x5F2F) returned verbatim when available. A value of 0xFFFF indicates the policy is not present or has not yet been read.	0x0000–0xFFFF
0x01	Selected AID Length	Length of the selected AID. 0x00 indicates no application is currently selected.	0x00–0x10
0x00-0x10	Selected AID	Application Identifier read from the credential (for example from the FCI template). Present only when the length is non-zero.	0x00–0xFF

Each accepted Type 0x01 status request **SHALL** complete with either osdp_CARDSTATUSR or osdp_PIVERROR. If a failed SM security check has blocked credential operations, the PD **SHALL** terminate a Status Type 0x01 request without sending further commands or data to the credential. The PD **SHALL** return osdp_PIVERROR 0x1023 (credential security conditions not satisfied) on the next poll. Otherwise, the standard status reply includes the selected AID when available and any previously read PIN Usage Policy bytes from Discovery Object tag 0x5F2F, unchanged. The status request itself requires no Discovery Object read. The PD reports 0xFFFF when the policy is unavailable. Security status reports the result of credential preparation. With no credential present, the PD reports credential type and security status zero, AID length zero, and PIN Usage Policy 0xFFFF. Removal during preparation returns osdp_PIVERROR 0x1026 (credential absent or removed). A preparation timeout returns osdp_PIVERROR 0x0002 (smart-card timeout). osdp_ABORT cancels the request with osdp_ACK and clears its pending completion reply.

SM diagnostics (Status Type 0x02) The ACU uses this status type to determine why SM setup failed and identify the CVC received by the PD. The PD **SHALL** accept this request while credential operations are blocked

by a failed security check. The PD **SHALL** return the most recent setup outcome and the CVC hash for the current credential and selected application. This request reads retained PD information; the PD **SHALL NOT** send a command to the credential in response to the diagnostic request.

The PD acknowledges the request with `osdp_ACK`, then returns `osdp_CARDSTATUSR` on subsequent polls. Status Type `0x02` uses the OSDP 3.0 multi-part format in Table 15. Every diagnostic fragment **SHALL** contain Status Type `0x02` immediately after the reply code, followed by MpSizeTotal, MpOffset, and MpFragmentSize. Status Type identifies the layout before the ACU reads the multi-part fields. The repeated Status Type byte is outside the diagnostic record and **SHALL NOT** be counted in the multi-part lengths or offsets.

Table 15: SM Diagnostic Reply (CARDSTATUSR, Status Type 0x02)

Size (bytes)	Name	Meaning	Value
0x01	CMND	Reply identifier.	`0x85 (tentative)`
0x01	Status Type	SM diagnostics. Repeated in every fragment, outside the diagnostic record.	`0x02`
0x02	MpSizeTotal	Total diagnostic record length, least-significant byte first.	`0x0023`
0x02	MpOffset	Offset within the diagnostic record, least-significant byte first.	`0x0000–0xFFFF`
0x02	MpFragmentSize	Diagnostic bytes in this reply, least-significant byte first.	`0x0001–0xFFFF`
0x01	Outcome	`0x00`: setup not attempted; `0x01`: SM established; `0x02`: SM unavailable; `0x03`: security check or SM processing failed, credential operations blocked.	`0x00–0x03`
0x01	Detail	Reason from Table 16.	`0x00–0x07`
0x01	CVC Hash Present	`0x01` when the PD received a complete CVC; otherwise `0x00`.	`0x00–0x01`
0x20	CVC Hash	SHA-256 of the received CVC, including tag `0x7F21`, its encoded length, and value, unchanged. All zero when CVC Hash Present is `0x00`.	`0x20 bytes`

Outcome values `0x04`-`0xFF` and Detail values `0x08`-`0xFF` are reserved for future use.

Table 16: SM Diagnostic Detail Codes

Value	Description
0x00	No failure detail. Used with Outcome 0x00 or 0x01.
0x01	Trust store empty. Outcome 0x02; CVC Hash Present is zero because setup was skipped.
0x02	No matching trust path through the loaded anchors. Outcome 0x02. The ACU can inspect the signer object to determine whether an anchor or intermediate CVC is missing.
0x03	CVC missing, malformed, or invalid. Outcome 0x03. CVC Hash Present is zero when no complete CVC was received.
0x04	Authentication cryptogram invalid. Outcome 0x03.
0x05	No credential SM algorithm supported by both the PD and credential. Outcome 0x02.
0x06	Other SM processing failure. Outcome 0x03.
0x07	SM disabled by PIVMODE. Outcome 0x02.

The diagnostic record consists of Outcome, Detail, CVC Hash Present, and CVC Hash, in that order. The PD **SHALL** assemble the complete record before dividing the record into reply fragments. The multi-part fields describe that complete record. A fragment may end within any field. The ACU **SHALL** reassemble the record before interpreting the fields.

The PD **SHALL** compute and retain the hash of a complete received CVC even when certificate or cryptogram validation fails. The hash identifies the received bytes; the hash provides no evidence that the CVC is trusted. The diagnostic record requires only the outcome, detail, presence flag, and 32-byte hash. The PD has no obligation to retain the CVC or other credential objects for later retrieval. PIVGETDATA reads the requested object from the credential through the active interface. If no complete CVC was received, the PD **SHALL** set CVC Hash Present to zero, fill CVC Hash with zero bytes, and report the applicable reason.

The PD **SHALL** keep the diagnostic record until credential removal or reset, PD reset, successful selection of a different application, or reapplication of PIVMODE. A later SM failure updates Outcome and Detail while retaining the CVC hash from that session. Loading a trust anchor leaves the previous outcome unchanged. Before a setup attempt, the record has Outcome 0x00, Detail 0x00, CVC Hash Present zero, and an all-zero CVC Hash.

The ACU **SHALL** receive any pending PIV Auto result before requesting diagnostics. While a diagnostic reply is pending, the ACU sends only POLL or ABORT. A competing request receives osdp_NAK 0x09 (unable to process command). ABORT cancels the transfer with ACK. Credential removal during transfer terminates the transfer with osdp_PIVERROR 0x1026 (credential absent or removed); the ACU discards any partial record.

3.6 PIV Secure PIN Entry (osdp_PIVSPE)

The osdp_PIVSPE command (command code 0xAA, tentative) controls Secure PIN Entry for the Enhanced PIV profile. A PD **SHALL** advertise both Function Code 12 bit 0x04 (Enhanced PIV) and Function Code 15 ~~Secure PIN Entry support~~ bit 0x02 (explicit PIV SPE) before an ACU uses this command. The OSDP 2.2 osdp_XWR Mode-01 SPE command and osdp_PR01SPER reply remain unchanged.

The ACU is responsible for conveying a clear indication that PIN entry is in progress. While the PD is acquiring a PIN, it **SHALL** suppress osdp_KEYPAD reports so that no PIN digit is sent to the ACU.

Table 17: PIV Secure PIN Entry (osdp_PIVSPE)

Size (bytes)	Name	Meaning	Value
0x01	CMND	Command identifier.	0xAA (tentative)
0x01	SPE Mode	Secure PIN Entry operating mode. See Table 18.	0x00–0xFF
0x01	Start Timeout	Seconds allowed to begin key entry; 0x00 selects the documented PD default (a value from 0x01 through 0xFF seconds).	0x00–0xFF
0x01	Inter-key Timeout	Seconds allowed between keystrokes; 0x00 selects the documented PD default (a value from 0x01 through 0xFF seconds).	0x00–0xFF
0x01	Flags	Behaviour modifiers. See Table 19.	0x00–0xFF
0x02	Cache Validity	PIN cache validity in seconds (LSB first). Mode 0x04 requires 0x0001-0xFFFF; other modes transmit 0x0000.	0x0000–0xFFFF

Table 18: SPE Mode Values

Value	Description
0x00	Report the persistent SPE status without initiating PIN entry.
0x01	Acquire a PIN and verify it with the credential currently present.
0x02	Verify the cached PIN over established VCI.
0x03	Verify the cached PIN over the selected credential interface. Use established credential SM when present.
0x04	Acquire and cache a PIN.
0x05	Abort active PIN entry and clear the cached PIN.
0x06	Report attempts remaining without initiating PIN verification.

Value	Description
0x07–0x7F	Reserved for future use.
0x80–0xFF	Private use.

Table 19: SPE Flag Values

Mask	Description
0x01	Retain a cached PIN across credential removal until its validity expires or another clearing event occurs. Valid only with Mode 0x04.
0x02	Auto Execute When Credential Is Presented. Valid only with verification Modes 0x01-0x03.
0x04–0x80	Reserved flag bits; **SHALL** be transmitted as zero.

Modes are enumerated values. Modes 0x02 and 0x03 submit the cached PIN using PIV VERIFY. Mode 0x02 requires established VCI; Mode 0x03 uses the credential interface selected by the active PIV mode. Mode 0x01 acquires the PIN and uses the selected interface. The ACU selects Mode 0x02 to require VCI explicitly, or Mode 0x03 to permit verification on either the selected contact interface or established VCI. Both modes have the same behavior on contactless. Mode 0x02 returns 0x09 (VCI not established) on contact, even when contact SM is established; Mode 0x03 permits that workflow. When credential SM is established, the PD **SHALL** use that session for Modes 0x01-0x03 and 0x06. Contactless PIN verification requires VCI. Contact PIN verification may use the contact interface without SM. Mode 0x04 requires a *Cache Validity* from 0x0001 through 0xFFFF. All other modes **SHALL** transmit 0x0000 in that field. The PD **SHALL NOT** transmit an entered or cached PIN to the ACU.

A zero *Start Timeout* or *Inter-key Timeout* selects the PD's documented default. Each default **SHALL** be from 1 through 255 seconds. Nonzero values specify seconds and **SHALL** override the corresponding default. The ACU should supply explicit values when deployment or test requirements specify the timing. Fields that do not apply to a selected mode **SHALL** be transmitted as zero.

Only one osdp_PIVSPE operation may be outstanding. Modes 0x01-0x04 and 0x06 each initiate one operation. The PD returns osdp_ACK when it accepts the request, then returns one osdp_PIVSPER in response to a later osdp_POLL. While such an operation is pending, the ACU **SHALL** send only osdp_POLL, Mode 0x05, or osdp_ABORT. Returning osdp_PIVSPER ends the operation unconditionally, including after a PIN mismatch.

Mode 0x00 returns osdp_PIVSPER directly. It reports only persistent PIN state; it does not report transient active or waiting states. An ACU recovering from an interrupted session first uses Mode 0x05 or osdp_ABORT, then queries Mode 0x00. Mode 0x06 is accepted with osdp_ACK; the PD reads the present credential without performing PIN verification and returns the count in a subsequent osdp_PIVSPER. This query **SHALL NOT** decrement the credential retry counter. Auto Execute does not make the count available without a present credential. The PD **SHALL** return 0xFF (attempts unavailable) when the credential is absent or the active interface and security context do not permit the query. Contactless queries require VCI.

Mode 0x05 is idempotent. It aborts active or partial PIN entry, clears the cached PIN, and completes with osdp_ACK; the PD **SHALL NOT** send a later osdp_PIVSPER for the canceled operation. osdp_ABORT cancels active or partial PIN entry but preserves a cache that was completed before the active operation.

A successful verification **SHALL** preserve the cached PIN until expiration or a clearing event. A later explicit verification request may reuse that PIN. The PD **SHALL** clear the cache on Mode 0x05, PIN mismatch, PD reset, loss of the OSDP Secure Channel 3.0, or a change to the PIV mode or a change from one selected application to another. Initial application selection and initial SM establishment **SHALL** preserve a PIN cached before credential presentation. Credential removal also clears it unless flag 0x01 was supplied with Mode 0x04. Loss of credential SM while the credential remains present clears the cache. Removal follows the removal flag even when it also ends SM. PIN-verified status is bound to the current credential, selected application, interface, and credential security context. The PD **SHALL** clear that status on credential removal or reset, a change to the selected application, interface change, PIV mode change, or loss of credential SM or VCI, and **SHALL NOT** infer it only from an earlier successful reply.

PIN cache expiration The PIN cache holds a PIN in the PD for later verification by a credential. The PD sends the PIN to the credential in a VERIFY command. Successful verification allows operations that require the PIN, subject to the credential's access rules.

Cache Validity limits how long the PD stores the PIN for reuse. When the cache expires, the PD erases the stored PIN. A credential that has already accepted the PIN remains PIN verified until the credential's rules clear that state. Mode 0x00 reports both whether the PD has a usable cached PIN and whether the current credential has accepted PIN verification (Table 21).

Starting and expiring the cache. The PD **SHALL** start the cache timer when PIN acquisition completes successfully, using elapsed time unaffected by clock adjustments. Delivery of the completion reply, status queries, and verification leave the deadline unchanged. A new successful acquisition starts a new deadline. At expiration, the PD **SHALL** erase the cached PIN and mark the cache unavailable.

Expiration before VERIFY. Immediately before submitting VERIFY to the credential transport, the PD **SHALL** check the cache deadline. If the deadline has been reached, the PD **SHALL** send no VERIFY and return osdp_PIVSPER 0x0A (no cached PIN). The PD **SHALL** make the deadline check and submission decision as one indivisible step. At the deadline, the PIN is expired and the PD **SHALL NOT** submit VERIFY.

Expiration after VERIFY. Once VERIFY has been submitted, the PD **SHALL** finish the exchange even if the cache expires. The PD reports the credential's verification result, 0x0B (credential communication failed or timed out), or 0x04 (credential removed), as applicable. The PD may retain a command copy needed to finish the exchange under the credential transport rules and **SHALL** erase that copy when the exchange ends. Each new VERIFY requires a new ACU request.

VERIFY without a confirmed result. If communication fails after VERIFY submission, the PD cannot determine whether the credential accepted the PIN. The PD **SHALL** clear its reported PIN-verification state and cached PIN, end any SM session, and require credential preparation to be restarted before another data or authentication operation. Restarting preparation restores a known security context after an exchange whose outcome is unknown. A verified credential reply determines the reported result even if delivery to the ACU is delayed.

Reporting expiration to the ACU. While idle, the PD erases an expired cached PIN without sending a reply. The ACU learns the current state from a status query or the result of a later verification request. If Auto Execute is waiting for credential presentation, expiration **SHALL** end that request with `0x0A` on a later poll. An accepted verification request receives ACK followed by one terminal PIVSPER, unless the ACU cancels the request. Expiration after the operation finishes leaves its pending completion reply unchanged.

Figure 7 shows how expiration affects VERIFY. Appendix D.5 provides packet and timing examples.

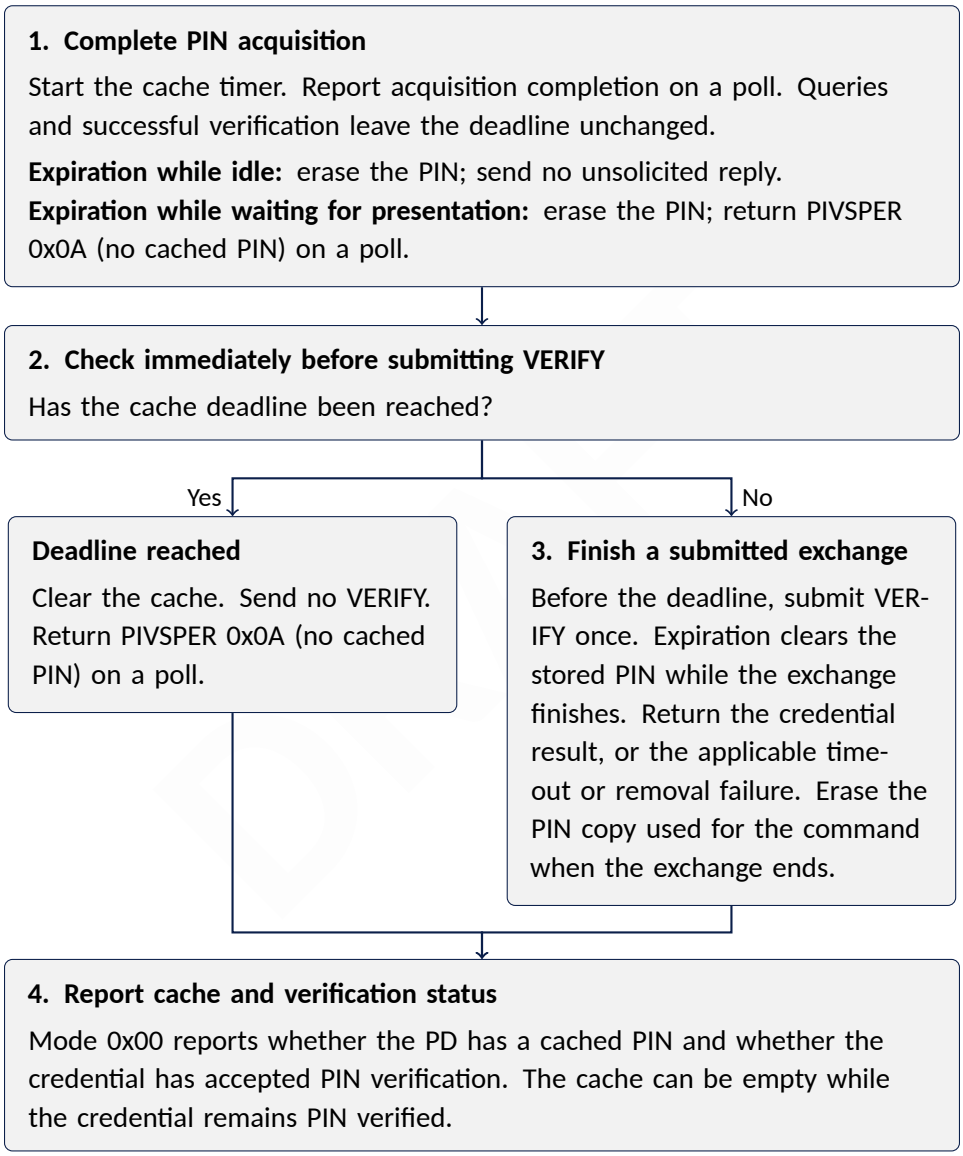


Figure 7: PIN cache lifetime and the VERIFY submission boundary. Idle expiration is silent. Expiration while waiting for presentation ends the pending request with PIVSPER 0x0A on a poll.

Flag `0x02` arms the verification selected by Mode `0x01`, `0x02`, or `0x03` to execute when a credential is presented. It **SHALL** be zero for Modes `0x00`, `0x04`, `0x05`, and `0x06`. A cached auto-execute workflow first acquires the PIN with Mode `0x04`, then selects normal or VCI verification with Mode `0x03` or `0x02` and flag `0x02`.

Without Auto Execute, verification begins after acceptance with the credential and required security context already available. An absent credential returns *Result* 0x04 (credential absent or removed). When Mode 0x02 is requested or verification uses contactless, missing VCI returns 0x09 (VCI not established). The ACU completes any required pairing before requesting verification.

Auto Execute requires Function Code 15 bit 0x04. The ACU **SHALL** check that bit before arming it; an unsupported Auto Execute request receives osdp_NAK 0x09 (unable to process command). With Auto Execute, the PD waits for presentation, completes credential preparation (Subsection 3.1), and attempts verification once. If preparation requires a pairing code, the PD completes the SPE operation with 0x09 (VCI not established). If the cached PIN expires while waiting, the PD completes with 0x0A (no cached PIN). A failed verification completes the operation without an automatic retry. The ACU may then perform setup and start a new SPE operation.

The ACU **SHALL** disable PIV Auto before arming SPE Auto Execute. A request to arm SPE Auto Execute while PIV Auto is enabled receives osdp_NAK 0x09 (unable to process command). This ensures that only one operation starts credential preparation and PIN verification when the credential is presented.

The PD **SHALL** reject an unsupported command with osdp_NAK 0x03, an incorrect command length with osdp_NAK 0x02, and an invalid mode, reserved bit, or field combination with osdp_NAK 0x09. Accepted credential-side outcomes are reported through osdp_PIVSPER, not osdp_PIVERROR.

Figure 8 summarizes reply timing and interpretation.

3.7 PIV Secure PIN Entry Reply (osdp_PIVSPER)

The osdp_PIVSPER reply (reply code 0x86, tentative) contains one mode-dependent *Result* byte.

Table 20: PIV Secure PIN Entry Reply (osdp_PIVSPER)

Size (bytes)	Name	Meaning	Value
0x01	CMND	Reply identifier.	0x86 (tentative)
0x01	Result	Mode-dependent SPE status, attempts count, or operation result.	0x00–0xFF

For Mode 0x00, *Result* is the status bitmap in Table 21. For Mode 0x06, it is the attempts value in Table 22. For Modes 0x01-0x04, it is the operation outcome in Table 23.

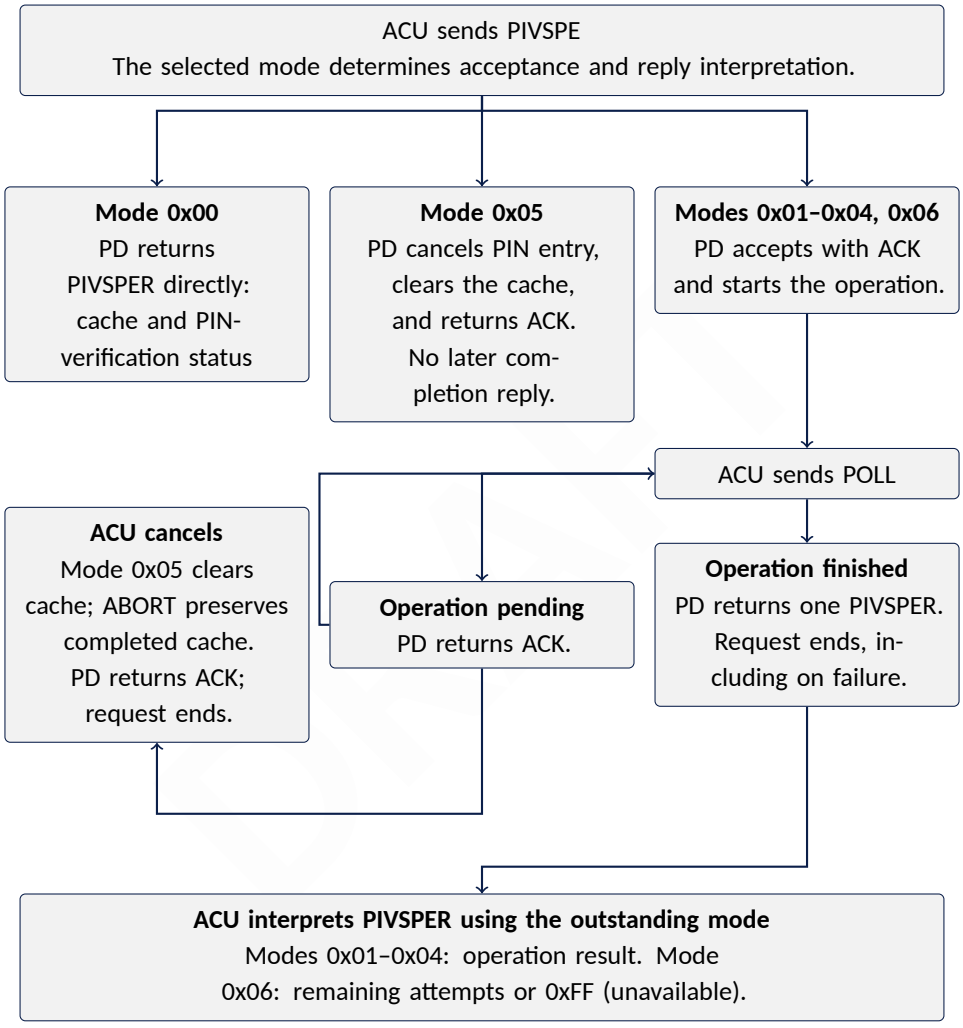


Figure 8: SPE request and reply lifecycle. While an operation is pending, the ACU sends only POLL, Mode 0x05, or ABORT. ABORT cancels the operation while preserving a previously completed cache. An unsupported or malformed request follows the NAK rules.

Table 21: PIVSPE Status Result (Mode 0x00)

Value	Description
0x00	No cached PIN is present and the current credential is not PIN verified.
0x01	A cached PIN is present.
0x02	The current credential and selected application are PIN verified.
0x03	A cached PIN is present and the current credential and selected application are PIN verified.
0x04–0x7F	Reserved for future use.
0x80–0xFF	Private use.

Table 22: PIVSPE Attempts Result (Mode 0x06)

Value	Description
0x00	The credential PIN is blocked.
0x01–0xFE	Number of PIN verification attempts remaining.
0xFF	Attempts remaining are unavailable, including when no credential is present.

Table 23: PIVSPE Result Codes

Value	Description
0x00	Requested operation completed.
0x01	Timeout – PIN entry did not begin within the allotted time.
0x02	Timeout – inter-digit interval exceeded.
0x03	PIN mismatch.
0x04	Credential absent or removed during operation.
0x05	Credential PIN blocked.
0x06	Wrong interface for the requested mode.
0x07	Incorrect parameters.
0x08	Security condition not met.
0x09	VCI not established for the requested mode.
0x0A	No cached PIN is available.
0x0B	Credential communication failed or timed out before a verified completion was received.
0x0C–0x7F	Reserved for future use.
0x80–0xFF	Private use.

3.8 PIV Biometric Match (osdp_PIVBIOMATCH)

The osdp_PIVBIOMATCH command (command code 0xAB, tentative) instructs the PD to acquire a live sample and compare it to the specified biometric template from the active application. Depending on the *Match Mode*, the comparison occurs either on the credential (on-card match) or within the PD (off-card match).

Table 24: PIV Biometric Match (osdp_PIVBIOMATCH)

Size (bytes)	Name	Meaning	Value
0x01	CMND	Command identifier.	0xAB (tentative)
0x01	Biometric Type	Selects the template type to compare. See Table 25.	0x00–0xFF
0x01	Match Mode	0x00 off-card match; 0x01 on-card match.	0x00–0x01

The command and its PIVBIOMATCHR reply **SHALL** use Secure Channel 3.0 with deployment-specific keys under paragraph 3.11.

Table 25: Biometric Type Values

Value	Description
0x01	Cardholder fingerprint template.
0x02	Cardholder facial image.
0x04	Cardholder iris template.

For contactless on-card comparison, the PD **SHALL** use credential SM. This operation can proceed with SM established and VCI absent; pairing and VCI are required only when the credential's access conditions require them. Other biometric operations use established SM when present.

Internal biometric-object reads **SHALL** use the explicit response-length and continuation rules in paragraph 2.1.2.

Controllers set *Match Mode* to select off-card or on-card comparison. Off-card flows require the PD to retrieve the appropriate template and, for encrypted TWIC biometric data on either contact or contactless, to have the appropriate TPK available. The ACU can supply the key using osdp_PIVLOADTPK. For a biometric operation against the TWIC application, if a required TPK is ~~missing or incorrect~~unavailable or reference decryption fails, the PD **SHALL** report osdp_PIVBIOMATCHR Result 0x07 (~~Invalid TPK~~ TPK/decryption failure). TPK failures **SHALL NOT** be reported as PIN-verification failures. On-card flows require the credential to support OCC. If the PD does not implement the command, it **SHALL** return osdp_NAK function-not-supported. Credential-side match outcomes (for example PIN verification required or credential removal during operation) **SHALL** be reported in the subsequent osdp_PIVBIOMATCHR result code per Table 27. Conditions that prevent the command from being attempted (for example unsupported

credential function, missing VCI when required, or invalid command parameters) **MAY** be reported via `osdp_PIVERROR` on a subsequent `osdp_POLL`.

TWIC reference decryption and availability The PD **SHALL** check reference availability before decryption, using Result `0x0A` below. For a present encrypted reference, it **SHALL** decipher the complete tag `0xBC` value using AES-128-ECB with the 16-byte TPK and no IV (TWIC Part 2, Section 4.4.1). It **SHALL** require a nonempty ciphertext length divisible by 16, decrypt with the AES library's no-padding option, then validate and remove PKCS#7 padding: the final byte is a count from 1 through 16 and every byte in that suffix equals the count (Sections 4.6.4 and 4.7.3, note 2). Invalid ciphertext length or outer TLV encoding produces PIVERROR `0x1028`. A missing TPK, invalid padding, or failure to decode the resulting CBEFF record produces PIVBIOMATCHR Result `0x07` (TPK unavailable or reference decryption failed).

Decoding **SHALL** check the CBEFF header version, security options, modality and format, and the declared biometric-data-block and signature-block lengths against the complete unpadded record. Fingerprint records **SHALL** contain a valid ANSI/INCITS 378 record, including its 464D5200 (FMR) identifier, version, record length, and finger-view boundaries. SP 800-76-2, Sections 4 and 9, defines these structures. A decoded reference proceeds to signature and identity validation under paragraph 3.8; validation failure rejects authentication through that validation path.

For PIV and TWIC comparisons, a missing object or reference element, an empty object, or an absent biometric reference produces PIVBIOMATCHR Result `0x0A`. A valid record with no usable reference for the requested modality produces Result `0x0B`. For TWIC fingerprints the PD **SHALL** check minutiae counts before interpreting quality values, following TWIC Part 2, Appendix G, and Part 3, Appendix F. A usable finger view permits comparison; the header quality value `-2` alone does not make a reference unusable. These outcomes end the biometric operation before live-sample comparison.

Off-card biometric validation For PIV and TWIC off-card ~~workflowsThe following~~ biometric authentication, the ACU **SHALL** accept match success only after validating the reference CBEFF record's signature and content-signer trust, and binding its signed identity to the verified identity of the presented credential. For PIV, the corresponding FASC-N or card UUID **SHALL** match the CHUID or another verified data element, following FIPS 201-3, Sections 6.2.1.1–6.2.1.2. For TWIC, the signed FASC-N **SHALL** match the identifier established by the selected TWIC reader mode, following TWIC Part 3, Sections 4.4.5–4.4.6.

The ACU **MAY** reuse enrollment-time validation of the exact signed reference record. Its enrollment record **SHALL** link the PIVBIOMATCHR hash to that verified record, its signed identity, signer evidence, and validation state. At each authentication, the ACU **SHALL** match the returned hash to the stored hash and perform the credential-identity binding above. Cached validation **SHALL** remain valid under the deployment's trust policy, with periodic rechecks as described in SP 800-116 Rev. 1, Sections 2.3 and 4.4. A missing hash entry or stale validation requires reference validation before acceptance. For TWIC, the stored hash identifies the original tag `0xBC` ciphertext associated with the verified decrypted CBEFF record, as defined below.

TWIC fingerprint and picture workflows The following application-specific workflows follow *TWIC Specification*, *Part 2*, Sections 4.1, 4.2, 4.6.4, and 4.7.3, and *Part 3*, Sections 4.4.5–4.4.8~~and 7.~~. They use off-card comparison (*Match Mode* `0x00`); the documented TWIC cards do not implement OCC or PIV credential

SM/VCI. Selecting a PIV application on a TWIC therefore does not make contactless PIN-protected facial retrieval available. The ACU **SHALL** verify the PD's advertised modality support before requesting a fingerprint (*Biometric Type* 0x01) or facial (0x02) comparison.

1. **Legacy TWIC fingerprint.** Select the TWIC application with Data Model 0x02. Obtain the credential's TPK from its magnetic stripe, its contact-interface key container, or the PACS record for that credential. If the ACU supplies the key, load it with osdp_PIVLOADTPK before requesting off-card fingerprint comparison. The PD reads object DFC103 and deciphers its tag 0xBC value using the TPK. No PIV PIN is required for this TWIC-application operation.

2. **Legacy TWIC facial.** The Legacy TWIC application contains no facial image object. For direct card retrieval, use the contact interface, select the PIV application with Data Model 0x01, and verify the PIV application PIN using osdp_PIVSPE before requesting off-card facial comparison. The reference is PIV object 5FC108; it is not encrypted with the TPK. If PIN verification has not been completed, the existing PIN-verification-required result applies~~, not Invalid TPK~~. Alternatively, the PACS can use a reference image collected during registration and indexed by the verified credential identifier. ~~That backend workflow does not imply that PIVBIOMATCH accepts a backend image payload~~The PACS retrieves and displays that image for picture verification. The contact-only/PIN requirement is stated in Part 3, Section 4.4.7, footnote 37; registration is described in *Part 4*, Section ~~2.5.1~~2.6.3.

3. **NextGen TWIC fingerprint.** Select the TWIC application with Data Model 0x02. Obtain the TPK from the PDF417 barcode, the contact-interface key container, or the credential's PACS record. Load the key when supplied by the ACU, then request off-card fingerprint comparison using object DFC103 and its encrypted tag 0xBC value. This workflow requires no PIV PIN.

4. **NextGen TWIC facial.** Select the TWIC application with Data Model 0x02, obtain the TPK as above, and request off-card facial comparison using object DFC108. The PD deciphers tag 0xBC using the TPK; no PIV PIN is required. Automated comparison by a facial-capable PD uses PIVBIOMATCH/PIVBIOMATCHR. Operator picture verification instead retrieves, verifies, and displays the reference image under TWIC Part 3, Section 4.4.7~~; PIVBIOMATCH does not return that image~~.

TWIC identity-verification workflows For electronic TWIC inspection under 33 CFR 101.520(c)(1), the ACU **SHALL** complete either the TWIC fingerprint workflow above (*Biometric Type* 0x01) or a match against a PACS-enrolled reference biometric linked to the presented TWIC's FASC-N. These are the card-template and PACS-held alternate-biometric paths in the definition of *biometric match* in 33 CFR 101.105.

For the PACS-held path, the PACS biometric subsystem captures the live sample and compares it with the enrolled reference. The ACU **SHALL** bind the reference's FASC-N and the comparison result to the current TWIC authentication transaction before completing identity verification. This workflow uses the PACS biometric subsystem's capture and comparison interface.

Picture verification uses the verified reference image and its bound FASC-N under TWIC Part 3, Sections 4.4.7–4.4.8: Mode 5 combines CHUID verification and Reference Picture Verification (RPV); Mode 6 adds Active Card Authentication. Fingerprint alternatives follow the operator's security plan under Part 3, Section 9, item 3. Face and iris comparisons against PACS-enrolled references use the PACS-held alternate-biometric path above.

The TPK is not readable from the credential over contactless. A Legacy magnetic stripe encodes the 16-byte key as 32 hexadecimal characters; the ACU decodes those characters before LOADTPK. A barcode or key-container read likewise supplies only the raw key value to LOADTPK ~~. Loading a correctly sized key does not prove that it belongs to the presented credential. A cleared or expired cache without another valid key source is a missing TPK for a TWIC comparison. The existing cache timeout and clear-on-removal flag remain applicable.~~ for the next biometric operation in the current presentation.

~~Before accepting TWIC biometric authentication, the system **SHALL** verify the decrypted CBEFF record's signature and authorized document-signer trust, and bind its signed FASC-N to the identifier verified through the CHUID or active card authentication for the same credential, as required by the selected TWIC reader mode. These checks are separate from TPK decryption and live-sample comparison. A signature or identifier-binding failure **SHALL NOT** be accepted merely because decryption or matching succeeded. An integrity failure alone does not establish that the TPK was incorrect. The ACU retains responsibility for the access decision and the checks assigned to it by the deployment; BIOMATCH success alone does not assert completed TWIC authentication.~~

For modes requiring active card authentication, Legacy TWIC uses its PIV application's card-authentication key, whereas NextGen provides a key in the TWIC application. An application switch clears the previous application's security state, so any subsequent PIV PIN verification must apply to the newly selected PIV context.

Informative TWIC Mode 4 composition TWIC Part 3, Section 4.4.6, combines Active Card Authentication and fingerprint verification. The ACU first completes TWIC ACA and card-validity checks, then selects the TWIC application using PIVSELECT, obtains the TPK, loads it when supplied by the ACU, and requests PIVBIOMATCH with Biometric Type `0x01` and Match Mode `0x00`. For Legacy cards this selection returns from the PIV application used for ACA to the TWIC application used for fingerprints. The ACA profile's presentation-continuity rules apply throughout this sequence.

The ACU uses the enrollment record identified by the PIVBIOMATCHR hash, validates its signed biometric reference and document-signer trust under paragraph 3.8, and binds its FASC-N to the ACA-certified FASC-N. For initial or updated enrollment, PIVGETDATA supplies the CHUID document-signer certificate from object `5FC102` and the encrypted fingerprint record from object `DFC103`. This composes the biometric checks with ACA for the card-template path in 33 CFR 101.520(c)(1). The ACU combines successful card authentication, expiration/CCL checks, and fingerprint verification before applying access policy. For a holder unable to provide a valid live sample, the operator-managed path in 33 CFR 101.520(c)(2) requires PIN entry and visual TWIC inspection.

3.9 PIV Biometric Match Reply (osdp_PIVBIOMATCHR)

After `osdp_PIVBIOMATCH` completes, the PD reports the result as a Secure Channel 3.0-protected poll response using reply code `0x87` (tentative). Paragraph 3.11 governs channel loss before delivery.

Table 26: PIV Biometric Match Reply (osdp_PIVBIOMATCHR)

Size (bytes)	Name	Meaning	Value
0x01	CMND	Reply identifier.	0x87 (tentative)
0x01	Result	Outcome of the biometric match. See Table 27.	0x00–0xFF
0x00 or 0x20	Hash	0x20-byte SHA-256 hash of the biometric template tag 0xBC data field. The PD **SHALL** include this field only when *Result* is 0x00 (off-card match success). The PD **SHALL** omit this field for on-card match success (0x09) and all failure result codes.	0x00–0xFF

When *Result* is 0x00 (off-card match success), the PD **SHALL** include the 32-byte SHA-256 hash of the biometric template tag 0xBC data field in the reply. For on-card match success (*Result* 0x09) and for all failure result codes, the PD **SHALL** omit the hash field; ACUs **SHALL NOT** rely on a hash received alongside any result other than 0x00.

The hash **SHALL** identify the exact reference record used for that comparison. The hash input is the tag 0xBC value as read from the credential, excluding its tag and length bytes. For the encrypted TWIC objects, this is the ciphertext, not the decrypted CBEFF record or a derived live-sample template. Decryption supplies the reference used for comparison; it does not change the reply's hash input.

~~Off-card biometric comparison demonstrates only that the live sample matched template data presented to the PD. Because~~ For off-card ~~biometric objects can be copied or duplicated, an off-card biometric match does not by itself prove authenticity of the credential. On-card biometric comparison is a credential-controlled assertion; the biometric object may be attacker controlled and the PD does not receive the template data being matched~~authentication, the ACU validates the signed reference and binds its identity under paragraph 3.8. To obtain cryptographic proof associated with an on-card biometric success, the ACU **SHALL** perform an additional authentication using an appropriately verified credential key after the successful on-card comparison.

Table 27: PIVBIOMATCH Result Codes

Value	Description
0x00	Biometric match success (off-card match).
0x01	Biometric match timeout (no entry started).
0x02	Biometric match failed (no match).
0x03	PIN verification required.
0x04	Credential removed during operation.
0x05	PIN verification failed.
0x06	PIN verification timeout.
0x07	~~Invalid TPK (the required TPK is missing or incorrect~~TPK unavailable or reference decryption failed (padding or decoded CBEFF structure).
0x08	Reserved (PD/credential errors are reported via osdp_PIVERROR).
0x09	On-card match success.
0x0A~~–0x7F~~	Biometric reference absent or empty.
0x0B	Biometric reference contains no usable data for the requested comparison.
0x0C–0x7F	Reserved for future use.
0x80–0xFF	Private use.

3.10 TWIC Privacy Key Load (osdp_PIVLOADTPK)

The osdp_PIVLOADTPK command (command code 0xAC, tentative) ~~instructs the PD to cache~~ supplies a TWIC Privacy Key (TPK) for ~~use with subsequent biometric operations against a~~ the next accepted biometric operation against the currently presented TWIC credential. Function Code 14 is the biometric-support compliance byte; before using this command, controllers **SHOULD** verify that Function Code 14 advertises the OSDP 3.0 biometric command set and the relevant biometric modality bits for the intended TWIC operation.

Table 28: TWIC Privacy Key Load (osdp_PIVLOADTPK)

Size (bytes)	Name	Meaning	Value
0x01	CMND	Command identifier.	0xAC (tentative)
~~0x01 Cache Timeout Number of seconds to retain the cached TPK. clears any cached key immediately; permit caching for up to 0xFF seconds. 0x00-0xFF 0x01 Control Flags Bit 0 set requests automatic clearing when the credential is removed; all other bits **SHALL** be zero. 0x00-0xFF~~ 0x10	TPK	Raw AES-128 TWIC Privacy Key value ~~to cache~~ for the next TWIC biometric operation in the current presentation.	0x00–0xFF

The command and its completion **SHALL** use Secure Channel 3.0 with deployment-specific keys under paragraph 3.11. The PD checks this condition before storing, clearing, or using the supplied TPK.

The *TPK* field **SHALL** contain exactly 16 bytes of raw AES-128 key material, as specified in *TWIC Specification*, *Part 2*, Section 4.6.2. It carries only the key value, excluding TLV tags and lengths, the algorithm identifier, the key index, and any hexadecimal text encoding. The command payload, including *CMND* and the 16-byte key, ~~*Cache Timeout*, and *Control Flags*,~~ is exactly ~~bytes. The TWIC key container permits up to 32 bytes for future algorithms; that capacity does not change this command's 16-byte key-field length.~~ 0x11 bytes.

~~Controllers set *Cache Timeout* to define how long (in seconds) the PD may retain the key . A value of clears any cached TPK immediately; the 16-byte *TPK* field remains present but is ignored. Values -allow caching for up to 255 seconds. Bit 0 of *Control Flags* directs the PD to clear the cached TPK when the credential is removed; all other bits are reserved and~~ The PD **SHALL** ~~be transmitted as zero. If the PD does not support TWIC TPK caching it **SHALL** return~~ accept the key only with a credential present and its TWIC application selected with Data Model `0x02`. A missing credential produces PIVERROR `0x1026`; an invalid payload or selected application produces `0x0005`. A PD implementing the command acknowledges completion after validating and loading the key into volatile memory; an unsupported command receives `osdp_NAK` function-not-supported. ~~Invalid payloads (including an incorrect TPK field length or reserved bits set)~~

The loaded key belongs to that uninterrupted presentation and the next accepted TWIC biometric operation. The PD **SHALL** ~~elicit `osdp_PIVERROR` (invalid parameter).~~ zeroize it when that operation completes or fails, on ABORT (including while idle), credential removal, application or PIVMODE change, Secure Channel 3.0 loss or replacement, PD reset, or tamper detection. Loading a replacement zeroizes the previous key. Each subsequent biometric operation requires a fresh key load or retrieval from a permitted source for the current credential. Decryption and result handling follow paragraph 3.8.

3.11 PIV VCI Pairing Code Transmit (osdp_PIVXMITPAIRING)

The `osdp_PIVXMITPAIRING` command (command code `0xAD`, tentative) submits an eight-digit pairing code to the selected contactless PIV application over established credential SM. The ACU **SHALL** send it over OSDP Secure Channel ~~2.0~~3.0. Credential SM, PIVMODE flag `0x02`, and credential support for VCI are prerequisites.

Table 29: PIV VCI Pairing Code Transmit (osdp_PIVXMITPAIRING)

Size (bytes)	Name	Meaning	Value
0x01	CMND	Command identifier.	`0xAD (tentative)`
0x08	Pairing Code	Eight ASCII digits (`'0'-'9'`) used by the PD during VCI establishment for the currently selected PIV application.	`0x30-0x39`

The pairing code contains eight ASCII digits (`'0'-'9'`). The PD accepts a valid request with `osdp_ACK`. When pairing is required, the PD submits the code using PIV `VERIFY`, reference `0x98`, protected by the established SM context. Success completes with `osdp_CARDSTATUSR` on a later `osdp_POLL`, reporting security status `0x01` (SM and VCI). If VCI is already available, the PD completes with the same status reply without submitting the code again.

An accepted operation completes with `osdp_PIVERROR` on failure: `0x0005` (invalid parameter) for malformed digits or no selected PIV application, `0x1023` (security conditions not satisfied) for missing SM, disabled SM/VCI preparation, the wrong interface, or a rejected pairing code, `0x1020` (function not supported by credential) when VCI is unsupported, `0x1026` (credential absent or removed), or `0x0002` (smart-card timeout). The PD includes the credential's failure status word as specified in Subsection 2.2.1. If a submitted pairing VERIFY times out without a confirmed result, the PD **SHALL** end SM, clear dependent VCI, PIN-verification, and cached-PIN state, and require an explicit preparation restart before further credential

data or authentication commands. The PD **SHALL** return 0x0002 (smart-card timeout) without SW1/SW2 and **SHALL NOT** retry VERIFY automatically. While pairing is pending, the ACU sends only osdp_POLL or osdp_ABORT. Competing commands receive osdp_NAK 0x09 (unable to process command).

The PD **SHALL** clear the transient pairing code on completion, failure, cancellation, or loss of its credential context. osdp_ABORT completes with osdp_ACK and clears the pending completion. For an OSDP retransmission, the PD **SHALL** replay the previous reply without resubmitting the pairing code.

OSDP security preconditions Trust-anchor load/status, pairing-code transmit, credential SM establishment, and PIV operations that depend on SM or VCI **SHALL** require ~~OSDP~~ Secure Channel ~~2.0~~ 3.0 between the ACU and PD with deployment-specific keys. ~~OSDP Secure Channelprotects the management commands and their results~~Secure Channel 3.0 provides confidentiality and integrity for the protected OSDP commands and replies. Credential SM protects the PD-to-card exchange. A security-precondition failure returns osdp_PIVERROR 0x0006 (~~secure channel required)where supported; otherwise the command's existing failure reply applies~~Secure Channel 3.0 required).

Protection of TPKs and biometric data on the OSDP link The ACU and PD **SHALL** use an active Secure Channel 3.0 session with deployment-specific keys for osdp_PIVLOADTPK, osdp_PIVBIOMATCH, and every osdp_PIVBIOMATCHR, including failure results. Factory/default keys do not satisfy this requirement. For osdp_PIVGETDATA, the same requirement applies to a request for any biometric or privacy-key object and every corresponding PIVGETDATAR fragment. The complete object identity determines protection, including requests for one child element, a partial byte range, or an empty result.

Protected PIV objects include Cardholder Fingerprints (5FC103), Cardholder Facial Image (5FC108), and Cardholder Iris Images (5FC121). Protected TWIC objects include Personal Information (DFC001), Handwritten Signature (DFC002), the TPK Container (DFC101), Fingerprints (DFC103), Facial Image (DFC108), Printed Information (DFC109), and Iris Image (DFC121). The rule covers ciphertext, decoded data, and equivalent object identifiers in the active data model. An implementation exposing another biometric or privacy-key object **SHALL** apply the same protection to that object. SP 800-73-5 Part 1, Appendix A, and TWIC Part 2, Sections 4.6–4.7, identify these objects and their container identifiers.

The ACU **SHALL** protect each request before transmitting its payload. The PD **SHALL** check both session state and protected-message status before processing a protected command and before releasing each reply or multipart fragment. An absent Secure Channel 3.0 session, an unprotected request or poll, or use of factory/default keys produces PIVERROR 0x0006, with the command's normal error timing and without credential work or sensitive reply data. The security precondition takes precedence over a command-specific biometric result. A local channel failure has no credential status word. A failed per-message security check terminates the protected operation and discards pending sensitive results and their retransmission state. The ACU **SHALL** discard a partial protected result on PIVERROR 0x0006, including after data delivery has begun.

If the session is lost or replaced before delivery completes, the PD **SHALL** cancel the protected operation, zeroize its loaded TPK and pending biometric data, and discard retransmission replies containing protected data. The ACU **SHALL** discard partial results from the ended session. Both parties **SHALL** require a fresh

protected request in the new session. These requirements protect the ACU–PD link against line tapping, including when the card's biometric object is already encrypted with the TPK.

Biometric data lifetime The PD **SHALL** keep reference records, decrypted CBEFF data, live samples, and derived probes in volatile working memory for the current operation and zeroize them on completion, failure, ABORT, credential removal, application or PIVMODE change, PD reset, tamper detection, or Secure Channel 3.0 loss or replacement. Logs **SHALL** contain only outcomes and non-biometric transaction metadata. Sensitive reply bytes needed for multipart delivery or OSDP retransmission **SHALL** be retained only for that delivery/replay lifetime in the active protected session, then zeroized.

The ACU/PACS maintains enrollment references and their validation evidence under the deployment's privacy, retention, and destruction policy. FIPS 201-3, Sections 2.5–2.6, defines issuer biometric retention and enrollment records; SP 800-76-2, Section 9, defines retained-record protection. TWIC Part 3, Sections 4.4.5–4.4.6, permits verified enrollment references in the back end.

3.12 Set PIV Mode (osdp_PIVMODE)

The `osdp_PIVMODE` command (command code `0xAE`, tentative) defines the operating context for PIV, TWIC, and related credential transactions. OSDP 3.0 replaces the OSDP 2.2 "Application + AID" fields with a configuration payload that includes a required Global Interface Mask byte, PIV Auto challenge parameters when autonomous operation is enabled, and a TLV list of configuration descriptors so controllers can specify multiple application profiles, preferred interfaces, and optional Application Identifiers. PDs **SHALL** advertise support via Function Code 12 bit `0x04` (Enhanced PIV).

Table 30: Set PIV Mode (osdp_PIVMODE)

Size (bytes)	Name	Meaning	Value
0x01	CMND	Command identifier.	`0xAE (tentative)`
0x02	MpSizeTotal	Total size of the reassembled configuration data beginning with PIV Mode Flags, least-significant byte first.	`0x0008–0xFFFF`
0x02	MpOffset	Offset of this fragment within the reassembled configuration data, least-significant byte first.	`0x0000–0xFFFF`
0x02	MpFragmentSize	Number of configuration data bytes following the multi-part fields in this fragment, least-significant byte first.	`0x0000–0xFFFF`
0x01	PIV Mode Flags	First byte of the reassembled configuration data; global behaviour flags (Section 3.12).	`0x00–0xFF`
0x01	Config Count	Number of Configuration Entry TLVs (tag `0x01`); excludes Resume.	`0x01–0x20`

Size (bytes)	Name	Meaning	Value
0x01	Global Interface Mask	Bitmask limiting interfaces available to all configurations (bit 0 contact, bit 1 contactless, bit ~~3 Bluetooth; bit 2~~ 4 Bluetooth; other bits reserved and set to zero).	~~0x00–0x0B~~ 0x00–0x13
0x04	Series Counter	Present only with Auto. Fresh ACU-assigned Series for ordinary configuration; installed Series for Resume. Least-significant byte first.	0x00000000 –0xFFFFFFFF
0x04	Sequence Counter	Present only with Auto. Initial PD allocation value for ordinary configuration; exact next unused value for Resume. Least-significant byte first.	0x00000000 –0xFFFFFFFF
0x20	Supplemental Entropy	Conditional fixed-width field present only when PIV Auto is enabled. All zeroes are permitted but not recommended.	0x20 octets
0x00-n	TLV Data	Configuration Entry TLVs (tag 0x01), followed by optional Resume (0208 and previous Series/Sequence). Auto entries have a mandatory trailing Authentication Key Reference; see Section 3.12.	–

The *PIV Mode Flags* byte enables autonomous operation and credential SM/VCI preparation (Table 31). Bits not listed remain reserved and **SHALL** be transmitted as zero.

Table 31: PIV Mode Flags

Mask	Description
0x01	Enable autonomous ~~PIV~~ authentication (PIV Auto). Data Model 0x01 selects PIV authentication; 0x02 selects TWIC Mode 2 Active Card Authentication. The PD prepares the credential, derives ~~a KMAC256 challenge from the configured counters and entropy~~ the profile-specific challenge, and reports the ~~authentication~~ result using osdp_PIVSTATUS.
0x02	Enable credential Secure Messaging preparation over contact and contactless interfaces. On contactless, also enable VCI when the credential permits VCI without pairing. Applies to autonomous and commanded operation. For PUT DATA, setting this flag requires an established SM session before writing; clearing it selects ordinary PUT DATA.
0x04	Enforce FICAM strict mode (PIV applications only).

Global and per-entry interface masks use the bit positions in Table 32; the Global Interface Mask bounds the interfaces a PD may use when processing all configuration entries. The contact (0x01), contactless (0x02), and Bluetooth (0x10) positions match Function Code 13. PIVMODE reserves bit 0x08; credential SM/VCI preparation uses PIV Mode Flags 0x02.

Table 32: PIV Mode Interface Masks

Mask	Description
0x01	Permit ISO 7816 contact credential interface use.
0x02	Permit ISO 14443 contactless credential interface use.
0x04	Reserved and **SHALL** be transmitted as zero.
0x08	Reserved and **SHALL** be transmitted as zero.
0x10	Permit Bluetooth credential interface use.
~~0x10–0x80~~ 0x20–0x80	Reserved and **SHALL** be transmitted as zero.

The reassembled configuration data begins at *MpOffset* 0x0000 with *PIV Mode Flags*, followed by the configuration count (1-32) and the required Global Interface Mask byte. *MpSizeTotal* counts all configuration data from *PIV Mode Flags* through the final TLV byte. If and only if PIV Auto is enabled in *PIV Mode Flags*, the Global Interface Mask is followed by the Series Counter, Sequence Counter, and fixed 32-byte Supplemental Entropy field before the TLV list. When PIV Auto is not enabled, those three conditional fields are absent and the TLV list begins immediately after the Global Interface Mask. Each configuration appears as a TLV:

The configuration count **SHALL** equal the number of complete Configuration Entry TLVs in the reassembled payload. Consequently, *MpSizeTotal* **SHALL** be at least eight bytes when PIV Auto is disabled and at least 49 bytes when it is enabled, and **SHALL** equal the exact encoded length of the required header, any conditional PIV Auto fields, all configuration TLVs, and any Resume TLV. Tags and lengths in this list are one byte each. The optional Resume TLV does not contribute to the configuration count. Truncated fields, unknown tags, trailing bytes, or a count that does not match the Configuration Entry sequence **SHALL** be treated as malformed configuration data.

- **Global Interface Mask (required)** — One-byte mask (no tag) that limits the interfaces the PD may use for all configurations (see Table 32). A value of 0x00 indicates no global restriction; reserved bits **SHALL** be zero.
- **Configuration Entry (Tag 0x01)** — Contains the following fields in order:
  - *Data Model* (0x01 PIV, 0x02 TWIC, 0x03 CAC, values 0x04-0x7F reserved, 0x80-0xFF private use). CAC entries use the same interface and AID encoding as PIV and TWIC entries.
  - *Interface Mask* — Uses the same bit positions as the global mask (Table 32) and **SHALL NOT** assert interfaces excluded by the Global Interface Mask. A value of 0x00 means “any interface permitted by the global mask.”
  - *AID Length* (0-16) and *AID* bytes when provided.

- *Authentication Key Reference* — One trailing byte present if and only if PIV Auto is enabled. For Data Model 0x01, the permitted values are 0x9A (PIV Authentication) and 0x9E (asymmetric Card Authentication). For Data Model 0x02, the permitted value is 0x9E (TWIC active card authentication). CAC and private-use Auto profiles are not defined here and **SHALL NOT** be accepted as this Auto profile. Non-Auto entries are unchanged.

TWIC ACA profile selection With Auto flag 0x01 enabled, Data Model 0x02 and key reference 0x9E select the TWIC Mode 2 Active Card Authentication (ACA) profile specified in *TWIC Specification, Part 3*, Section 4.4.4. Before configuring this profile, the ACU **SHALL** verify Function Code 12 flags 0x04, 0x08, and 0x10. A supporting PD **SHALL** advertise all three. An Auto configuration containing a TWIC entry **SHALL** contain TWIC entries only. A PD **SHALL** reject a mixed Auto configuration or a TWIC ACA configuration it does not support with PIVERROR 0x0005, preserving the installed configuration. Non-Auto configurations retain their existing rules.

The Configuration Entry value length **SHALL** be exactly 0x03 plus AID Length without Auto, or 0x04 plus AID Length with Auto. The key reference belongs to that entry; there is no configured algorithm byte and no implicit key default. The earlier Auto entry layout without the key reference **SHALL** be rejected, not guessed or interpreted as selecting 0x9E.

PDs process configuration entries in order, applying the global mask before honouring each entry's mask. Once an application has been selected for an Auto attempt, failure of discovery, a security prerequisite, or authentication **SHALL NOT** cause selection of another entry, key, or interface. Invalid masks, unknown data-model identifiers, unsupported Auto models, invalid key references, or malformed TLVs return osdp_PIVERROR 0x0005; unmet channel requirements return 0x0006; multi-part framing errors return 0x0007; insufficient configuration storage returns 0x1021. The PD **SHALL** validate the complete reassembled command before replacing a configuration or clearing its credential context. Rejected configuration preserves the installed configuration and valid security state. The final fragment receives osdp_ACK only after complete validation and acceptance; credential work for the newly accepted Auto attempt begins after that ACK.

Per-credential algorithm discovery For each Auto attempt, the PD **SHALL** retrieve the certificate corresponding to the configured key in the authentication application: object 5FC105 for PIV 0x9A, or 5FC101 for asymmetric 0x9E. TWIC DFC101 is a TPK container, not an authentication certificate. The PD **SHALL** extract certificate tag 0x70 and its single CertInfo byte in tag 0x71: 0x00 denotes uncompressed DER; 0x01 denotes GZIP-compressed DER. Missing, duplicated, or invalid required fields, failed decompression, and malformed DER certificate data produce PIVSTATUS Result 0x03. The PD **SHALL** discover the algorithm from the certificate's *subjectPublicKeyInfo*, following SP 800-73-5 Part 1, Appendix C.1, using the subject RSA modulus size or named EC curve, not the issuer's certificate-signature algorithm. It **SHALL NOT** reuse another credential's algorithm or send trial authentication commands to probe algorithms. RSA profiles require rsaEncryption SubjectPublicKeyInfo; EC profiles require id-ecPublicKey with the named P-256 or P-384 curve. Other subject-key profiles, including RSA-PSS-restricted keys, **SHALL** produce Result 0x04, not a guessed algorithm or a different signature encoding.

Table 33: PIV Auto Subject Public-Key Profiles

Subject public key	Algorithm	Digest profile	Digest bytes
RSA-2048	0x07	SHA-256	0x20
RSA-3072	0x05	SHA-384	0x30
EC named curve P-256	0x11	SHA-256	0x20
EC named curve P-384	0x14	SHA-384	0x30

This table fixes this proposal's Auto digest choice, independently of the issuer's signature. A missing or malformed certificate produces Result 0x03; an unsupported subject-key algorithm, size, curve, or Auto profile produces Result 0x04. Neither permits another key or symmetric authentication as fallback. The PD reports the discovered PIV identifier when one can be determined, and 0x00 otherwise. No GENERAL AUTHENTICATE is sent until discovery succeeds. The PD verifies the returned key proof with this certificate's public key; the ACU retains certificate-path, validity, revocation, identity-binding, algorithm-acceptance, and access-decision responsibilities. ~~PD success does not prove trust.~~

TWIC ~~Auto application routing~~Mode 2 Active Card Authentication For a TWIC entry with AID Length zero, the PD **SHALL** select the common nine-byte prefix A00000036720000001; a supplied AID is honoured instead. Full selected AIDs A000000367200000010103 and ~~identify NextGen and Legacy, respectively~~ A000000367200000010104 identify NextGen V8 and V10, respectively; A000000367200000010101 identifies Legacy. A prefix, absent certificate, or failed authentication **SHALL NOT** be treated as evidence of Legacy. The PD **SHALL** obtain positive full-AID generation evidence from the SELECT response; unavailable evidence produces Result 0x03, and an unsupported ~~generation~~ major version produces Result 0x04, without guessing. For NextGen V8 and V10, the PD **SHALL** read the certificate and perform 0x9E authentication in the TWIC application. For another valid, non-reserved minor version within major version 0x01, the PD **SHALL** retain the TWIC application and attempt the same certificate-based route, subject to the existing certificate, algorithm-discovery, and authentication requirements. An unknown minor version alone **SHALL NOT** produce Result 0x04 or authorize a transition to the PIV application. Only for positively identified Legacy TWIC, it **SHALL** select the credential's PIV application and use its asymmetric 0x9E key and 5FC101 certificate. This is the configured TWIC route, not failure-driven fallback. ~~Before that Legacy transition, the PD **SHALL** read the TWIC CHUID and retain its identity only for this attempt, then bind it to the PIV application's certificate below.~~ The configured entry and TWIC authentication purpose remain unchanged; the actual selected application's access and state-clearing rules apply. CARDSTATUSR reports the actual selected AID. These routes follow TWIC Part 2, Sections 4.1, 4.4.2, 4.7.1, and 7.5, Appendices C and J, and TWIC Part 3, Appendix D.3.

For ~~NextGen TWIC~~ GET DATA, the PD **SHALL** use ~~Le = 0xFF, not absent Le or , and~~ the explicit response-length and continuation rules in paragraph 2.1.2 to retrieve the complete object~~using GET RESPONSE as specified in~~ . TWIC Part 2, Section 5.2 and Appendix E, provide the application-specific data-object and response-processing descriptions. Authentication command and response chaining follow Section 5.3 and Appendix E. A partial Dynamic Authentication Template is not success. For TWIC ACA, the PD **SHALL** obtain the 25-byte FASC-N from the Card Authentication Certificate's subjectAltName

twicFASC-N entry (OID 1.3.6.1.4.1.29138.6.6), as specified by TWIC Part 3, Section 4.4.4, items 10 and 14. Missing, malformed, or multiple FASC-N entries produce Result 0x03. The PD **SHALL** ~~obtain~~ encode the UUID field as sixteen zero bytes in both the attempt template and PIVSTATUS. PIV entries obtain their FASC-N and nonzero UUID/GUID from CHUID object 5FC102. The TWIC ACA subject-key profiles are RSA-2048 (0x07, SHA-256) and ECC P-384 (0x14, ~~tags and . PIV and NextGen TWIC success require a nonzero UUID. For a positively identified Legacy TWIC only, an actually retrieved 16-byte nil GUID is permitted when a well-formed, nonzero 25-byte FASC-N matches the FASC-N in the authentication certificate's subjectAltName. An absent or unreadable GUID **SHALL NOT** be fabricated as nil. The PD and~~ SHA-384); another subject-key profile produces Result 0x04. TWIC Part 2, Section 5.3, specifies the GENERAL AUTHENTICATE algorithm and Card Authentication key references.

On successful proof verification, the PD reports Result 0x00 and the complete returned Dynamic Authentication Template. During the same uninterrupted credential presentation, the ACU **SHALL** ~~enforce this Legacy binding; the ACU permits the exception only for its configured TWIC purpose.~~ retrieve the current Card Authentication Certificate through osdp_PIVGETDATA from the application used for ACA, decode it, discover its subject-key algorithm, and verify that its certified FASC-N and algorithm match PIVSTATUS. It **SHALL** reconstruct the challenge and verify the signature using that certificate's public key. Following TWIC Part ~~2, Sections 4.3 and 6, define the relevant identity representations.~~

~~TWIC Auto success reports a certificate-associated key proof, not completed TSA reader authentication, CCL checking, biometric comparison, or TSA reader conformance. In particular, TWIC Part~~ 3, Section 4.4.4~~, specifies a random or pseudorandom challenge of at least 127 bytes. The 32-~~, items 6–16, the ACU **SHALL** validate the issuer signature and certificate path, validity period, digitalSignature key usage, id-TWIC-cardAuth extended key usage (OID 1.3.6.1.4.1.29138.6.8), and critical extensions. It **SHALL** check credential expiration and the certified FASC-N against TSA's Canceled Card List (CCL), and perform CRL checking when configured. These steps implement the card-authentication and card-validity portions of 33 CFR 101.520(a)–(b). The ACU **SHALL** accept the ACA factor only when the certificate and proof checks succeed, the credential is unexpired, and its FASC-N is absent from the CCL.

The PD and ACU **SHALL** retain acceptance state only for that uninterrupted presentation. Credential removal, replacement, or ~~48-byte KMAC digest and RSA encoding here are not claimed to establish that separate requirement; deployments requiring it need the corresponding TWIC authentication profile and qualification~~a communication interruption that prevents confirmation of continuity ends the transaction. The selected application and security state follow normal PIVSELECT rules; the controller's ACA evidence remains scoped to the same presentation through an intentional application selection for subsequent commanded verification.

When PIV Auto is enabled, the PD prepares the presented credential as described in Subsection 3.1, derives the challenge, issues PIV GENERAL AUTHENTICATE, and reports completion through osdp_PIVSTATUS. The PD **SHALL** use the configured key, the algorithm discovered from that credential's role certificate, and the selected interface for that attempt. If SM is unavailable, the PD continues that same authentication only when the credential's access rules permit the authentication without SM, as for key 0x9E. If the configured authentication requires SM, the PD reports the SM failure. A failed signature or cryptogram check ends the attempt even when the authentication could otherwise proceed without SM.

PIV Auto can establish VCI when the credential permits VCI without a pairing code. For an authentication that requires SM, the PD first checks whether SM is established. If SM is unavailable, the PD reports Result 0x02

(SM failure) with the applicable detail code. The PD then checks any VCI and PIN-verification requirements, in that order. If the selected authentication requires VCI and pairing is required, the PD completes with Result 0x05, Detail 0x01 (pairing code required). The ACU can then begin commanded pairing. Required PIN verification that has not been completed returns Result 0x05, Detail 0x02 (PIN verification required). PIV Auto performs one automatic attempt per presentation, subject to counter availability. Keeping the credential presented does not restart the attempt. After a terminal prerequisite failure is delivered, the ACU may complete commanded pairing and PIVSPE PIN verification while Auto remains enabled, then explicitly resume as defined below. SPE Auto Execute remains prohibited while PIV Auto is enabled. Removal and a new presentation, or ordinary PIVMODE reapplication, starts a new attempt when counters are available. Ordinary reapplication follows mode-change clearing; loading an anchor alone does not restart Auto.

PIV Auto configuration **SHALL** be accepted only when OSDP Secure Channel ~~2.0~~ 3.0 is active and the channel is not using a factory/default SCBK or other factory key material. A PD **SHALL** reject PIV Auto configuration or execution with `osdp_PIVERROR` 0x0006 (~~secure channel~~ Secure Channel 3.0 required) when these preconditions are not met.

Counter ownership and exhaustion Ordinary Auto PIVMODE supplies a Series, an initial Sequence, and fixed-width 32-byte Supplemental Entropy. All-zero entropy is permitted but not recommended. The ACU **SHALL** allocate a fresh Series for every ordinary Auto configuration and prevent its reuse for the same PD across ACU or PD restart. Within that Series, the PD **SHALL** reserve the next unused Sequence before every attempt, including presentation preparation and accepted Resume. The supplied initial Sequence is used first; each reservation advances the allocator by one. Failure, cancellation, timeout, and removal do not release a reservation; removal does not reset the allocator. PIVSTATUS reports the reserved pair even if no challenge was generated. No new attempt starts while another operation or undelivered completion is pending.

Sequence 0xFFFFFFFF may be reserved once. That attempt may finish normally; afterward Auto is suspended until ordinary PIVMODE installs a fresh Series with normal clearing. The PD **SHALL NOT** wrap, start another Auto attempt, or issue further Auto credential commands or repeated exhaustion statuses. The terminal status carrying 0xFFFFFFFF informs the ACU to install a fresh Series before another attempt. Resume cannot change Series to avoid that clearing rule.

Explicit prerequisite continuation An optional Resume TLV is 0208, followed by the previous terminal PIVSTATUS Series and Sequence, four bytes each, least-significant byte first. It **SHALL** occur once, after all Configuration Entry TLVs, and only with Auto enabled. Its presence requests Resume; absence retains ordinary PIVMODE clearing. It is transient metadata, not installed configuration. Header Series **SHALL** equal the installed Series; header Sequence **SHALL** equal its exact next unused value. Supplemental Entropy may change; flags, global mask, count, entry order, models, entry masks, AIDs, and key references **SHALL** otherwise match the installed configuration exactly.

Resume requires complete delivery of a terminal PIVSTATUS 0x05/0x01 or 0x05/0x02 whose counter pair matches the TLV, with the same continuously present credential, selected application, interface, configured entry, and OSDP session. The PD **SHALL** recheck these conditions after reassembly and before acceptance. Removal, credential reset, context-changing selection, ordinary mode replacement, an active anchor change

invalidating SM, blocked SM, or OSDP session replacement invalidates continuation. Delivering the final fragment of an old status after context loss **SHALL NOT** restore eligibility. Successful commanded pairing, PIN verification, permitted reads, and diagnostics preserve eligibility.

Acceptance **SHALL** atomically consume prior eligibility and reserve the supplied pair, then ACK before credential work. It preserves completed pairing, SM/VCI, PIN verification, and any otherwise valid completed PIN cache. The PD **SHALL NOT** select another entry, change interface or key, submit VERIFY automatically, or retry authentication automatically. The accepted attempt returns exactly one polled PIVSTATUS. A still-missing prerequisite returns 0x05/0x01 or 0x05/0x02 with the newly reserved pair. Removal after acceptance returns Result 0x01; SM failure follows existing blocked-state rules.

Before acceptance, malformed Resume, changed configuration, invalid counters, or exhaustion return PIVERROR 0x0005; unmet channel requirements return 0x0006; an absent credential returns 0x1026; stale or invalidated context returns 0x1023; another operation or undelivered completion returns NAK 0x09 (busy). Rejection starts no attempt and consumes no counters or valid security state. Normal OSDP retransmission of the same authenticated command bytes in the same session and OSDP sequence context replays the cached ACK or completion without another reservation or card operation. The same logical Resume sent with a new OSDP sequence is a new request and cannot reuse consumed eligibility.

The PIV Auto challenge **SHALL** be derived from the active Secure Channel ~~2.0~~ 3.0 session key material using two KMAC256 operations. First, the PD derives a 32-byte PIV Auto KDK with key material formed by concatenating S-ENC, S-MAC1, and S-MAC2; empty input data; output length 32 bytes; and customization string OSDP-PIV-AUTO-KDK-v1. Second, the PD derives the PIV-entry challenge digest using the PIV Auto KDK as the KMAC key, the fixed-width template as input data, output length L, and customization string OSDP-PIV-AUTO-CHALLENGE-v1. The template **SHALL** be exactly 0x61 bytes in Table 34 order, with no implicit padding or length prefixes. Output length L and digest profile follow Table 33. PIV entries **SHALL** use the successfully read nonzero UUID/GUID; TWIC ACA entries **SHALL** ~~have been read successfully; only the positive Legacy TWIC exception above permits nil~~use sixteen zero bytes. PIV Auto challenge derivation **SHALL NOT** use static factory keys. ~~This serialization does not redefine the Secure Channel 2.0 key-binding requirement; its independent validation remains separate.~~

Table 34: Normative PIV Auto KMAC Challenge Template

Field, in order	Bytes	Encoding
Template label and version	0x0E	ASCII OSDP-PIV-AUTO, followed by byte 0x01.
Series Counter	0x04	Reserved Series, least-significant byte first.
Sequence Counter	0x04	Reserved Sequence, least-significant byte first.
FASC-N	0x19	Raw credential bytes.
UUID/GUID	0x10	~~Raw credential bytes, including an allowed actual Legacy nil GUID.~~ PIV: successfully read, nonzero credential UUID/GUID. TWIC ACA: sixteen zero bytes.
Authentication Key Reference	0x01	Configured key reference for this attempt.
Algorithm Identifier	0x01	Discovered subject-key identifier.

Field, in order	Bytes	Encoding
Supplemental Entropy	0x20	Exact configured or resumed ~~32-byte~~ 0x20-byte value.

For PIV entries using RSA, the PD **SHALL** encode the KMAC digest with EMSA-PKCS1-v1_5, the profile's SHA-256 or SHA-384 DigestInfo, and a block matching the discovered modulus length. The KMAC output is already the message hash~~and **SHALL NOT** be hashed again. For~~ . For PIV entries using EC, it **SHALL** submit the 32- or 48-byte digest directly~~, without rehashing.~~ . TWIC entries use the challenge construction below. GENERAL AUTHENTICATE uses discovered P1, configured key reference P2, and tag 0x7C containing empty 0x82 and 0x81 carrying the encoded input. The returned 0x7C template is preserved whole, excluding transport status words.

TWIC Auto challenge construction For Data Model 0x02, the PD **SHALL** derive a 0x7F-byte pseudorandom challenge C using KMAC256 with the PIV Auto KDK as key, the exact Table 34 template as input, output length 127 bytes, and customization string OSDP-TWIC-AUTO-CHALLENGE-v1. The Series and Sequence allocation rules provide a fresh input for each attempt.

The PD **SHALL** compute D=SHA-256(C) for RSA-2048 or D=SHA-384(C) for ECC P-384. Following TWIC Part 2, Section 5.3, it **SHALL** encode D with SHA-256 DigestInfo and EMSA-PKCS1-v1_5 padding for RSA-2048, or supply D as the prehashed ECDSA input for ECC P-384. GENERAL AUTHENTICATE tag 0x81 carries the resulting 0x100-byte RSA block or 0x30-byte EC digest, with key reference 0x9E. The PD **SHALL** verify the signature against D. The ACU **SHALL** independently reconstruct C and D and verify the returned signature. The configured TWIC data model selects this construction for both the Legacy PIV application route and the NextGen TWIC application route.

Standards-review note: TWIC challenge interpretation This proposal interprets the 0x7F-byte pseudorandom value as the ACA challenge message and applies the selected signature algorithm's hashing and encoding before GENERAL AUTHENTICATE. TWIC Part 3, Section 4.4.4, item 12, specifies transmission of at least 127 bytes of unique challenge data; Part 2, Section 5.3, specifies the card command's padded hash input and algorithm references. Standards review must resolve the relationship between that literal transmission requirement and the proposed RSA/ECC prehash construction before accepting conformance to item 12.

3.13 PIV Auto Status Reply (osdp_PIVSTATUS)

When a PIV Auto attempt completes, including a failure during preparation, the PD reports the result as a poll response using reply code 0x89 (tentative). The first osdp_POLL response after completion **SHALL** contain osdp_PIVSTATUS; subsequent fragments, when required, use the same reply and advance *MpOffset* until the complete status payload is delivered. There is no corresponding osdp_PIVSTATUS request command.

Table 35: PIV Auto Status Reply (osdp_PIVSTATUS)

Size (bytes)	Name	Meaning	Value
0x01	CMND	Reply identifier.	0x89 (tentative)
0x02	MpSizeTotal	Total size of the complete PIV Auto status payload, least-significant byte first.	0x0000–0xFFFF
0x02	MpOffset	Offset of this fragment within the complete status payload, least-significant byte first.	0x0000–0xFFFF
0x02	MpFragmentSize	Length of the status payload bytes in this fragment, least-significant byte first.	0x0000–0xFFFF
0x01	Result	Result of the PIV Auto authentication attempt. See Table 36.	0x00–0xFF
0x01	Detail	Result-specific detail from Table 36. Set to 0x00 where no detail is defined.	0x00–0xFF
0x04	Series Counter	Reserved Series, even on failure before challenge generation. Least-significant byte first.	0x00000000 –0xFFFFFFFF
0x04	Sequence Counter	Reserved Sequence; also binds an eligible Resume. Least-significant byte first.	0x00000000 –0xFFFFFFFF
0x19	FASC-N	PIV: FASC-N ~~read from the credential~~ from CHUID. TWIC ACA: FASC-N from the Card Authentication Certificate.	200-bit FASC-N
0x10	UUID	~~Credential~~ PIV: credential UUID/GUID. ~~Success is nonzero except for the positively identified, FASC-N-bound Legacy TWIC exception~~, nonzero on success. TWIC ACA: sixteen zero bytes. Failure may report zero when unavailable.	0x10 octets
0x01	Key Reference	Configured authentication key reference, including failure before challenge generation.	0x9A or 0x9E

Size (bytes)	Name	Meaning	Value
0x01	Algorithm Identifier	Discovered GENERAL AUTHENTICATE identifier; 0x00 on failure when none could be determined. Never configured by the ACU.	For example 0x07
0x02	Signed Response Length	Total length of the signed response, least-significant byte first.	0x0000–0xFFFF
0x00-n	Signed Response	Complete response bytes returned by the credential. This field is part of the reassembled status payload.	Variable
0x00 or 0x02	Status Word	SW1 followed by SW2 when the credential returned the reported failure. Omitted in the other cases specified below. Part of the reassembled payload.	–

Result and detail codes follow Table 36. When *Result* is 0x00, *Detail* is reserved and **SHALL** be 0x00. When *Result* is not 0x00, the PD **SHALL** set *Signed Response Length* to 0x0000 unless a credential response was received and is useful for ACU diagnostics.

The PD **SHALL** append SW1 followed by SW2 when a credential command returned the failure being reported. For a verified protected reply, the PD uses the protected status. The PD **SHALL** omit this field on success, local failure, timeout, credential removal without a failure reply, or failure to verify a protected reply. An earlier command's status **SHALL NOT** be used. These are the status-word rules of Subsection 2.2.1. The status word follows Signed Response in the reassembled payload and may cross a fragment boundary. After accounting for the fixed fields from Result through Signed Response Length in Table 35 (0x37 bytes) and *Signed Response Length*, the ACU **SHALL** accept exactly zero or two remaining bytes; any other length is malformed. Result 0x06 uses Detail 0x00; the complete status word carries the credential's error. The PD returns one terminal PIVSTATUS, with no follow-up PIVERROR for that attempt.

Table 36: PIV Auto Status Result Codes

Value	Description
0x00	Key-proof success. *Detail* is 0x00~~;~~ . PIV UUID is nonzero~~except for the defined, FASC-N-bound Legacy TWIC case~~; TWIC ACA UUID is zero and FASC-N comes from the Card Authentication Certificate. Signed Response carries the complete returned Dynamic Authentication Template. ~~ACU trust~~ The ACU completes the profile-specific certificate and acceptance checks~~remain required~~.
0x01	Credential not present or removed before PIV Auto completed.

Value	Description
0x02	Credential SM unavailable or failed. Detail: 0x01 trust store empty; 0x02 no matching trust path; 0x03 missing or invalid CVC; 0x04 invalid authentication cryptogram; 0x05 required SM unsupported; 0x06 other SM processing failure; 0x07 required SM disabled. Detail assignments are tentative.
0x03	Required ~~credential dataobject, FASC-N, UUID/GUID,~~ profile data, key, certificate, or application-generation evidence is missing, malformed, or cannot be obtained. PIV requires CHUID FASC-N and nonzero UUID; TWIC ACA requires certificate FASC-N.
0x04	Configured key, credential generation, or discovered subject public-key algorithm/profile is not supported for Auto.
0x05	Required credential security condition is absent. Detail: 0x01 pairing code required; 0x02 PIN verification required; 0x03 required VCI unsupported; 0x04 multiple contactless credentials; 0x00 other unmet condition. Detail assignments are tentative.
0x06	Credential GENERAL AUTHENTICATE operation failed. Detail is 0x00; the Status Word field carries the credential failure status when available under the rules below.
0x07	Signed response verification failed.
0x08	Timeout or PD internal failure before a definitive credential result was available.
0x09–0x7F	Reserved for future standard result codes.
0x80–0xFF	Private use.

For Result 0x02, Details 0x03, 0x04, and 0x06 block further credential operations until the context is restarted, as specified in Subsection 3.1. Details 0x01, 0x02, 0x05, and 0x07 leave operations permitted without SM available.

For failure before challenge generation, the PD **SHALL** report the reserved series and sequence counters and configured key reference. It reports the discovered algorithm identifier, or 0x00 if none could be determined. It **SHALL** set unavailable FASC-N and UUID fields to zero and *Signed Response Length* to zero. These identity fields describe data actually obtained from the credential. The PD **SHALL** set *Detail* to zero unless the result defines a detail value.

The reply reports the exact reserved counters, credential identity, and returned signed response. The ACU **SHALL** check the key reference against its configured authentication purpose, independently discover the algorithm from the role certificate, and reject a mismatching reported algorithm. It verifies counter uniqueness, identity binding, the proof, and certificate path under its trust policy. ~~Success~~ For PIV entries, success **SHALL** include a nonzero UUID~~, except for the bound Legacy TWIC case in~~ . For TWIC ACA, the UUID field **SHALL** be zero and verification follows Subsection 3.12. The PD **SHALL** preserve returned proof bytes, excluding transport status words. PIVSTATUS's wire layout is unchanged by key discovery and Resume.

If the complete signed response does not fit in one OSDP response, the PD **SHALL** fragment osdp_PIVSTATUS using the OSDP 3.0 multi-part convention. *MpSizeTotal* is the total status payload size, *MpOffset* is the zero-based offset of the fragment, and *MpFragmentSize* is the number of payload bytes carried in that fragment. The logical status payload begins at offset zero with the fixed fields; each

fragment carries the next contiguous bytes of that payload. A fragment boundary may occur within the fixed fields or within the signed response.

3.14 VCI Trust Anchor Load (osdp_PIVVCILOADTA)

The `osdp_PIVVCILOADTA` command (command code `0xAF`, tentative) loads, replaces, or deletes a credential SM trust anchor. The existing command name and record format apply to both contact and contactless SM. The ACU **SHALL** validate and authorize each anchor under its deployment certificate policy before loading it. The PD uses the loaded public key to validate the credential's SM CVC.

Table 37: VCI Trust Anchor Load (osdp_PIVVCILOADTA)

Size (bytes)	Name	Meaning	Value
0x01	CMND	Command identifier.	`0xAF (tentative)`
0x02	MpSizeTotal	Total size of the reassembled operation data, including the 0x0C-byte metadata block and Anchor Data, least-significant byte first.	`0x000C–0x024C`
0x02	MpOffset	Offset of this fragment within the reassembled operation data beginning with Operation, least-significant byte first.	`0x0000–0xFFFF`
0x02	MpFragmentSize	Number of operation data bytes following the multi-part fields in this fragment, least-significant byte first.	`0x0000–0xFFFF`
0x01	Operation	`0x00` load/replace anchor; `0x01` delete anchor. Anchor inventory is requested with `osdp_PIVVCITASTATUS`.	`0x00–0x01`
0x01	Trust Anchor ID	Controller-assigned identifier for the anchor.	`0x01–0x80`
0x08	Issuer Identifier Number	For load/replace, the leftmost eight bytes of the subjectKeyIdentifier from the content-signing X.509 certificate and a match for tag `0x42` in the trust anchor record. For delete, this value **SHALL** match the stored anchor's IIN.	`0x00–0xFF`
0x02	Anchor Length	Total size of the variable-length VCI trust anchor record, least-significant byte first. Set to `0x0000` when deleting.	`0x0000–0x0240`

Size (bytes)	Name	Meaning	Value
0x00-n	Anchor Data	Fragment bytes of the OSDP-defined VCI trust anchor record. Omitted for delete operations.	0x00–0xFF

The reassembled operation data begins at *MpOffset* 0x0000 with a 12-byte metadata block containing *Operation*, *Trust Anchor ID*, *Issuer Identifier Number*, and *Anchor Length*, followed immediately by *Anchor Data*. These fields may cross fragment boundaries and are interpreted only after the complete logical data has been reassembled. Controllers set *Operation* to 0x00 to load or replace the anchor identified by *Trust Anchor ID*. Set *Operation* to 0x01 with *Anchor Length* 0x0000 and *MpSizeTotal* 0x000C to delete the anchor. For a delete operation, the supplied IIN **SHALL** match the IIN of the stored anchor selected by *Trust Anchor ID*; a mismatch **SHALL** be rejected with osdp_PIVERROR 0x0005 (invalid parameter). Inventory is requested with osdp_PIVVCITASTATUS, not with osdp_PIVVCILOADTA. The *Issuer Identifier Number (IIN)* is an OSDP-defined anchor selector: it **SHALL** be the leftmost eight bytes of the subjectKeyIdentifier from the content-signing X.509 certificate associated with this anchor. For load/replace, this value **SHALL** match the IIN field (tag 0x42) in the VCI trust anchor record.

The *Anchor Length* declares the exact size of the variable-length VCI trust anchor record. For every operation, *MpSizeTotal* **SHALL** equal 0x000C plus *Anchor Length*. For an operation accepted for processing, the sum of all accepted *MpFragmentSize* values **SHALL** equal *MpSizeTotal*, and no fragment **SHALL** extend beyond *MpSizeTotal*. For load operations, the reassembled bytes following the metadata block **SHALL** contain exactly *Anchor Length* bytes. A conforming VCI trust anchor record **SHALL NOT** exceed 576 bytes (0x0240), making the maximum *MpSizeTotal* 0x024C. The PD **SHALL** accept any valid record length from the minimum BER-TLV encoding through 0x0240, subject to supported-key and storage-capacity checks.

PD storage accounting for VCI trust anchors is variable-length. A PD **SHALL NOT** treat every anchor as consuming the maximum 0x0240 bytes unless that is the PD's documented physical allocation model. When reporting capacity, loading, replacing, or deleting anchors, the PD **SHALL** account for stored anchors by their actual *Anchor Length* values and **SHALL** use the same accounting rule for both osdp_PIVVCITASTATUSR *Available Bytes* and osdp_PIVVCILOADTA storage acceptance. A replacement operation **SHALL** be evaluated after releasing the storage occupied by the existing anchor with the same *Trust Anchor ID*; if the replacement cannot be completed, the prior anchor **SHALL** remain available. A delete operation **SHALL** release the storage associated with the deleted anchor.

Table 38: VCI Trust Anchor Record Format

Tag	Length	Name	Value
0x7F50	Variable	VCI Trust Anchor Record	OSDP-defined record used to verify a secure messaging CVC or an Intermediate CVC.
0x80	0x01	Record Profile Identifier	0x01.
0x42	0x08	Issuer Identifier Number	Anchor selector. This value **SHALL** match the IIN in the osdp_PIVVCILOADTA metadata.
0x7F49	0x01–0x0226	SubjectPublicKeyInfo	DER-encoded X.509 SubjectPublicKeyInfo for the public key trusted by the ACU. Supported public-key algorithms are RSA public keys up to 4096 bits and EC public keys for P-256 or P-384.

The VCI trust anchor record ~~is intentionally not a NIST SP 800-73 Intermediate CVC and does not carry X.509 path-validation state~~supplies a public key for credential CVC verification. The ACU is responsible for selecting, loading, unloading, and validating the VCI anchors it permits a PD to use before the PD relies on them. The PD uses the record only as a local public-key verifier selected by IIN. The maximum record size is derived from the largest supported key case: a 550-byte RSA-4096 SubjectPublicKeyInfo plus the fixed fields above and BER-TLV length overhead. The resulting worst case is 573 bytes; this specification rounds the maximum accepted record size to 576 bytes.

Use during credential SM establishment When the PD receives the secure messaging CVC from the credential, it **SHALL** first extract the Issuer Identification Number from tag 0x42. If a loaded VCI trust anchor has a matching IIN, the PD **SHALL** use that anchor's SubjectPublicKeyInfo to verify the secure messaging CVC signature. If verification succeeds, the PD may continue credential SM establishment using the public key in the secure messaging CVC.

If no loaded VCI trust anchor matches the secure messaging CVC IIN, the PD **SHALL** retrieve the Secure Messaging Certificate Signer data object (object identifier 0x5FC122, container identifier 0x1017) from the credential and inspect it for one Intermediate CVC (tag 0x7F21). If the object is absent or no Intermediate CVC is present, validation fails. If an Intermediate CVC is present, the PD **SHALL** verify all of the following before continuing:

- The secure messaging CVC IIN (tag 0x42) equals the Subject Identifier (tag 0x5F20) in the Intermediate CVC.
- The Intermediate CVC Role Identifier (tag 0x5F4C) is 0x12.
- A loaded VCI trust anchor exists whose IIN equals the Intermediate CVC IIN (tag 0x42).
- The Intermediate CVC signature verifies using the matching VCI trust anchor public key.
- The secure messaging CVC signature verifies using the public key carried in the Intermediate CVC.

The PD validates only the CVC chain needed for credential SM establishment: either the loaded trust anchor directly verifies the secure messaging CVC, or the loaded trust anchor verifies the Intermediate CVC that then verifies the secure messaging CVC. The PD **SHALL NOT** establish credential SM unless this CVC validation succeeds. The PD **SHALL NOT** perform full X.509 path validation, validity-period evaluation, certificate-policy evaluation, or revocation processing for VCI trust anchors; those checks are completed by the ACU before loading anchors and again as an acceptance policy after credential SM state is reported.

An absent signer object, an absent intermediate CVC, or the absence of a matching loaded anchor leaves the PD without a matching trust path. The PD reports SM Detail `0x02` for that condition. A malformed CVC, incorrect role or linkage, or failed signature check reports Detail `0x03` and blocks credential operations. These distinctions apply to both PIV Auto results and SM diagnostics.

ACU acceptance of credential SM After receiving `osdp_CARDSTATUSR` with credential SM established, the ACU **SHALL** validate the content-signing certificate corresponding to the reported *SM Anchor IIN* before treating the credential SM session as accepted for the deployment. ~~This validation is an additional ACU acceptance check and does not replace the PD 's required~~ The PD performs CVC validation during credential SM establishment; the ACU performs certificate-path acceptance checks. The ACU **SHALL** perform all of the following:

- Obtain the content-signing X.509 certificate. If the ACU does not already possess a cached copy whose subjectKeyIdentifier leftmost-eight-bytes equals the *SM Anchor IIN*, the ACU **SHALL** use `osdp_PIVGETDATA` to retrieve the Secure Messaging Certificate Signer data object (object identifier `0x5FC122`, container identifier `0x1017`) and extract the certificate from inner tag `0x70`.
- Verify that the leftmost eight bytes of the certificate's subjectKeyIdentifier equal the *SM Anchor IIN* reported in `osdp_CARDSTATUSR`.
- Verify that the certificate's SubjectPublicKeyInfo matches the SubjectPublicKeyInfo stored in the loaded VCI trust anchor record for that IIN.
- Verify that the certificate includes the PIV content-signing extended key usage OID (`2.16.840.1.101.3.6.7`) or the PIV-I equivalent (`2.16.840.1.101.3.8.7`).
- Verify that the current time falls within the certificate's validity period (`notBefore` $\leq$ now $\leq$ `notAfter`). If the certificate has expired or is not yet valid, the ACU **SHALL** reject the credential SM session and **SHALL** unload the corresponding trust anchor.
- Perform X.509 path validation and revocation checking in accordance with the deployment's certificate validation policy.

Table 39: VCI Trust Anchor Examples from NIST Vectors

Vector	Content Signer	Anchor IIN	SPKI	Record
card_2, card_5	Test PIV Content Signer 4 (EC P-384, direct)	B6103792270FBD08	0x78 bytes	0x8C bytes
card_3	Test PIV Content Signer 1 (RSA-2048, intermediate)	9BC5E7735F14EC7F	0x126 bytes	0x13D bytes
card_4	Test PIV Content Signer 3 (EC P-256, direct)	1905976BDDCA823D	0x5B bytes	0x6E bytes
card_16	Test PIV-I Content Signer 1 (RSA-2048, intermediate)	4CC9C7B3A4F1C83E	0x126 bytes	0x13D bytes

In the card_16 intermediate example, the secure messaging CVC IIN is 6F8FF48F42B3E22E; the Intermediate CVC Subject Identifier (tag 0x5F20) is also 6F8FF48F42B3E22E; and the Intermediate CVC IIN (tag 0x42) is 4CC9C7B3A4F1C83E, matching the loaded RSA trust anchor.

PDs **SHALL** reassemble the complete operation data before processing it. For a load or replacement, the PD **SHALL** verify that the reassembled *Anchor Data* portion is exactly *Anchor Length* bytes and is a single 0x7F50 record before committing it to storage. A successful operation returns osdp_ACK. Failures return:

- osdp_NAK function-not-supported when the PD does not support trust-anchor caching.
- osdp_PIVERROR 0x0005 (invalid parameter) for malformed metadata (unknown operation, reserved bits, invalid lengths, MpSizeTotal not equal to 0x000C plus *Anchor Length*, or a reassembled Anchor Data length not equal to *Anchor Length*).
- osdp_PIVERROR 0x0007 (invalid multi-part transfer) when multi-part sequencing fails.
- osdp_PIVERROR 0x1021 (insufficient PD memory or storage) when the PD lacks storage space for the anchor.

PDs are not required to maintain a real-time clock. PDs may retain either the supplied VCI trust anchor record or an equivalent internal representation so long as the public key and supplied IIN remain available for the verification flow above.

Trust anchors may be stored in volatile or non-volatile memory at the PD's discretion. Controllers should not assume retention beyond the device's documented behaviour.

Replacing or deleting the anchor used by an active SM session **SHALL** invalidate that session, its VCI state, and PIN-verified status. The PD **SHALL** clear security-dependent cached material and trust evidence for that session. A failed replacement preserves the existing anchor and session. Changes to unrelated anchors leave the active session unchanged. Trust-store updates take effect in the next explicitly started preparation attempt.

3.15 VCI Trust Anchor Status (osdp_PIVVCITASTATUS)

The osdp_PIVVCITASTATUS command (command code 0xA6, tentative) enumerates cached trust anchors and reports remaining storage capacity. The request carries no payload. The reply uses the standard multi-part fields (MpSizeTotal, MpOffset, MpFragmentSize) so PDs can stream the inventory across one or more fragments using reply code 0x88 (tentative).

Table 40: VCI Trust Anchor Status Request (osdp_PIVVCITASTATUS)

Size (bytes)	Name	Meaning	Value
0x01	CMND	Command identifier.	0xA6 (tentative)

Table 41: VCI Trust Anchor Status Reply (osdp_PIVVCITASTATUSR)

Size (bytes)	Name	Meaning	Value
0x01	CMND	Reply identifier.	0x88 (tentative)
0x02	MpSizeTotal	Length of the complete status payload, least-significant byte first.	0x0000–0xFFFF
0x02	MpOffset	Offset of this fragment within the status payload, least-significant byte first.	0x0000–0xFFFF
0x02	MpFragmentSize	Length of the data block in this fragment, least-significant byte first.	0x0000–0xFFFF
0x01	Total Anchor Count	Number of cached trust anchors.	0x00–0xFF
0x02	Available Bytes	Remaining storage capacity for additional variable-length trust anchor records, least-significant byte first.	0x0000–0xFFFF
0x00-n	Anchor Records	Zero or more 0x0C-byte inventory records. Each record contains: TrustAnchorID (0x01 byte for load/delete management), anchor selector IIN (0x08 bytes from the stored trust anchor record, not necessarily the secure CVC IIN), AnchorLength (0x02 bytes LSB first, maximum 0x0240), and Attributes (0x01 byte, reserved). Records that span fragments continue in subsequent replies.	–

The PD first reports the total number of anchors and the remaining free bytes. *Available Bytes* is the amount of trust-anchor storage that remains for additional variable-length VCI trust anchor record data after accounting for currently stored anchors by their actual *Anchor Length* values. Each record thereafter lists a trust anchor by management ID, anchor selector IIN, anchor length, and reserved attributes (transmitted as zero). The IIN in this inventory is the IIN from the stored VCI trust anchor record and is not necessarily the secure messaging CVC IIN reported in `osdp_CARDSTATUSR`. Error responses follow the standard pattern: `osdp_NAK` function-not-supported when the command is not implemented, and `osdp_PIVERROR` `0x0007` (invalid multi-part transfer) for multi-part errors.

Table 42: Trust Anchor Record Structure

Field	Size	Meaning	`Value Range`
Trust Anchor ID	0x01 byte	Identifier assigned by the ACU; reused to replace or delete an anchor.	`0x01–0x80`
Issuer Identifier Number (IIN)	0x08 bytes	Anchor selector supplied in the VCI trust anchor record.	`0x00–0xFF`
Anchor Length	0x02 bytes (LSB first)	Total size of the stored variable-length VCI trust anchor record supplied for this anchor.	`0x0000–0x0240`
Attributes	0x01 byte	Reserved for future use; transmitted as zero.	`0x00`

A References

A.1 Normative References

- Security Industry Association, *Open Supervised Device Protocol (OSDP) v2.2*, 2020.
- Transportation Security Administration, *TWIC Specification*, *Part 2 — TWIC Card Application*, ~~December 2024~~October 2026, Sections 4.1–4.~~4~~.5, 4.6.2, 4.6.4, 4.7.1, 4.7.3, 4.7.8, 4.8, 5.2–5.~~3~~.4, 6, 7.5, and Appendices ~~C–E~~ C–G and J (application selection, authentication, identity, APDU chaining, biometric objects, E-Stickers, and TWIC Privacy Key encoding).
- Transportation Security Administration, *TWIC Specification*, *Part 3 — TWIC Reader Requirements*, ~~July 2024~~October 2026, Sections 4.4.4–4.4.8~~and 7~~, 4.6, 8–9, and Appendices D.3 and F (TWIC card authentication, biometric, and application-specific processing).
- National Institute of Standards and Technology, *NIST Special Publication 800-73-5*, *Interfaces for Personal Identity Verification (PIV) — Part 1: PIV Card Application Namespace, Data Model, and Representation*, 2024.
- National Institute of Standards and Technology, *NIST Special Publication 800-73-4*, *Interfaces for Personal Identity Verification (PIV) — Part 2: PIV Card Application Card Command Interface*, February 2015.
- National Institute of Standards and Technology, *NIST Special Publication 800-73-5*, *Interfaces for Personal Identity Verification (PIV) — Part 2: PIV Card Application Card Command Interface*, 2024.
- National Institute of Standards and Technology, *NIST Special Publication 800-78-5*, *Cryptographic Algorithms and Key Sizes for Personal Identity Verification*, 2024.
- National Institute of Standards and Technology, *Federal Information Processing Standards Publication 201-3, Personal Identity Verification (PIV) of Federal Employees and Contractors*, January 2022.
- National Institute of Standards and Technology, *NIST Special Publication 800-116 Revision 1, Guidelines for the Use of PIV Credentials in Facility Access*, May 2018, Sections 2.3 and 4.4 (signature validation and cached validation results).
- National Institute of Standards and Technology, *NIST Special Publication 800-76-2, Biometric Specifications for Personal Identity Verification*, July 2013, Sections 4 and 9 (fingerprint and signed CBEFF record encoding).
- National Institute of Standards and Technology, *NIST Special Publication 800-185*, *SHA-3 Derived Functions: cSHAKE, KMAC, TupleHash, and ParallelHash*, December 2016.
- International Organization for Standardization and International Electrotechnical Commission, *ISO/IEC 7816-4:2020*, *Identification cards — Integrated circuit cards — Part 4: Organization, security and commands for interchange*, Sections 5.2 and 11.8.1 (response-length encoding and GET RESPONSE).

- Internet Engineering Task Force, *RFC 5280, Internet X.509 Public Key Infrastructure Certificate and Certificate Revocation List (CRL) Profile*, May 2008.
- Internet Engineering Task Force, *RFC 1951, DEFLATE Compressed Data Format Specification version 1.3*; and *RFC 1952, GZIP file format specification version 4.3*, May 1996.
- Internet Engineering Task Force, *RFC 2119, Key words for use in RFCs to Indicate Requirement Levels*, March 1997; and *RFC 8174, Ambiguity of Uppercase vs Lowercase in RFC 2119 Key Words*, May 2017.

A.2 Informative References

- Transportation Security Administration, *TWIC Specification, Part 4 — TWIC uses and registration by a PACS*, ~~July 2024, Sections 2.5.1 and 2.5.2~~ October 2026, Sections 2.6–2.6.3 and 3.1 (Legacy and NextGen registration guidance and E-Sticker PACS authorizations).
- Transportation Security Administration, *TWIC Technical Advisories TA-2026-TWIC001, TA-2026-TWIC002, and TA-2026-TWIC003 V1* (EC384 migration, CA signing certificates, and issuance timeframe), 2026.
- National Institute of Standards and Technology, *NIST Internal Report 8347, Personal Identity Verification Cards, Version 2*, 2021.
- U.S. General Services Administration, *Federal Identity, Credential, and Access Management (FICAM) Architecture*, Revision 3, 2023.
- United States Coast Guard, *33 CFR 101.105, Definitions*, biometric match, `https://www.ecfr.gov/current/title-33/section-101.105`.
- United States Coast Guard, *33 CFR 101.520, Electronic TWIC inspection*, paragraphs (a)–(c), `https://www.ecfr.gov/current/title-33/section-101.520`.
- Open Physical, *High Speed Autonomous PIV: Concept Paper*, 2024.

B PIVGETDATA Worked Examples (Informative)

These examples show the enhanced request defined in Section 2.1.3, the OSDP 3.0 multi-part response defined in Section 2.1.4, and the error reply defined in Section 2.2.1.

~~The~~ In the frame exchanges below, the ACU and PD are simulated. Credential data comes from the identified existing card captures; constructed error cases are identified separately. These examples omit OSDP Secure Channel 3.0 for clarity. CRC detects transmission errors. The exchanges provide neither authentication nor confidentiality. Some source captures used credential-side secure messaging; the examples use the data recovered after that processing.

The files under `worked-examples/pivdata/` contain the complete frame exchanges, expected results, source hashes, and validation instructions. Each example ID also appears in the data file, so the printed example can be matched to its complete exchange. Long responses are shortened in this appendix only.

B.1 Reading the exchanges

Field values use hexadecimal with a `0x` prefix. Packet dumps use space-separated hexadecimal byte pairs under *Packet bytes (hex)*. Two-byte field values are shown as numbers in the request table and in least-significant-byte-first order in the packet bytes.

An accepted PIVGETDATA command receives ACK. The ACU then sends POLL until the PD returns a data fragment or PIVERROR. The first data fragment has offset zero, even when Offset in the command is nonzero. Further polls retrieve the rest of the data in order, until the ACU has received the advertised total. Polls during processing receive ACK. The examples use a maximum frame size of `0x0080` bytes, including header and CRC; this leaves `0x0072` data bytes per PIVGETDATAR fragment when no security block is present.

A whole-object read includes the object's outer tag, length, and value. An inner-tag read includes the selected element's tag, length, and value, and excludes the enclosing object's tag and length. Offset and Requested Length then determine which bytes to return. Offset zero starts at the selected tag; skipping its tag and length returns only value bytes.

The PD divides the requested data across replies. Each TLV header occurs once in the reassembled data. Each reply has its own OSDP framing, reply identifier, and multi-part fields; the ACU joins only the Card Data bytes. Certificate and biometric values retain their original encoding. Selecting the FASC-N and UUID elements individually allows the ACU to read them regardless of their positions within the CHUID.

All exchanges use PD address 0x01 and CRC-16. Each request table gives the field values; the packet bytes below it show how those values are encoded.

Complete machine-readable exchanges include every frame. The fixture manifest records the inputs, expected results, source locations, and hashes. Hex includes the OSDP header and CRC, without an optional leading mark byte. Long replies show their first 0x18 and last 0x10 bytes; ellipsis text is not transmitted.

B.2 Source fixtures

S1: NIST SD33 card02, contact card capture.

S2: NIST SD33 card01, contact card capture.

The 19 examples below are a representative subset of the 37 checked exchanges in the linked JSON file.

B.3 Complete Discovery Object

Read the complete Discovery Object. Tag Length 0x00 keeps its outer tag 0x7E and its length, followed by all its contents.

Example ID: `discovery-whole`. Source: S1.

Table 43: Request fields

Field	Value	Notes
Command	0xA3	osdp_PIVGETDATA
Identifier Length	0x01	Bytes in Object Identifier.
Object Identifier	0x7E	Object to read.
Tag Length	0x00	Select the whole object.
Element Tag	Omitted	Tag Length is zero.
Offset	0x0000	Start at the selected tag.
Requested Length	0x0000	All remaining bytes.

Result: 0x0014 returned bytes in 0x01 data reply.

Table 44: Command and reply exchange

Message	Packet bytes (hex)	Notes
ACU PIVGETDATA	53 01 0F 00 05 A3 01 7E 00 00 00 00 00 4A 01	Read request.
PD ACK	53 81 08 00 05 40 39 35	Command accepted.
ACU POLL	53 01 08 00 06 60 D8 66	Poll 0x01.
PD PIVGETDATAR	53 81 22 00 06 80 14 00 00 00 14 00 7E 12 4F 0B A0 00 00 03 08 00 00 10 00 01 00 5F 2F 02 68 20 DB F9	Total 0x0014 Offset 0x0000 Fragment 0x0014

B.4 Discovery PIN Usage Policy (0x5F2F)

Read the PIN Usage Policy from the Discovery Object. The reply contains tag 0x5F2F, length 0x02, and the two policy bytes.

Example ID: `discovery-pin-policy`. Source: S1.

Table 45: Request fields

Field	Value	Notes
Command	0xA3	osdp_PIVGETDATA
Identifier Length	0x01	Bytes in Object Identifier.
Object Identifier	0x7E	Object to read.
Tag Length	0x02	Bytes in Element Tag.
Element Tag	0x5F2F	Element to select.
Offset	0x0000	Start at the selected tag.
Requested Length	0x0000	All remaining bytes.

Result: 0x0005 returned bytes in 0x01 data reply.

Table 46: Command and reply exchange

Message	Packet bytes (hex)	Notes
ACU PIVGETDATA	53 01 11 00 05 A3 01 7E 02 5F 2F 00 00 00 00 03 F0	Read request.
PD ACK	53 81 08 00 05 40 39 35	Command accepted.
ACU POLL	53 01 08 00 06 60 D8 66	Poll 0x01.
PD PIVGETDATAR	53 81 13 00 06 80 05 00 00 00 05 00 5F 2F 02 68 20 31 67	Total 0x0005 Offset 0x0000 Fragment 0x0005

B.5 Read the FASC-N element (0x30)

Read the FASC-N element from the CHUID without depending on its position. The returned element begins with tag 0x30 and length 0x19, followed by the FASC-N.

Example ID: `chuid-fascn`. Source: S1.

Table 47: Request fields

Field	Value	Notes
Command	0xA3	osdp_PIVGETDATA
Identifier Length	0x03	Bytes in Object Identifier.
Object Identifier	0x5FC102	Object to read.
Tag Length	0x01	Bytes in Element Tag.
Element Tag	0x30	Element to select.
Offset	0x0000	Start at the selected tag.
Requested Length	0x0000	All remaining bytes.

Result: 0x001B returned bytes in 0x01 data reply.

Table 48: Command and reply exchange

Message	Packet bytes (hex)	Notes
ACU PIVGETDATA	53 01 12 00 05 A3 03 5F C1 02 01 30 00 00 00 00 15 89	Read request.
PD ACK	53 81 08 00 05 40 39 35	Command accepted.
ACU POLL	53 01 08 00 06 60 D8 66	Poll 0x01.
PD PIVGETDATAR	53 81 29 00 06 80 1B 00 00 00 1B 00 30 19 D6 50 18 58 55 E5 6D C8 12 79 45 A1 6C DB 90 6E 78 80 C0 82 86 50 18 43 F5 E1 F1	Total 0x001B Offset 0x0000 Fragment 0x001B

B.6 Read the FASC-N value

Read just the FASC-N bytes. Offset 0x0002 skips the selected element's 0x30 0x19 header; Requested Length 0x0019 limits the result to the FASC-N value.

Example ID: `chuid-fascn-value`. Source: S1.

Table 49: Request fields

Field	Value	Notes
Command	0xA3	osdp_PIVGETDATA
Identifier Length	0x03	Bytes in Object Identifier.
Object Identifier	0x5FC102	Object to read.
Tag Length	0x01	Bytes in Element Tag.
Element Tag	0x30	Element to select.
Offset	0x0002	Bytes to skip from the selected tag.
Requested Length	0x0019	Maximum bytes to return.

Result: 0x0019 returned bytes in 0x01 data reply.

Table 50: Command and reply exchange

Message	Packet bytes (hex)	Notes
ACU PIVGETDATA	53 01 12 00 05 A3 03 5F C1 02 01 30 02 00 19 00 96 DD	Read request.
PD ACK	53 81 08 00 05 40 39 35	Command accepted.
ACU POLL	53 01 08 00 06 60 D8 66	Poll 0x01.
PD PIVGETDATAR	53 81 27 00 06 80 19 00 00 00 19 00 D6 50 18 58 55 E5 6D C8 12 79 45 A1 6C DB 90 6E 78 80 C0 82 86 50 18 43 F5 6C 73	Total 0x0019 Offset 0x0000 Fragment 0x0019

B.7 Read the UUID element (0x34)

Read the UUID (also called the GUID) element from the CHUID. The returned element begins with tag 0x34 and length 0x10, followed by the UUID.

Example ID: `chuid-guid`. Source: S1.

Table 51: Request fields

Field	Value	Notes
Command	0xA3	osdp_PIVGETDATA
Identifier Length	0x03	Bytes in Object Identifier.
Object Identifier	0x5FC102	Object to read.
Tag Length	0x01	Bytes in Element Tag.
Element Tag	0x34	Element to select.
Offset	0x0000	Start at the selected tag.
Requested Length	0x0000	All remaining bytes.

Result: 0x0012 returned bytes in 0x01 data reply.

Table 52: Command and reply exchange

Message	Packet bytes (hex)	Notes
ACU PIVGETDATA	53 01 12 00 05 A3 03 5F C1 02 01 34 00 00 00 00 13 00	Read request.
PD ACK	53 81 08 00 05 40 39 35	Command accepted.
ACU POLL	53 01 08 00 06 60 D8 66	Poll 0x01.
PD PIVGETDATAR	53 81 20 00 06 80 12 00 00 00 12 00 34 10 45 85 34 70 A4 03 48 44 AE 15 C6 49 59 59 FA D5 FA C7	Total 0x0012 Offset 0x0000 Fragment 0x0012

B.8 Read the UUID value

Read just the UUID bytes. Offset 0x0002 skips the selected element's 0x34 0x10 header; Requested Length 0x0010 limits the result to the UUID value.

Example ID: `chuid-guid-value`. Source: S1.

Table 53: Request fields

Field	Value	Notes
Command	0xA3	osdp_PIVGETDATA
Identifier Length	0x03	Bytes in Object Identifier.
Object Identifier	0x5FC102	Object to read.
Tag Length	0x01	Bytes in Element Tag.
Element Tag	0x34	Element to select.
Offset	0x0002	Bytes to skip from the selected tag.
Requested Length	0x0010	Maximum bytes to return.

Result: 0x0010 returned bytes in 0x01 data reply.

Table 54: Command and reply exchange

Message	Packet bytes (hex)	Notes
ACU PIVGETDATA	53 01 12 00 05 A3 03 5F C1 02 01 34 02 00 10 00 08 EE	Read request.
PD ACK	53 81 08 00 05 40 39 35	Command accepted.
ACU POLL	53 01 08 00 06 60 D8 66	Poll 0x01.
PD PIVGETDATAR	53 81 1E 00 06 80 10 00 00 00 10 00 45 85 34 70 A4 03 48 44 AE 15 C6 49 59 59 FA D5 75 CB	Total 0x0010 Offset 0x0000 Fragment 0x0010

B.9 Bounded CHUID prefix

Read the first 0x0010 bytes of the CHUID, starting with its outer tag and length. In this capture, the range ends partway through the FASC-N element.

Example ID: `chuid-prefix`. Source: S1.

Table 55: Request fields

Field	Value	Notes
Command	0xA3	osdp_PIVGETDATA
Identifier Length	0x03	Bytes in Object Identifier.
Object Identifier	0x5FC102	Object to read.
Tag Length	0x00	Select the whole object.
Element Tag	Omitted	Tag Length is zero.
Offset	0x0000	Start at the selected tag.
Requested Length	0x0010	Maximum bytes to return.

Result: 0x0010 returned bytes in 0x01 data reply.

Table 56: Command and reply exchange

Message	Packet bytes (hex)	Notes
ACU PIVGETDATA	53 01 11 00 05 A3 03 5F C1 02 00 00 00 10 00 A1 F4	Read request.
PD ACK	53 81 08 00 05 40 39 35	Command accepted.
ACU POLL	53 01 08 00 06 60 D8 66	Poll 0x01.
PD PIVGETDATAR	53 81 1E 00 06 80 10 00 00 00 10 00 53 82 07 C8 EE 02 07 C4 30 19 D6 50 18 58 55 E5 5A 2E	Total 0x0010 Offset 0x0000 Fragment 0x0010

B.10 Complete certificate object

Read the complete certificate object, including its outer 0x53 header and all elements inside it. The result is larger than one reply, so the ACU polls for the remaining fragments.

Example ID: `certificate-whole`. Source: S1.

Table 57: Request fields

Field	Value	Notes
Command	0xA3	osdp_PIVGETDATA
Identifier Length	0x03	Bytes in Object Identifier.
Object Identifier	0x5FC105	Object to read.
Tag Length	0x00	Select the whole object.
Element Tag	Omitted	Tag Length is zero.
Offset	0x0000	Start at the selected tag.
Requested Length	0x0000	All remaining bytes.

Result: 0x066F returned bytes in 0x0F data replies.

Table 58: Command and reply exchange

Message	Packet bytes (hex)	Notes
ACU PIVGETDATA	53 01 11 00 05 A3 03 5F C1 05 00 00 00 00 00 93 3F	Read request.
PD ACK	53 81 08 00 05 40 39 35	Command accepted.
ACU POLL	53 01 08 00 06 60 D8 66	Poll 0x01.
PD PIVGETDATAR	53 81 80 00 06 80 6F 06 00 00 72 00 53 82 06 6B 70 82 06 62 30 82 06 5E ... 0x58 bytes omitted ... 65 73 74 20 43 41 31 33 30 31 06 03 55 04 4F 7E	Total 0x066F Offset 0x0000 Fragment 0x0072
POLL/reply	Omitted	0x0D exchanges; complete frames are in the JSON file.
ACU POLL	53 01 08 00 05 60 8B 33	Poll 0x0F.

Message	Packet bytes (hex)	Notes
PD PIVGETDATAR	53 81 41 00 05 80 6F 06 3C 06 33 00 49 D9 67 65 56 F7 94 17 43 D7 84 02 9F 61 BF D6 D3 99 B6 DC D1 DE 0D 87 F5 F8 1A A9 CD 4D B8 BB D0 E6 6E 4B F5 75 87 9D AB F1 EF 91 D2 1A 71 01 00 FE 00 F7 26	Total 0x066F Offset 0x063C Fragment 0x0033

B.11 Read the certificate element (0x70)

Read element 0x70, including its tag, encoded length, and certificate bytes. This omits the enclosing 0x53 header and the other elements in the object. The 0x70 header appears once in the returned data.

Example ID: `certificate-element`. Source: S1.

Table 59: Request fields

Field	Value	Notes
Command	0xA3	osdp_PIVGETDATA
Identifier Length	0x03	Bytes in Object Identifier.
Object Identifier	0x5FC105	Object to read.
Tag Length	0x01	Bytes in Element Tag.
Element Tag	0x70	Element to select.
Offset	0x0000	Start at the selected tag.
Requested Length	0x0000	All remaining bytes.

Result: 0x0666 returned bytes in 0x0F data replies.

Table 60: Command and reply exchange

Message	Packet bytes (hex)	Notes
ACU PIVGETDATA	53 01 12 00 05 A3 03 5F C1 05 01 70 00 00 00 00 39 81	Read request.
PD ACK	53 81 08 00 05 40 39 35	Command accepted.
ACU POLL	53 01 08 00 06 60 D8 66	Poll 0x01.
PD PIVGETDATAR	53 81 80 00 06 80 66 06 00 00 72 00 70 82 06 62 30 82 06 5E 30 82 04 46 ... 0x58 bytes omitted ... 43 41 31 33 30 31 06 03 55 04 03 13 2A 54 2D 25	Total 0x0666 Offset 0x0000 Fragment 0x0072

Message	Packet bytes (hex)	Notes
POLL/reply	Omitted	0x0D exchanges; complete frames are in the JSON file.
ACU POLL	53 01 08 00 05 60 8B 33	Poll 0x0F.
PD PIVGETDATAR	53 81 38 00 05 80 66 06 3C 06 2A 00 56 F7 94 17 43 D7 84 02 9F 61 BF D6 D3 99 B6 DC D1 DE 0D 87 F5 F8 1A A9 CD 4D B8 BB D0 E6 6E 4B F5 75 87 9D AB F1 EF 91 D2 1A 8B 7C	Total 0x0666 Offset 0x063C Fragment 0x002A

B.12 Read a compressed certificate element

Read element 0x70 from a card that stores a compressed certificate. The PD returns the original tag, length, and compressed bytes. The ACU can read the object's certificate-information element to determine how to interpret the value.

Example ID: `certificate-compressed-element`. Source: S2.

Table 61: Request fields

Field	Value	Notes
Command	0xA3	osdp_PIVGETDATA
Identifier Length	0x03	Bytes in Object Identifier.
Object Identifier	0x5FC105	Object to read.
Tag Length	0x01	Bytes in Element Tag.
Element Tag	0x70	Element to select.
Offset	0x0000	Start at the selected tag.
Requested Length	0x0000	All remaining bytes.

Result: 0x05EA returned bytes in 0x0E data replies.

Table 62: Command and reply exchange

Message	Packet bytes (hex)	Notes
ACU PIVGETDATA	53 01 12 00 05 A3 03 5F C1 05 01 70 00 00 00 00 39 81	Read request.
PD ACK	53 81 08 00 05 40 39 35	Command accepted.
ACU POLL	53 01 08 00 06 60 D8 66	Poll 0x01.
PD PIVGETDATAR	53 81 80 00 06 80 EA 05 00 00 72 00 70 82 05 E6 1F 8B 08 00 00 00 00 00 ... 0x58 bytes omitted ... CE F4 47 FF F4 CF 5D BE EF 9C 7B CF 39 F7 79 A2	Total 0x05EA Offset 0x0000 Fragment 0x0072
POLL/reply	Omitted	0x0C exchanges; complete frames are in the JSON file.
ACU POLL	53 01 08 00 07 60 E9 55	Poll 0x0E.
PD PIVGETDATAR	53 81 2E 00 07 80 EA 05 CA 05 20 00 7F C4 30 BC D9 5C B8 05 0A 09 18 39 F8 6D A5 80 16 10 4D DC 8D 38 FC 05 BF 05 DC 1B AF 07 00 00 96 99	Total 0x05EA Offset 0x05CA Fragment 0x0020

B.13 Read a range within the certificate element

Read 0x0104 bytes starting at offset 0x0100 of the certificate element, counting from tag 0x70. The reply offsets are 0x0000, 0x0072, and 0x00E4; each identifies a position within the 0x0104 returned bytes.

Example ID: `certificate-offset256`. Source: S1.

Table 63: Request fields

Field	Value	Notes
Command	0xA3	osdp_PIVGETDATA
Identifier Length	0x03	Bytes in Object Identifier.
Object Identifier	0x5FC105	Object to read.
Tag Length	0x01	Bytes in Element Tag.
Element Tag	0x70	Element to select.
Offset	0x0100	Bytes to skip from the selected tag.
Requested Length	0x0104	Maximum bytes to return.

Result: 0x0104 returned bytes in 0x03 data replies.

Table 64: Command and reply exchange

Message	Packet bytes (hex)	Notes
ACU PIVGETDATA	53 01 12 00 05 A3 03 5F C1 05 01 70 00 01 04 01 EC 6A	Read request.
PD ACK	53 81 08 00 05 40 39 35	Command accepted.
ACU POLL	53 01 08 00 06 60 D8 66	Poll 0x01.
PD PIVGETDATAR	53 81 80 00 06 80 04 01 00 00 72 00 14 30 12 06 03 55 04 0B 13 0B 54 65 ... 0x58 bytes omitted ... 00 03 82 01 0F 00 30 82 01 0A 02 82 01 01 CE C4	Total 0x0104 Offset 0x0000 Fragment 0x0072
ACU POLL	53 01 08 00 07 60 E9 55	Poll 0x02.
PD PIVGETDATAR	53 81 80 00 07 80 04 01 72 00 72 00 00 C3 C3 53 BF 0F 03 48 37 48 A4 BA ... 0x58 bytes omitted ... 4A A1 F7 74 E7 A2 DE BC F3 87 AA 94 76 21 24 7E	Total 0x0104 Offset 0x0072 Fragment 0x0072
ACU POLL	53 01 08 00 05 60 8B 33	Poll 0x03.

Message	Packet bytes (hex)	Notes
PD PIVGETDATAR	53 81 2E 00 05 80 04 01 E4 00 20 00 FA 54 AA 2C DF 26 69 45 7E 6D CD 3A 48 EE 5E 4B 31 FD A9 4A D8 73 01 A7 5C C8 50 D2 0A 60 D1 CC 14 CD	Total 0x0104 Offset 0x00E4 Fragment 0x0020

B.14 Complete Key History object

Read the complete Key History object. Its outer 0x53 header and all elements inside it are returned unchanged.

Example ID: `key-history`. Source: S1.

Table 65: Request fields

Field	Value	Notes
Command	0xA3	osdp_PIVGETDATA
Identifier Length	0x03	Bytes in Object Identifier.
Object Identifier	0x5FC10C	Object to read.
Tag Length	0x00	Select the whole object.
Element Tag	Omitted	Tag Length is zero.
Offset	0x0000	Start at the selected tag.
Requested Length	0x0000	All remaining bytes.

Result: 0x000A returned bytes in 0x01 data reply.

Table 66: Command and reply exchange

Message	Packet bytes (hex)	Notes
ACU PIVGETDATA	53 01 11 00 05 A3 03 5F C1 0C 00 00 00 00 00 71 77	Read request.
PD ACK	53 81 08 00 05 40 39 35	Command accepted.
ACU POLL	53 01 08 00 06 60 D8 66	Poll 0x01.
PD PIVGETDATAR	53 81 18 00 06 80 0A 00 00 00 0A 00 53 08 C1 01 04 C2 01 00 FE 00 F3 05	Total 0x000A Offset 0x0000 Fragment 0x000A

B.15 BIT Group with no biometric templates

Read a Biometric Information Template (BIT) Group with no biometric templates. The reply keeps the outer 0x7F61 tag and the element that reports a count of zero.

Example ID: `bit-empty`. Source: constructed example.

Table 67: Request fields

Field	Value	Notes
Command	0xA3	osdp_PIVGETDATA
Identifier Length	0x02	Bytes in Object Identifier.
Object Identifier	0x7F61	Object to read.
Tag Length	0x00	Select the whole object.
Element Tag	Omitted	Tag Length is zero.
Offset	0x0000	Start at the selected tag.
Requested Length	0x0000	All remaining bytes.

Result: 0x0006 returned bytes in 0x01 data reply.

Table 68: Command and reply exchange

Message	Packet bytes (hex)	Notes
ACU PIVGETDATA	53 01 10 00 05 A3 02 7F 61 00 00 00 00 00 3E 14	Read request.
PD ACK	53 81 08 00 05 40 39 35	Command accepted.
ACU POLL	53 01 08 00 06 60 D8 66	Poll 0x01.
PD PIVGETDATAR	53 81 14 00 06 80 06 00 00 00 06 00 7F 61 03 02 01 00 35 7C	Total 0x0006 Offset 0x0000 Fragment 0x0006

B.16 Empty 0x53 container

Read an object with no contents. The reply still contains two bytes, 0x53 0x00, because whole-object selection includes the outer tag and length.

Example ID: `empty-5300`. Source: constructed example.

Table 69: Request fields

Field	Value	Notes
Command	0xA3	osdp_PIVGETDATA
Identifier Length	0x03	Bytes in Object Identifier.
Object Identifier	0x5FC105	Object to read.
Tag Length	0x00	Select the whole object.
Element Tag	Omitted	Tag Length is zero.
Offset	0x0000	Start at the selected tag.
Requested Length	0x0000	All remaining bytes.

Result: 0x0002 returned bytes in 0x01 data reply.

Table 70: Command and reply exchange

Message	Packet bytes (hex)	Notes
ACU PIVGETDATA	53 01 11 00 05 A3 03 5F C1 05 00 00 00 00 00 93 3F	Read request.
PD ACK	53 81 08 00 05 40 39 35	Command accepted.
ACU POLL	53 01 08 00 06 60 D8 66	Poll 0x01.
PD PIVGETDATAR	53 81 10 00 06 80 02 00 00 00 02 00 53 00 5D 38	Total 0x0002 Offset 0x0000 Fragment 0x0002

B.17 Zero-byte successful response

Handle a successful card response with zero data bytes. The PD sends one PIVGETDATAR with all three multi-part fields zero.

Example ID: `empty-success`. Source: constructed example.

Table 71: Request fields

Field	Value	Notes
Command	0xA3	osdp_PIVGETDATA
Identifier Length	0x01	Bytes in Object Identifier.
Object Identifier	0x7E	Object to read.
Tag Length	0x00	Select the whole object.
Element Tag	Omitted	Tag Length is zero.
Offset	0x0000	Start at the selected tag.
Requested Length	0x0000	All remaining bytes.

Result: 0x0000 returned bytes in 0x01 data reply.

Table 72: Command and reply exchange

Message	Packet bytes (hex)	Notes
ACU PIVGETDATA	53 01 0F 00 05 A3 01 7E 00 00 00 00 00 4A 01	Read request.
PD ACK	53 81 08 00 05 40 39 35	Command accepted.
ACU POLL	53 01 08 00 06 60 D8 66	Poll 0x01.
PD PIVGETDATAR	53 81 0E 00 06 80 00 00 00 00 00 00 C0 05	Total 0x0000 Offset 0x0000 Fragment 0x0000

B.18 Existing child with an empty value

Read an element whose value is empty. The reply contains its tag and zero length, 0x30 0x00, for a total of two bytes.

Example ID: `zero-length-child`. Source: constructed example.

Table 73: Request fields

Field	Value	Notes
Command	0xA3	osdp_PIVGETDATA
Identifier Length	0x01	Bytes in Object Identifier.
Object Identifier	0x7E	Object to read.
Tag Length	0x01	Bytes in Element Tag.
Element Tag	0x30	Element to select.
Offset	0x0000	Start at the selected tag.
Requested Length	0x0000	All remaining bytes.

Result: 0x0002 returned bytes in 0x01 data reply.

Table 74: Command and reply exchange

Message	Packet bytes (hex)	Notes
ACU PIVGETDATA	53 01 10 00 05 A3 01 7E 01 30 00 00 00 00 DC 37	Read request.
PD ACK	53 81 08 00 05 40 39 35	Command accepted.
ACU POLL	53 01 08 00 06 60 D8 66	Poll 0x01.
PD PIVGETDATAR	53 81 10 00 06 80 02 00 00 00 02 00 30 00 24 66	Total 0x0002 Offset 0x0000 Fragment 0x0002

B.19 Missing direct child

Request an element that is absent from the object. The PD accepts the command, then reports PIVERROR on the next poll.

Example ID: `missing-child`. Source: constructed example.

Table 75: Request fields

Field	Value	Notes
Command	0xA3	osdp_PIVGETDATA
Identifier Length	0x01	Bytes in Object Identifier.
Object Identifier	0x7E	Object to read.
Tag Length	0x01	Bytes in Element Tag.
Element Tag	0x34	Element to select.
Offset	0x0000	Start at the selected tag.
Requested Length	0x0000	All remaining bytes.

Result: osdp_PIVERROR 0x1024 (data object or tag not found); see Section 2.2.1.

Table 76: Command and reply exchange

Message	Packet bytes (hex)	Notes
ACU PIVGETDATA	53 01 10 00 05 A3 01 7E 01 34 00 00 00 00 DA BE	Read request.
PD ACK	53 81 08 00 05 40 39 35	Command accepted.
ACU POLL	53 01 08 00 06 60 D8 66	Poll 0x01.
PD PIVERROR	53 81 0A 00 06 8A 24 10 A3 74	0x1024 (data object or tag not found)

B.20 Element request against an empty response

Request an element when the card returns no data. The PD reports that the tag was not found. A whole-object request for the same response would succeed with zero bytes.

Example ID: `empty-missing-child`. Source: constructed example.

Table 77: Request fields

Field	Value	Notes
Command	0xA3	osdp_PIVGETDATA
Identifier Length	0x01	Bytes in Object Identifier.
Object Identifier	0x7E	Object to read.
Tag Length	0x01	Bytes in Element Tag.
Element Tag	0x34	Element to select.
Offset	0x0000	Start at the selected tag.
Requested Length	0x0000	All remaining bytes.

Result: osdp_PIVERROR 0x1024 (data object or tag not found); see Section 2.2.1.

Table 78: Command and reply exchange

Message	Packet bytes (hex)	Notes
ACU PIVGETDATA	53 01 10 00 05 A3 01 7E 01 34 00 00 00 00 DA BE	Read request.
PD ACK	53 81 08 00 05 40 39 35	Command accepted.
ACU POLL	53 01 08 00 06 60 D8 66	Poll 0x01.
PD PIVERROR	53 81 0A 00 06 8A 24 10 A3 74	0x1024 (data object or tag not found)

B.21 Ambiguous direct-child selection

Request a tag that occurs twice directly inside the outer container. Element selection requires a unique match, so the PD reports an error.

Example ID: `duplicate-child`. Source: constructed example.

Table 79: Request fields

Field	Value	Notes
Command	0xA3	osdp_PIVGETDATA
Identifier Length	0x01	Bytes in Object Identifier.
Object Identifier	0x7E	Object to read.
Tag Length	0x01	Bytes in Element Tag.
Element Tag	0x30	Element to select.
Offset	0x0000	Start at the selected tag.
Requested Length	0x0000	All remaining bytes.

Result: osdp_PIVERROR 0x1027 (ambiguous element selection); see Section 2.2.1.

Table 80: Command and reply exchange

Message	Packet bytes (hex)	Notes
ACU PIVGETDATA	53 01 10 00 05 A3 01 7E 01 30 00 00 00 00 DC 37	Read request.
PD ACK	53 81 08 00 05 40 39 35	Command accepted.
ACU POLL	53 01 08 00 06 60 D8 66	Poll 0x01.
PD PIVERROR	53 81 0A 00 06 8A 27 10 F0 21	0x1027 (ambiguous element selection)

B.22 TWIC NEXGEN E-Sticker: captured write, read, and erase

This card-command trace was captured on 2026-11-11 from a NEXGEN V8 TWIC using an Identiv SCR3500 C contact reader and PC/SC. The selected TWIC AID is A000000367200000010103. The APDUs and responses are recorded in `worked-examples/twic-esticker/trace.json`. Each response below includes SW1 and SW2. The initial inspection returned E1006282, an uninitialized E-Sticker. The verified sequence below starts with the empty encoding E1028000 established by the first erase.

1. **Select the TWIC application.** Command: 00A4040009A000000367200000001FF.

   Response: 612A4F0BA000000367200000010103790 74F05A000000367501 24E455847454E2052454C4541534520312E307F6608020203F8020280019000.

2. **Verify E1 is empty.** Command: 00CB3FFF035C01E1FF.

   Response: E10280006282.

3. **Write the E-Sticker.** Command: 00DB3FFF07E1058103414243.

   Response: 9000. The native E1 object contains child tag 0x81 with value ABC.

4. **Read back the exact object.** Command: 00CB3FFF035C01E1FF.

   Response: E10581034142436282.

5. **Erase the E-Sticker.** Command: 00DB3FFF04E1028000.

   Response: 9000.

6. **Verify E1 is empty again.** Command: 00CB3FFF035C01E1FF.

   Response: E10280006282.

TWIC Part 2, Sections 5.2 and 5.4 and Appendices E–F, define the transfers, short-read status, and empty E-Sticker encoding. Here Le is 0xFF; the complete objects contain fewer than 255 bytes, and each read returns 0x6282. Under the response-length rules in paragraph 2.1.2, PIVGETDATA A301E10000000000 returns the captured object bytes in PIVGETDATAR: E1058103414243 after writing and E1028000 after erasing. The OSDP mapping uses TWIC Data Model 0x02 and the selected TWIC application, as in the E-Sticker example in the PIVPUTDATA subsection.

C High-Speed Autonomous PIV Data Templates (Informative)

This appendix illustrates the per-entry authentication key selector, per-card certificate discovery, and explicit resumption of PIV Auto. The configuration, discovery, resume, and TWIC sequences added in this revision are illustrative; they have not been executed or used to regenerate cryptographic vectors, and TWIC authentication. The normative rules are in Subsections 3.12, 3.12, and 3.12.

The existing APDU fixtures and generated signing examples are retained separately. Their use a prototype KDK input uses the SC1-style consisting of the 48-byte prototype session-key setand does not establish Secure Channel 2.0 interoperability. SC2 remains a required upstream dependency; its transport and final PIV Auto binding are outside these examples. .

C.1 Vector Sources

Table 81: PIV Auto Vector Sources

Source	Use	Notes
NIST SD 33 / IR 8347	Live-card APDU fixtures	Cards 2, 3, 4, 5, and 16 provide VCI contactless enrolment, OPACITY, and key-challenge transcripts.
NIST SM/VCI captures	CVC and trust anchors	Provides validator-ready 7F21, 5FC122, and 7F50 objects.
GSA FICAM card-builder	Local keys and certificates	Used only as reusable local test material for source-backed simulation.
Generated YubiKey PIV	Missing algorithm profiles	Reusable RSA-1024, RSA-3072, and P-384 test keys are committed as public fixtures and loaded through `ykmanpiv`; RSA-1024 is informative legacy test material only.

C.2 Key Selection and Certificate Discovery

An Auto-enabled Configuration Entry ends with one Authentication Key Reference byte, after the AID bytes. There is no configured algorithm byte. With Auto disabled, the key-reference byte is absent and the previous entry encoding is unchanged. For example, the default PIV application with contact and contactless permitted is 01040103009E: tag `0x01`, length `0x04`, Data Model `0x01`, Interface Mask `0x03`, AID Length `0x00`, and key `0x9E`. Changing only the last byte to `0x9A` selects PIV Authentication without fixing the card's key type.

The logical configuration begins with three header bytes, two four-byte counters, and 32 entropy bytes. The shortest Auto-enabled configuration is therefore `0x31` bytes (49 decimal), including its one six-byte Configuration Entry. Configuration Count counts Configuration Entry TLVs only.

Table 82: Illustrative PIV Auto Configuration and Resume Bytes

Message	Field	Bytes
Card Authentication	Flags, Count, Global Mask, Series, Sequence	010103070000002A000000
Card Authentication	Supplemental Entropy	00000000000000000000000000000000000000000000000000000000000000000000000000000000
Card Authentication	Configuration Entry	01040103009E
Contactless PIV Authentication	Flags, Count, Global Mask, Series, Sequence	030102070000002A000000
Contactless PIV Authentication	Supplemental Entropy	00000000000000000000000000000000000000000000000000000000000000000000000000000000
Contactless PIV Authentication	Configuration Entry	01040102009A
Resume after pairing and PIN	Flags, Count, Global Mask, Series, Sequence	030102070000002B000000
Resume after pairing and PIN	Supplemental Entropy	00000000000000000000000000000000000000000000000000000000000000000000000000000000
Resume after pairing and PIN	Unchanged Configuration Entry	01040102009A
Resume after pairing and PIN	Final Resume TLV	0208070000002A000000

Concatenate the rows belonging to one message. The two initial logical payloads are 0x31 bytes; the resume payload is 0x3B bytes. For a single fragment, prefix the initial payload with AE310000003100, or the resume payload with AE3B0000003B00. These prefixes contain CMND, MpSizeTotal, MpOffset, and MpFragmentSize. ~~OSDP packet headers and SC2 protectionare not shown. All-zero entropy is permitted but not recommended for deployment~~These are logical payloads before OSDP framing and Secure Channel 3.0 protection. Deployment entropy follows the requirements in Subsection 3.12.

Table 83: Illustrative Authentication Certificate Reads

Configured Key	Logical Card APDU	Certificate Object
PIV 0x9A	00CB3FFF055C035FC105FF	0x5FC105; contactless requires VCI.
PIV 0x9E	00CB3FFF055C035FC101FF	0x5FC101; no PIV PIN required.
NextGen TWIC 0x9E	00CB3FFF055C035FC101FF	0x5FC101 in the selected TWIC application.

These are logical APDUs; an established credential SM session protects applicable PIV commands. The PD reassembles the returned certificate object, extracts tag 0x70, and interprets its one-byte CertInfo in tag 0x71: 0x00 identifies uncompressed DER and 0x01 identifies GZIP-compressed DER. It decompresses when indicated before parsing SubjectPublicKeyInfo. Missing or malformed certificate evidence ends the attempt with Result 0x03; a recognized but unsupported subject-key profile uses Result 0x04. Before an algorithm is determined, PIVSTATUS reports Algorithm Identifier 0x00; that failure value never selects a default card algorithm.

Table 84: Mixed PIV Subject Keys under the Same Configuration

Subject Public Key	Configured Key	Discovered Algorithm	Challenge Profile
RSA-2048	0x9A or 0x9E	0x07	~~32-byte~~ 0x20-byte digest; SHA-256 DigestInfo.
RSA-3072	0x9A or 0x9E	0x05	~~48-byte~~ 0x30-byte digest; SHA-384 DigestInfo.
Named curve P-256	0x9A or 0x9E	0x11	~~32-byte~~ 0x20-byte prehashed ECDSA input.
Named curve P-384	0x9A or 0x9E	0x14	~~48-byte~~ 0x30-byte prehashed ECDSA input.

The four subject-key types at each of the two references give eight PIV profiles. Within either reference, successive RSA and ECC cards use identical configuration bytes. The certificate changes the GENERAL AUTHENTICATE P1 byte, challenge encoding, and reported algorithm. For example, an RSA CA can sign a P-384 subject certificate using SHA-256: the PD still discovers 0x14 and uses the 48-byte P-384 profile. The issuer's signature algorithm does not select the card's authentication algorithm. No alternate key is tried if discovery fails.

C.3 Illustrative PIV Authentication Resume

In the contactless 0x9A example above, attempt 0x00000007/ 0x0000002A ends with Result 0x05, Detail 0x01 because pairing is required. The ACU receives the complete result, then sends osdp_PIVXMITPAIRING, for example AD3132333435363738, and waits for confirmed VCI establishment. It requests commanded PIN acquisition and verification with AA010000000000: Mode 0x01, default finite

timeouts, Auto Execute clear, and Cache Validity zero. The ACU waits for the successful PIVSPER before sending the resume PIVMODE payload above.

The final 0208070000002A000000 identifies the completed prerequisite failure, not the counters of the new attempt. The header retains Series 0x00000007 and supplies its exact next unused Sequence 0x0000002B. Configuration Count remains one. The entropy may change; the flags, masks, ordered entries, AIDs, and key references remain unchanged. For this eligible resume, completed pairing and PIN verification remain bound to the same presentation, application, interface, and security contexts.

The PD reads certificate 0x5FC105, discovers its subject-key profile, and performs one new key proof. Pairing or PIN completion alone does not restart Auto. Credential removal, a context change, or an interrupted security session invalidates resumption; an ordinary mode replacement still clears state. Duplicate OSDP delivery does not repeat pairing, PIN verification, or the proof.

C.4 Illustrative TWIC ~~Auto Key Proof~~Active Card Authentication

~~The~~ After confirming Function Code 12 flags 0x04, 0x08, and 0x10, the ACU enables Auto flag 0x01 and supplies the TWIC Configuration Entry 010D020209A000000367200000019E~~selects~~ : Data Model 0x02, contactless interface, the nine-byte TWIC AID prefix, and ~~asymmetric~~ Card Authentication key 0x9E. ~~It does not configure an algorithm or a TPK.~~ The configuration's Series, Sequence, and Supplemental Entropy fields supply the attempt inputs. All its Configuration Entries select TWIC ACA. The PD issues ~~and examines the full AID returned in~~ 00A4040009A000000367200000001FF and examines tag 0x4F:

- **Legacy:** A000000367200000010101. ~~Read the TWIC CHUID with , retaining its identity only for this attempt. Then select the~~ Select the PIV application with ~~and retrieve certificate~~ 00A404000BA 000000308000010000100FF, then retrieve Card Authentication Certificate 0x5FC101 ~~. Use that application's key while retaining the configured TWIC acceptance policy. A present, correctly sized nil GUID remains zero in the challenge and PIVSTATUS only when the valid nonzero FASC-N matches the Card Authentication certificate. Missing identity data does not receive the nil-GUID exception~~using the PIV application.

- ~~NextGen~~**NexGen V8 or V10:** A000000367200000010103 or A000000367200000010104, respectively. Remain in the TWIC application ~~, read CHUID and~~ and read certificate 0x5FC101 ~~using Le , and retain the populated GUID. Concatenate the initial data and subsequent GET RESPONSE data~~ with 00CB3FFF 055C035FC101FF. Reassemble the returned data using GET RESPONSE when status 0x61XX indicates more bytes. ~~Le can return with no data and is not a successful certificate read.~~

These application routes follow TWIC Part 3, Appendix D.3, and Part 2, Section 4.1 and Appendices C and J.

The ~~documented NextGen profile uses~~ PD obtains the FASC-N from the certificate's twicFASC-N subjectAltName entry and uses sixteen zero bytes for UUID in the challenge template and status reply. It discovers RSA-2048 ~~: discover~~ (0x07) or ECC P-384 (0x14) from the certificate~~SPKI and use GENERAL AUTHENTICATE with P1 and P2~~ 's subject public key and uses key reference 0x9E. ~~Neither workflow submits a PIV PIN or uses the TPK for this proof. The Legacy application transition preserves the card-authentication purpose; it does not authorize fallback from PIV Authentication to Card Authentication . A TWIC-model entry is invalid.~~ For each TWIC attempt, the PD derives a 0x7F-byte pseudorandom

challenge from the PIV Auto KDK and the fixed-width attempt template, using KMAC256 with customization `OSDP-TWIC-AUTO-CHALLENGE-v1`. It hashes the complete challenge with SHA-256 for RSA-2048 or SHA-384 for ECC-384. For RSA, tag `0x81` carries the SHA-256 DigestInfo and EMSA-PKCS1-v1_5 padding in a `0x100`-byte block. For ECC, tag `0x81` carries the `0x30`-byte SHA-384 digest. The card performs the private-key operation and returns the signature in tag `0x82`.

~~These new TWIC examples contain no captured card responses or newly computed signatures. Implementation fixtures must use public synthetic keys, certificates, and invented identifiers, with their provenance kept separate from physical-card captures. PIVSTATUS success reports the OSDP-defined key proof, not completion of a TSA reader mode or an access decision. Certificate trust, identity binding, expiration, and applicable cancellation checks remain required by deployment policy.~~ Following the proposed interpretation in Subsection 3.12, the ACU reconstructs the same `0x7F`-byte challenge from the reported counters and identity, configured entropy, and session KDK. It computes the digest and verifies the returned signature with the freshly retrieved Card Authentication Certificate's public key. It validates the certificate and card status as specified in TWIC Part 3, Section 4.4.4~~requires at least 127 bytes of unique challenge data; these illustrative examples do not establish that fixed RSA padding and a shorter KMAC digest satisfy that requirement. TSA reader qualification remains a separate verification item~~, and 33 CFR 101.520(a)–(b). For Mode 4, it continues with the commanded fingerprint workflow described in Subsection 3.8.

C.5 Supported Profiles

Table 85: PIV Auto Supported Profiles

Profile	`Alg`	`Key Ref`	PIV Auto	Coverage
RSA-1024	`06`	`9A`, `9E`	No	Informative legacy/non-current test-only material.
RSA-2048	`07`	`9A`, `9E`	Yes	NIST SD 33 cards 2, 3, 16; generated-card; GSA-backed simulation.
RSA-3072	`05`	`9A`, `9E`	Yes	Committed YubiKey material; requires YubiKey firmware with RSA-3072 support.
ECC P-256	`11`	`9A`, `9E`	Yes	NIST SD 33 cards 4 and 5; generated-card coverage.
ECC P-384	`14`	`9A`, `9E`	Yes	Committed YubiKey material. NIST card 5 also provides 9C/9D APDU examples.

C.6 Fixed-Width KMAC Challenge Template

Table 86: PIV Auto KMAC Challenge Template

Field	Size	Encoding	Notes
Template label	0x0E	ASCII plus version byte	`OSDP-PIV-AUTO` followed by 01.
Series Counter	0x04	LSB first	ACU-provided series, for example 07000000.
Sequence Counter	0x04	LSB first	PD-allocated attempt counter within that series, for example 2A000000; see Subsection 3.12.
FASC-N	0x19	Raw bytes	~~Credential~~ PIV: CHUID FASC-N. TWIC ACA: certificate FASC-N~~from CHUID~~.
UUID	0x10	Raw bytes	~~Credential UUID; the recognized Legacy TWIC nil-GUID exception follows~~ PIV: credential UUID. TWIC ACA: sixteen zero bytes; see Subsection 3.12. ~~Unavailable UUID otherwise causes failure.~~
PIV Key Reference	0x01	Raw byte	Challenged key reference, normally 9A or 9E.
PIV Algorithm	0x01	Raw byte	Discovered supported identifier: 05, 07, 11, or 14; 06 remains informative legacy test material only.
Supplemental Entropy	0x20	Raw bytes	Fixed-width ACU entropy field; all zeroes are permitted but not recommended.

The fixed-width encoding and profile mapping are normative in Tables 34 and 33; this appendix illustrates them. The ~~retained~~ prototype vectors derive ~~their KDK as . This is SC1-style prototype input , not evidence of an SC2 key binding~~a 32-byte KDK using KMAC256 with key `S-ENC||S-MAC1||S-MAC2`, empty input data, and customization string `OSDP-PIV-AUTO-KDK-v1`. This key input contains 48 bytes. The challenge stage ~~is . uses~~ uses KMAC256 with the PIV Auto KDK as key, the fixed-width template as input, output length `L`, and customization string `OSDP-PIV-AUTO-CHALLENGE-v1`. For PIV entries, `L` is 32 bytes for SHA-256 profiles and 48 bytes for SHA-384 profiles. TWIC entries use the `0x7F`-byte challenge derivation and SHA-256 or SHA-384 hashing in Subsection 3.12. The supplemental entropy field is always present and fixed at 32 bytes; all zeroes are valid for test vectors but not recommended for deployments.

C.7 Dynamic Authentication Templates

C.7.1 RSA Key Challenge Input

Table 87: RSA Dynamic Authentication Template

Tag / Bytes	Length	Name	Value
7C	Variable	Dynamic Authentication Template	Outer template for the PIV GENERAL AUTHENTICATE command data.
82	00	Response request	Empty value requesting the card response in tag 82.
81	Modulus length	Challenge block	EMSA-PKCS1-v1_5 encoded digest block with the same length as the RSA modulus.
0001FF...FF00	Variable	Padding prefix	EMSA-PKCS1-v1_5 block prefix before the digest information.
30...	Variable	DigestInfo	ASN.1 DigestInfo for SHA-256 or SHA-384 followed by the KMAC-derived digest; SHA-256 uses prefix 3031300D060960864801650304020105000420 and SHA-384 uses prefix 3041300D060960864801650304020205000430.

Table 88: RSA Dynamic Authentication Examples

Profile	Alg	Key Ref	Tag 81	Response 82	Command APDU
RSA-1024	06	9A/9E	0x80 bytes	0x80 bytes	Informative legacy/non-current test-only APDU.
RSA-2048	07	9A/9E	0x100 bytes	0x100 bytes	~~0x112-byte~~ 0x113-byte extended APDU.
RSA-3072	05	9A/9E	0x180 bytes	0x180 bytes	~~0x192-byte~~ 0x193-byte extended APDU.

C.7.2 ECC Signature Input

Table 89: ECC Dynamic Authentication Template

Tag / Bytes	Length	Name	Value
7C	Variable	Dynamic Authentication Template	Outer template for the PIV GENERAL AUTHENTICATE command data.
82	00	Response request	Empty value requesting the card response in tag 82.
81	Digest length	Challenge digest	KMAC-derived digest supplied directly to the ECC signing operation.
Digest bytes	0x20 or 0x30 bytes	Digest value	SHA-256-width digest for P-256; SHA-384-width digest for P-384.

Table 90: ECC Dynamic Authentication Examples

Profile	Alg	Key Ref	Tag 81	Response 82	Command APDU
ECC P-256	11	9A/9E	0x20 bytes	DER ECDSA signature, about 0x46-0x48 bytes	0x2C-byte short APDU.
ECC P-384	14	9A/9E	0x30 bytes	DER ECDSA signature, about 0x66-0x68 bytes	0x3C-byte short APDU.

C.7.3 Worked Dynamic Authentication Examples

Worked examples are generated from `test-vectors/piv-auto-demo/generated/piv-auto-simulation-report.json`.

Table 91: PIV Auto Worked Example: PIV Authentication RSA-1024

Field	Value
Profile	9a-rsa1024
Source	Committed YubiKey loader material in test-vectors/piv-auto-yubikey-material/9a-rsa1024. Informative legacy/non-current test-only coverage.
Algorithm ID	0x06
Key Reference	0x9A
KMAC Challenge Digest	5B2FC08860E3999EEF8F6CF5D2B0C9AE05CBB28A5B423BFE4F646F6F47FCA4F2

Field	Value
Tag 81 Challenge Input	0001FFFF

Field	Value
Response Tag 0x82 Length	0x0080 bytes

Table 93: PIV Auto Worked Example: PIV Authentication RSA-2048

Field	Value
Profile	9a-rsa2048
Source	Committed RSA-2048 card-authentication key in test-vectors/piv-auto-demo/source-p12, originally imported from GSA FICAM card-builder. NIST SD 33 / IR 8347 and generated-card coverage.
Algorithm ID	0x07
Key Reference	0x9A
KMAC Challenge Digest	E5D733C7FBBA8344701EE8909F1E6642E040CCF3265E0D5D28A26D73C344C19C
Tag 81 Challenge Input	0001FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF FFF

Table 94: PIV Auto Worked Example: Card Authentication RSA-2048

Field	Value
Profile	9e-rsa2048
Source	Committed RSA-2048 card-authentication key in test-vectors/piv-auto-demo/source-p12, originally imported from GSA FICAM card-builder. NIST SD 33 / IR 8347 and generated-card coverage.
Algorithm ID	0x07
Key Reference	0x9E
KMAC Challenge Digest	0864C776F2374124D3E63F0B0B29FC1C5F0E8FF8BB1FA80E2723293B86A0158E
Tag 81 Challenge Input	0001FFFF

Table 95: PIV Auto Worked Example: PIV Authentication RSA-3072

Field	Value
Profile	9a-rsa3072
Source	Committed YubiKey loader material in test-vectors/piv-auto-yubikey-material/9a-rsa3072. Generated-card coverage.
Algorithm ID	0x05
Key Reference	0x9A
KMAC Challenge Digest	1503FD1E89AA91A5B37E6872AC3846F654B7B1D099B2F3AEB597472DFE38E714 42816BC43454622219F61A3C6DAFDB4F
Tag 81 Challenge Input	0001FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF FFFF

Field	Value
Dynamic Authentication Response	7C8201848282018019C03ACCABE74C9259FA5A2267BC4F1401AB352F210BEFE0F74B8B74F25C82E0A83E1635C19C341CBD97D71D856BF132A2C3CDD223AC7D37451E9083A0C5FDB8013577741520A8A2375896E3839252F10DD8826A00A065F06154EE7A51703D65CC00AAA66E28BE5683085B4045882AA17B925457D3B6B0BC0C4581D10ED4653A1D4713E4FB5A4C72054789C9A4C1E6BC2B6B497228A9C765365040D00C139671175C262BA2E6D6064F1F84AAC0D5279B91F9FDE6D6B02B2423C7CC3FA7408E80A8040B92B819F80D6AD23B86C8F339D50F332802CBED0511215B03DF9496DD64F77ADBE4AE844F0D708F230E6F06911DADCA6888B9C80C94A16FCB0DE7467B70766668CDEA0053289295256F8BEDC693F058F6244BB6B315C4C9F07D89432A479AF72C666181C7B47CAA72C7ABE8B323658BD681FA10A1E88E2AFAB03BFDE9B571CA6FEDA05A24CC6436061D633691822FDEDD3DAE4E130A1CCE50E65FDFC5EA451B7AA8F83D4385D40E005E368D064E105D6F5325B36D955F3D88B5BEF8407890000
Response Tag 0x82 Length	0x0180 bytes

Table 96: PIV Auto Worked Example: Card Authentication RSA-3072

Field	Value
Profile	9e-rsa3072
Source	Committed YubiKey loader material in test-vectors/piv-auto-yubikey-material/9e-rsa3072. Generated-card coverage.
Algorithm ID	0x05
Key Reference	0x9E
KMAC Challenge Digest	4C98712B25DB7F477096FA3D0AC93FA2CA6BBE89CB291AC26EDF0D8312CA61D0CA238F1656247B2DF38E8B72D9AF7F5E
Tag 81 Challenge Input	0001FFFF

Field	Value
GENERAL AUTHENTICATE APDU	0087059E00018A7C8201868200818201800001FFFF

Field	Value
Dynamic Authentication Response	7C4A824830460221009D25A44E56572983819019DF26A2720E462A11D2CB3CD86A9142DEF4BCC1271B022100D084982BB9D3E2D3E6B0965956C71DBA0FE44FE328C0A2BD72951DFB4D9296FE9000
Response Tag 0x82 Length	0x0048 bytes

Table 98: PIV Auto Worked Example: Card Authentication ECC P-256

Field	Value
Profile	9e-ecp256
Source	Committed P-256 signing key in test-vectors/piv-auto-demo/source-p12, originally imported from GSA FICAM card-builder. NIST SD 33 / IR 8347 and generated-card coverage.
Algorithm ID	0x11
Key Reference	0x9E
KMAC Challenge Digest	965AF088B02DF65E7C299ABF53A5AB2DCBB3C789458DE8D0C819B19A9266658D
Tag 81 Challenge Input	965AF088B02DF65E7C299ABF53A5AB2DCBB3C789458DE8D0C819B19A9266658D
GENERAL AUTHENTICATE APDU	0087119E267C2482008120965AF088B02DF65E7C299ABF53A5AB2DCBB3C789458DE8D0C819B19A9266658DFF
Dynamic Authentication Response	7C4A824830460221008B204FA48FAA27A57D6E90F985199F284D2A8E0D0F2E051BBFBCACBA824619B9022100C6E04880D322DE7D10823EC78C8EA4CCB69FE394F694F1D12C797FA7BCAD769599000
Response Tag 0x82 Length	0x0048 bytes

Table 99: PIV Auto Worked Example: PIV Authentication ECC P-384

Field	Value
Profile	9a-ecp384
Source	Committed YubiKey loader material in test-vectors/piv-auto-yubikey-material/9a-ecp384. Generated-card coverage.
Algorithm ID	0x14
Key Reference	0x9A
KMAC Challenge Digest	C813683447986D513FC17CED5DB4D335BFBEAFB1AF9E5200C6B7B1C1E3535D700C2E05F9011717E153998832B7588D28
Tag 81 Challenge Input	C813683447986D513FC17CED5DB4D335BFBEAFB1AF9E5200C6B7B1C1E3535D700C2E05F9011717E153998832B7588D28

Field	Value
GENERAL AUTHENTICATE APDU	0087149A367C3482008130C813683447986D513FC17CED5DB4D335BFBEAFB1AF9E5200C6B7B1C1E3535D700C2E05F9011717E153998832B7588D28FF
Dynamic Authentication Response	7C6A82683066023100D88ABABB9893D88341CFA2F1AB6E3D35E4AF02F4F8D085B5A3DE63574DCF694741D869066B8E6297429120614DCC9CE40231009A3F203BB5EA98839008B0D25F98C98DE8E87622F61279B2F2B29343912999A36311102B310F9985844BCC248350377B9000
Response Tag 0x82 Length	0x0068 bytes

Table 100: PIV Auto Worked Example: Card Authentication ECC P-384

Field	Value
Profile	9e-ecp384
Source	Committed YubiKey loader material in test-vectors/piv-auto-yubikey-material/9e-ecp384. Generated-card coverage.
Algorithm ID	0x14
Key Reference	0x9E
KMAC Challenge Digest	F3480C6C1DAD008D3E14D0C815D381F01420E9D3402A175BED097C6949E4BA447FA0A11745CE4D054A95B10A9D5CC689
Tag 81 Challenge Input	F3480C6C1DAD008D3E14D0C815D381F01420E9D3402A175BED097C6949E4BA447FA0A11745CE4D054A95B10A9D5CC689
GENERAL AUTHENTICATE APDU	0087149E367C3482008130F3480C6C1DAD008D3E14D0C815D381F01420E9D3402A175BED097C6949E4BA447FA0A11745CE4D054A95B10A9D5CC689FF
Dynamic Authentication Response	7C698267306502303 4C6B2659FBE26A78D2615F8C0EBF2E9674292D9DE3CE8A31D9DA297B5AE236AC886E9E9C288D42474564BF0C46AE40C023100CCBD59A047D73F9FD58F60218FBCB4F99B18EBD7EB70EFE5CF64D5EB1A65F1676A867E1B1E5EB4CF9F34B44BE905D29C9000
Response Tag 0x82 Length	0x0067 bytes

C.8 APDUs

Table 101: PIV Auto APDU Reference: Card Preparation

Operation	Command APDU	Success Response	Purpose
Select PIV Application	00A4040000BA0000003080000100 00100FF	9000	Selects the PIV application before reading objects or authenticating.
Get Discovery Object	00CB3FFF035C017EFF	7E...9000	Reads the Discovery Object so the PD can determine application and PIN/VCI policy.
Verify PIN (optional)	0020008008313233343536FFFF	9000	Verifies example PIN 123456 padded with FF; failure responses include 6 3CX and 6983.

Table 102: PIV Auto APDU Reference: GENERAL AUTHENTICATE Command

Field	Encoding	Source	Notes
CLA	00	ISO/IEC 7816-4	Logical command before credential SM wrapping. Apply SP 800-73-5 Part 2, Section 4.2, when SM is established.
INS	87	ISO/IEC 7816-4	GENERAL AUTHENTICATE instruction.
P1	Algorithm ID	Certificate SPKI	05, 07, 11, or 14 for supported PIV Auto profiles; 06 appears only in informative RSA-1024 legacy examples.
P2	Key Reference	Configuration Entry	9A or 9E for PIV; 9E for TWIC Auto.
Lc	Short or extended	Command length	Short APDUs use one length byte; extended APDUs use 00 followed by two length bytes.
Command data	7C...820081.....	Dynamic Authentication Template	Tag 82 is empty to request a response; tag 81 carries the exact challenge input.

Field	Encoding	Source	Notes
Le	FF or FFFF	Response request	~~Requests the maximum response supported by the card interface~~ Explicit short or extended response limit; see paragraph 2.1.2.

Table 103: PIV Auto APDU Reference: GENERAL AUTHENTICATE Response

Case	Response APDU	PD Handling	OSDP Payload
Complete response	7C...82...9000	Strip only the ISO status word.	Carry the complete 7C Dynamic Authentication Template in osdp_CRAUTHR or osdp_PIVSTATUS.
Additional response bytes	61XX	Issue GET RESPONSE with Le min(SW2count, 0xFF); 6100 uses 0xFF.	Concatenate returned data before reporting the final 7C template.
GET RESPONSE command	00C00000[Le]	Retrieve remaining card response bytes with an explicit nonzero length up to 0xFF.	Not exposed directly to the ACU.
Credential error	69.. or 6A..	Preserve command failure detail.	Auto reports terminal PIVSTATUS Result 0x06 with the applicable status word; commanded CRAUTH uses PIVERROR.

Table 104: PIV Auto APDU Reference: OPACITY / VCI Context

Operation	Command Shape	Key Ref	Relationship to PIV Auto
OPACITY establish	0087[27/2E]04[Lc]7C...81...8200	04	Establishes credential SM after CVC and cryptogram verification. Contactless VCI also depends on Discovery policy and pairing when required.
PIV Auto challenge	0087[alg][9A/9E][Lc]7C...820081.....[FF/FFFF]	9A/9E	Performs the autonomous key challenge after card preparation and any required VCI setup.

Operation	Command Shape	Key Ref	Relationship to PIV Auto
Response handling	7C...82...9000	9A/9E	The PD reports the returned signed response bytes exactly, excluding only SW1/SW2.

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D Credential SM and VCI Worked Examples (Informative)

These examples follow the preparation and PIN-verification flows in Subsection 3.1. A simulated ACU sends complete OSDP packets to a simulated PD. The PD replays existing card captures and reports the resulting credential state. These examples omit OSDP Secure Channel 3.0 for clarity; the test harness assumes that the OSDP Secure Channel 3.0 authorization requirements are satisfied.

The contact and contactless card01 captures supply the card commands and replies. The verifier checks the CVC signature against an existing trust anchor, recomputes SM key establishment, and checks protected command and response MACs and encryption. The replay retains intervening captured exchanges so the session counters and MAC chains remain intact. Those intervening exchanges preserve the original card session. The card captures include earlier provisioning and data-read commands; those commands impose no additional requirements on automatic card preparation. The examples replay recorded sessions; no connected card is used.

Table 105: Replay coverage

Case	Evidence	Purpose
Contact SM	Captured SM session and CVC validation	Report SM with VCI absent.
Incorrect PIN	Modeled authenticated rejection	Clear the failed PIN while preserving SM and VCI.
Commanded pairing and PIN	Captured protected VERIFY exchanges	Complete pairing before verifying the cached PIN.
PIV Auto failure	Modeled results and full status words	Report the cause across multipart replies.
PIN cache expiration	Modeled timing, with captured VERIFY for the submitted-command case	Separate cache availability from credential PIN verification.

Complete packets, APDUs, and source references are in the SM/VCI example file. The accompanying README gives the replay and independent-verification commands. Packet dumps use hexadecimal byte pairs, including the OSDP header and CRC. Each example starts at sequence `0x01` on an established link to PD address `0x01`. Standard status-query and pairing replies use the same `osdp_CARDSTATUSR` layout with Status Type `0x01`. Diagnostic replies carry Status Type `0x02` before the multi-part fields in every fragment. Timing examples identify the modeled elapsed time at each exchange; captured APDUs retain their original order.

D.1 Credential SM over contact

The PD prepares a contact credential and reports SM established without VCI. The reply identifies the CVC issuer and trust anchor used for that SM session.

Example ID: `contact-sm`. Credential exchanges come from the contact card01 capture. The complete replay retains its original APDU order.

Table 106: Reported security state

Field	Value	Notes
Status Type	0x01	Standard credential status.
Credential Security Status	0x02	SM established without VCI.
SM Trust Anchor ID	0x01	Loaded anchor used to validate SM.
SM CVC IIN	0xB6103792270FBD08	Issuer named by the credential's SM CVC.
SM Anchor IIN	0xB6103792270FBD08	Same issuer: direct trust path.

Table 107: OSDP command and reply exchange

Message	Packet bytes (hex)	Notes
ACU CARDSTATUS	53 01 09 00 05 A9 01 8C 9B	Read current state after automatic credential preparation.
PD ACK	53 81 08 00 05 40 39 35	Command accepted.
ACU POLL	53 01 08 00 06 60 D8 66	Poll for completion.
PD CARDSTATUSR	53 81 2A 00 06 85 01 01 02 01 B6 10 37 92 27 0F BD 08 B6 10 37 92 27 0F BD 08 48 00 0B A0 00 00 03 08 00 00 10 00 01 00 98 79	Status 0x02: SM established without VCI.

D.2 PIN before credential presentation

The PD collects the PIN locally and reports completion before the credential is presented. Initial selection and SM establishment preserve the cache. The ACU then completes pairing and requests PIN verification over VCI.

Example ID: `cache-before-card`. Credential exchanges come from the contactless card01 capture. The complete replay retains its original APDU order.

Table 108: OSDP command and reply exchange

Message	Packet bytes (hex)	Notes
ACU PIVSPE	53 01 0E 00 05 AA 04 00 00 00 3C 00 FE 9F	Collect a PIN locally; cache for 0x003C seconds.
PD ACK	53 81 08 00 05 40 39 35	Command accepted.
ACU POLL	53 01 08 00 06 60 D8 66	Poll for completion.
PD PIVSPER	53 81 09 00 06 86 00 05 10	Result 0x00: requested PIN operation completed.
ACU CARDSTATUS	53 01 09 00 07 A9 01 EC F5	Read current state after automatic credential preparation.
PD ACK	53 81 08 00 07 40 5B 53	Command accepted.
ACU POLL	53 01 08 00 05 60 8B 33	Poll for completion.
PD CARDSTATUSR	53 81 2A 00 05 85 01 02 02 01 B6 10 37 92 27 0F BD 08 B6 10 37 92 27 0F BD 08 48 00 0B A0 00 00 03 08 00 00 10 00 01 00 1E 84	Status 0x02: SM established without VCI.
ACU PIVXMITPAIRING	53 01 10 00 06 AD 30 30 30 30 30 30 30 32 EB 8B	Submit the captured pairing code.
PD ACK	53 81 08 00 06 40 6A 60	Command accepted.
ACU POLL	53 01 08 00 07 60 E9 55	Poll for completion.
PD CARDSTATUSR	53 81 2A 00 07 85 01 02 01 01 B6 10 37 92 27 0F BD 08 B6 10 37 92 27 0F BD 08 48 00 0B A0 00 00 03 08 00 00 10 00 01 00 56 3D	Status 0x01: SM and VCI established.
ACU PIVSPE	53 01 0E 00 05 AA 02 00 00 00 00 00 E7 52	Verify the cached PIN over established VCI.

Message	Packet bytes (hex)	Notes
PD ACK	53 81 08 00 05 40 39 35	Command accepted.
ACU POLL	53 01 08 00 06 60 D8 66	Poll for completion.
PD PIVSPER	53 81 09 00 06 86 00 05 10	Result 0x00: requested PIN operation completed.

D.3 PIV Auto needs commanded pairing

In this modeled case, the configured PIV Auto authentication requires VCI. SM is established, but pairing remains necessary. PIV Auto reports the unmet condition and ends the attempt. A later poll receives ACK; the ACU can begin commanded pairing.

Example ID: `auto-pairing-required`.

Table 109: OSDP command and reply exchange

Message	Packet bytes (hex)	Notes
ACU POLL	53 01 08 00 05 60 8B 33	Poll for completion.
PD PIVSTATUS	53 81 45 00 05 89 37 00 00 00 37 00 05 01 01 00 00 00 02 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 9A 07 00 00 FF 09	Result 0x05, Detail 0x01: pairing code required.
ACU POLL	53 01 08 00 06 60 D8 66	Poll for completion.
PD ACK	53 81 08 00 06 40 6A 60	No completion pending.

D.4 SM failure diagnostics

In this modeled case, PIV Auto reports Result 0x02, Detail 0x04: invalid authentication cryptogram. The ACU then requests the SM diagnostic record. The PD returns the failure reason and the hash of the received CVC, with no further credential commands. Three fragments carry the 35-byte record. The CVC bytes used for the hash come from the existing capture.

Example ID: `diagnostic-cvc-hash`.

Table 110: OSDP command and reply exchange

Message	Packet bytes (hex)	Notes
ACU CARDSTATUS	53 01 09 00 06 A9 02 BF F2	Request retained SM diagnostics; no credential command is sent.
PD ACK	53 81 08 00 06 40 6A 60	Command accepted.
ACU POLL	53 01 08 00 07 60 E9 55	Poll for completion.
PD CARDSTATUSR	53 81 1F 00 07 85 02 23 00 00 00 10 00 03 04 01 A5 50 00 73 BA 17 70 37 0A 1D 58 B1 5B 2F F2	Diagnostic: total 0x0023, offset 0x0000, fragment 0x0010.
ACU POLL	53 01 08 00 05 60 8B 33	Poll for completion.
PD CARDSTATUSR	53 81 1F 00 05 85 02 23 00 10 00 10 00 97 5D D1 50 7F 39 22 BB 44 20 69 9B 2D FB 42 83 52 B1	Diagnostic: total 0x0023, offset 0x0010, fragment 0x0010.
ACU POLL	53 01 08 00 06 60 D8 66	Poll for completion.
PD CARDSTATUSR	53 81 12 00 06 85 02 23 00 20 00 03 00 C7 6C E9 B4 96	Diagnostic: total 0x0023, offset 0x0020, fragment 0x0003.

D.5 PIN cache expires while idle

The PD acquires a PIN at t=0 with a ten-second validity period. At t=10 the PD erases the PIN. A poll receives ACK because no operation is pending. A status query reports an empty cache; a subsequent verification request ends with no cached PIN. The PD sends no VERIFY. This example models the timing and credential state.

Example ID: `pin-expiration-idle`.

Table 111: OSDP command and reply exchange

Message	Packet bytes (hex)	Notes
ACU PIVSPE	53 01 0E 00 05 AA 04 00 00 00 0A 00 CD 30	t=0 s. Acquire a PIN; Cache Validity is 0x000A seconds.
PD ACK	53 81 08 00 05 40 39 35	t=0 s. Command accepted.
ACU POLL	53 01 08 00 06 60 D8 66	t=0 s. Poll for completion.
PD PIVSPER	53 81 09 00 06 86 00 05 10	t=0 s. Result 0x00: requested PIN operation completed.
ACU POLL	53 01 08 00 07 60 E9 55	t=10 s. Poll for completion.
PD ACK	53 81 08 00 07 40 5B 53	t=10 s. No completion pending.
ACU PIVSPE	53 01 0E 00 05 AA 00 00 00 00 00 00 A7 D9	t=10 s. Report empty cache and no PIN verification.
PD PIVSPER	53 81 09 00 05 86 00 55 49	t=10 s. Cache empty; credential PIN verification absent.
ACU PIVSPE	53 01 0E 00 06 AA 03 00 00 00 00 00 32 DF	t=10 s. Request verification using the expired cache.
PD ACK	53 81 08 00 06 40 6A 60	t=10 s. Command accepted.
ACU POLL	53 01 08 00 07 60 E9 55	t=10 s. Poll for completion.
PD PIVSPER	53 81 09 00 07 86 0A 7F 86	t=10 s. Result 0x0A: no cached PIN.

D.6 Expiration while waiting for a credential

The ACU arms cached-PIN verification with Auto Execute flag 0x02. The credential is absent when the cache expires at t=10. The PD ends the pending request with PIVSPER 0x0A (no cached PIN). This example models the waiting state.

Example ID: `pin-expiration-waiting`.

Table 112: OSDP command and reply exchange

Message	Packet bytes (hex)	Notes
ACU PIVSPE	53 01 0E 00 05 AA 04 00 00 00 0A 00 CD 30	t=0 s. Acquire a PIN; Cache Validity is 0x000A seconds.
PD ACK	53 81 08 00 05 40 39 35	t=0 s. Command accepted.
ACU POLL	53 01 08 00 06 60 D8 66	t=0 s. Poll for completion.
PD PIVSPER	53 81 09 00 06 86 00 05 10	t=0 s. Result 0x00: requested PIN operation completed.
ACU PIVSPE	53 01 0E 00 07 AA 03 00 00 02 00 00 81 F6	t=0 s. Arm cached-PIN verification for credential presentation.
PD ACK	53 81 08 00 07 40 5B 53	t=0 s. Command accepted.
ACU POLL	53 01 08 00 05 60 8B 33	t=0 s. Poll for completion.
PD ACK	53 81 08 00 05 40 39 35	t=0 s. No completion pending.
ACU POLL	53 01 08 00 06 60 D8 66	t=10 s. Poll for completion.
PD PIVSPER	53 81 09 00 06 86 0A 4F B1	t=10 s. Result 0x0A: no cached PIN.

D.7 Expiration before VERIFY submission

The PD accepts verification at t=9. Before the PD submits VERIFY, the deadline is reached at t=10. The PD returns PIVSPER 0x0A (no cached PIN) and sends no VERIFY. Acceptance alone does not reserve the PIN beyond its deadline. Timing is modeled.

Example ID: `pin-expiration-boundary`.

Table 113: OSDP command and reply exchange

Message	Packet bytes (hex)	Notes
ACU PIVSPE	53 01 0E 00 05 AA 04 00 00 00 0A 00 CD 30	t=0 s. Acquire a PIN; Cache Validity is 0x000A seconds.
PD ACK	53 81 08 00 05 40 39 35	t=0 s. Command accepted.
ACU POLL	53 01 08 00 06 60 D8 66	t=0 s. Poll for completion.
PD PIVSPER	53 81 09 00 06 86 00 05 10	t=0 s. Result 0x00: requested PIN operation completed.
ACU PIVSPE	53 01 0E 00 07 AA 03 00 00 00 00 00 E1 98	t=9 s. Accept verification at t=9, before the cache deadline.
PD ACK	53 81 08 00 07 40 5B 53	t=9 s. Command accepted.
ACU POLL	53 01 08 00 05 60 8B 33	t=10 s. Poll for completion.
PD PIVSPER	53 81 09 00 05 86 0A 1F E8	t=10 s. Result 0x0A: no cached PIN.

D.8 Expiration after VERIFY submission

The PD submits VERIFY at t=9, before the deadline. The cache expires at t=10 while the exchange is completing. The PD reports successful verification, then reports an empty cache with the credential still PIN verified. Credential APDUs come from the contactless capture; the timing is modeled.

Example ID: `pin-expiration-submitted`.

Table 114: OSDP command and reply exchange

Message	Packet bytes (hex)	Notes
ACU PIVSPE	53 01 0E 00 05 AA 04 00 00 00 0A 00 CD 30	t=0 s. Acquire a PIN; Cache Validity is 0x000A seconds.
PD ACK	53 81 08 00 05 40 39 35	t=0 s. Command accepted.
ACU POLL	53 01 08 00 06 60 D8 66	t=0 s. Poll for completion.
PD PIVSPER	53 81 09 00 06 86 00 05 10	t=0 s. Result 0x00: requested PIN operation completed.
ACU PIVXMITPAIRING	53 01 10 00 07 AD 30 30 30 30 30 30 30 32 AE E4	t=0 s. Supply the captured pairing code after SM preparation.
PD ACK	53 81 08 00 07 40 5B 53	t=0 s. Command accepted.
ACU POLL	53 01 08 00 05 60 8B 33	t=0 s. Poll for completion.
PD CARDSTATUSR	53 81 2A 00 05 85 01 02 01 01 B6 10 37 92 27 0F BD 08 B6 10 37 92 27 0F BD 08 48 00 0B A0 00 00 03 08 00 00 10 00 01 00 D4 6F	t=0 s. Status 0x01: SM and VCI established.
ACU PIVSPE	53 01 0E 00 06 AA 02 00 00 00 00 00 92 9A	t=9 s. Accept verification at t=9, before the cache deadline.
PD ACK	53 81 08 00 06 40 6A 60	t=9 s. Command accepted.
ACU POLL	53 01 08 00 07 60 E9 55	t=10 s. Poll for completion.
PD PIVSPER	53 81 09 00 07 86 00 35 27	t=10 s. Result 0x00: requested PIN operation completed.
ACU PIVSPE	53 01 0E 00 05 AA 00 00 00 00 00 00 A7 D9	t=10 s. Empty cache; credential PIN verified.
PD PIVSPER	53 81 09 00 05 86 02 17 69	t=10 s. Cache empty; credential PIN verified.

D.9 PIN reuse preserves the original deadline

Two explicit contact verification requests use the same cached PIN at t=1 and t=2. Both succeed. At t=10, the cache expires and the credential remains PIN verified. The successes and timing are modeled; the example demonstrates the PD state rules.

Example ID: `pin-expiration-reuse`.

Table 115: OSDP command and reply exchange

Message	Packet bytes (hex)	Notes
ACU PIVSPE	53 01 0E 00 05 AA 04 00 00 00 0A 00 CD 30	t=0 s. Acquire a PIN; Cache Validity is 0x000A seconds.
PD ACK	53 81 08 00 05 40 39 35	t=0 s. Command accepted.
ACU POLL	53 01 08 00 06 60 D8 66	t=0 s. Poll for completion.
PD PIVSPER	53 81 09 00 06 86 00 05 10	t=0 s. Result 0x00: requested PIN operation completed.
ACU PIVSPE	53 01 0E 00 07 AA 03 00 00 00 00 00 E1 98	t=1 s. Explicit contact verification using the same cached PIN.
PD ACK	53 81 08 00 07 40 5B 53	t=1 s. Command accepted.
ACU POLL	53 01 08 00 05 60 8B 33	t=1 s. Poll for completion.
PD PIVSPER	53 81 09 00 05 86 00 55 49	t=1 s. Result 0x00: requested PIN operation completed.
ACU PIVSPE	53 01 0E 00 06 AA 03 00 00 00 00 00 32 DF	t=2 s. Explicit contact verification using the same cached PIN.
PD ACK	53 81 08 00 06 40 6A 60	t=2 s. Command accepted.
ACU POLL	53 01 08 00 07 60 E9 55	t=2 s. Poll for completion.
PD PIVSPER	53 81 09 00 07 86 00 35 27	t=2 s. Result 0x00: requested PIN operation completed.
ACU PIVSPE	53 01 0E 00 05 AA 00 00 00 00 00 00 A7 D9	t=10 s. Cache expired; credential PIN verification remains valid.
PD PIVSPER	53 81 09 00 05 86 02 17 69	t=10 s. Cache empty; credential PIN verified.

D.10 Credential failure status across two fragments

PIV Auto reports a modeled GENERAL AUTHENTICATE rejection: Result 0x06, Detail 0x00. The credential status is 0x6982 (security status not satisfied). The first fragment ends with SW1, 0x69; the second carries SW2, 0x82. The ACU reassembles the complete record before interpreting the status word.

Example ID: `auto-status-word`.

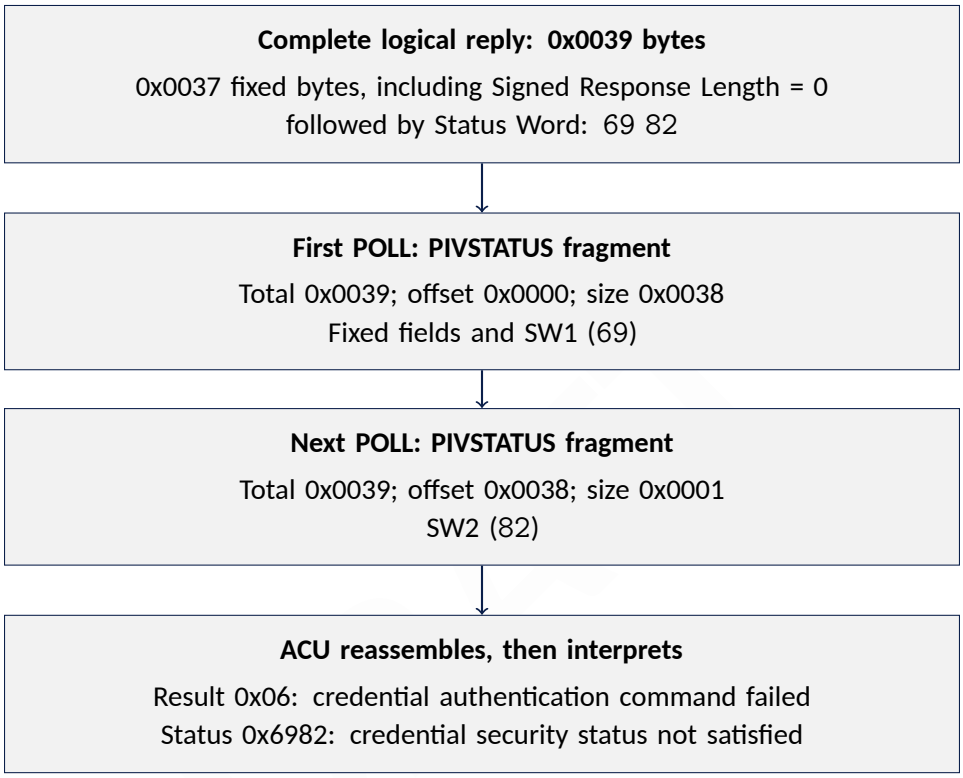


Figure 9: OSDP 3.0 multipart fields describe the entire logical reply. A fragment boundary can fall between SW1 and SW2.

Table 116: OSDP command and reply exchange

Message	Packet bytes (hex)	Notes
ACU POLL	53 01 08 00 05 60 8B 33	Poll for completion.
PD PIVSTATUS	53 81 46 00 05 89 39 00 00 00 38 00 06 00 01 00 00 00 02 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 9A 07 00 00 69 E4 69	PIV Auto credential rejection; SW1/SW2 cross the fragment boundary.
ACU POLL	53 01 08 00 06 60 D8 66	Poll for completion.

Message	Packet bytes (hex)	Notes
PD PIVSTATUS	53 81 0F 00 06 89 39 00 38 00 01 00 82 9D EB	PIV Auto credential rejection; SW1/SW2 cross the fragment boundary.

D.11 Incorrect PIN preserves SM and VCI

This modeled case starts with SM, VCI, and a cached PIN available. The credential returns an authenticated 0x63C2 (incorrect PIN, two attempts remain). The PD reports PIN mismatch and clears the cache and PIN-verification state. SM and VCI remain established. The SPE reply carries the mismatch result; a separate Mode 0x06 query retrieves remaining attempts. The ACU can acquire a corrected PIN and request verification again using the existing session.

Example ID: `sm-credential-rejection`.

Table 117: OSDP command and reply exchange

Message	Packet bytes (hex)	Notes
ACU PIVSPE	53 01 0E 00 05 AA 02 00 00 00 00 00 E7 52	Verify a cached PIN over an established VCI; initial state is modeled.
PD ACK	53 81 08 00 05 40 39 35	Command accepted.
ACU POLL	53 01 08 00 06 60 D8 66	Poll for completion.
PD PIVSPER	53 81 09 00 06 86 03 66 20	Result 0x03: PIN mismatch.
ACU PIVSPE	53 01 0E 00 07 AA 00 00 00 00 00 00 01 56	Cache erased and PIN-verification state cleared after PIN mismatch.
PD PIVSPER	53 81 09 00 07 86 00 35 27	Cache empty; credential PIN verification absent.
ACU CARDSTATUS	53 01 09 00 05 A9 01 8C 9B	Read credential status after ordinary rejection.
PD ACK	53 81 08 00 05 40 39 35	Command accepted.
ACU POLL	53 01 08 00 06 60 D8 66	Poll for completion.
PD CARDSTATUSR	53 81 2A 00 06 85 01 02 01 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 FF FF 0B A0 00 00 03 08 00 00 10 00 01 00 A9 03	Status 0x01: SM and VCI established.

D.12 Recovering trust for contactless PIV Auto

This example uses a credential that supports SM and requires a pairing code for VCI. PIV Auto is configured for contactless authentication with key 0x9A, but the PD's trust store is empty. The ACU already holds the roots and validation policy authorized for the deployment. Figure 10 shows the recovery sequence.

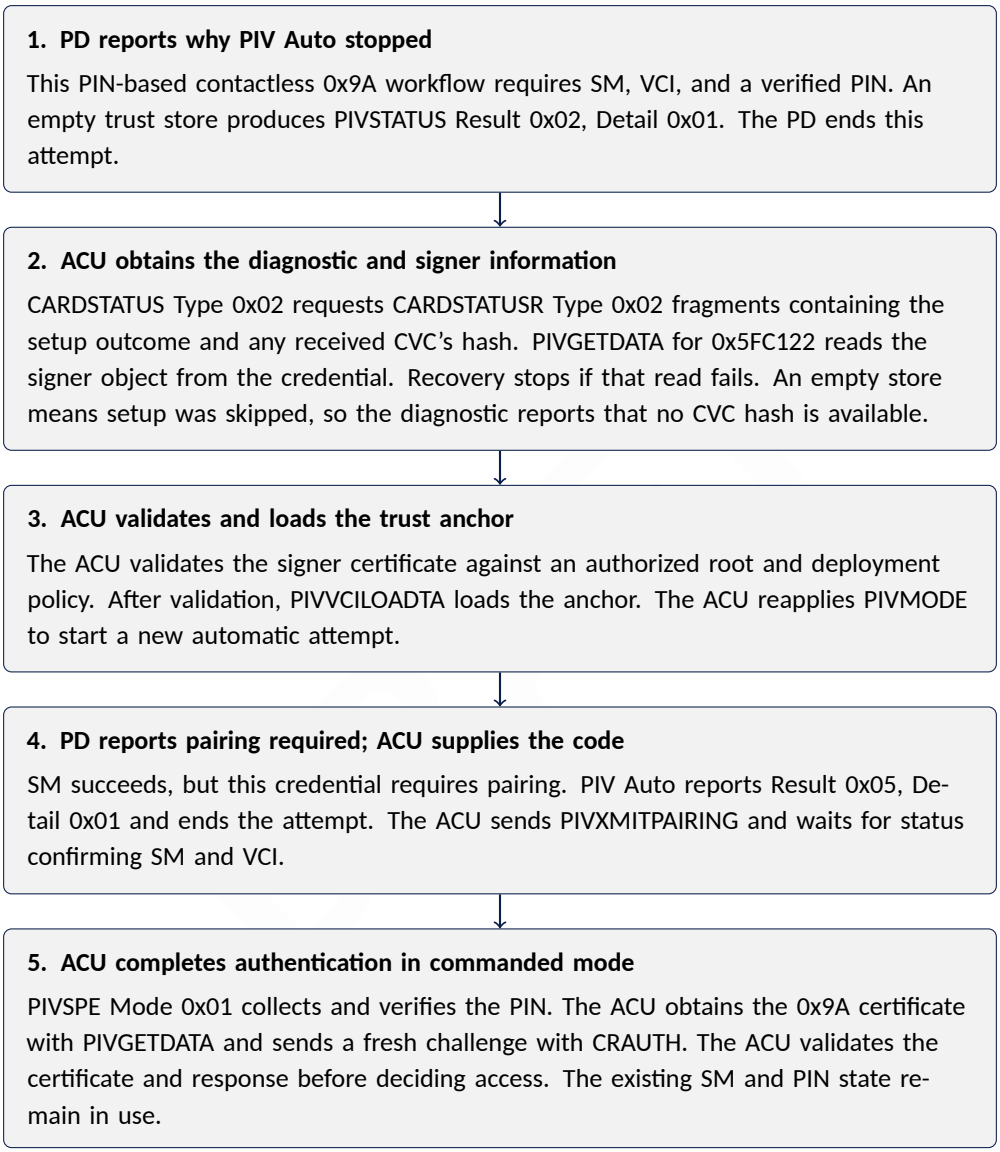


Figure 10: Recovery from an empty PD trust store for contactless PIV authentication with key 0x9A. Every step depends on a successful result from the preceding step. Unsupported SM or VCI requires a different ACU-selected workflow.

The first PIV Auto attempt ends with Result 0x02, Detail 0x01 (trust store empty). The ACU requests the SM diagnostic record, then reads the Secure Messaging Certificate Signer object with PIVGETDATA, Object Identifier 0x5FC122, Tag Length zero, Offset zero, and Requested Length zero. When the read succeeds, the reply contains the complete object, including signer certificate tag 0x70 and any intermediate CVC in

tag 0x7F21. The PD reads the credential through the contactless interface. If the object is absent or the credential denies access, the PD reports the corresponding PIVERROR. The ACU then selects another supported workflow or ends the transaction. Recovery relies on the credential returning the object.

The ACU validates the signer certificate's chain to an authorized root, validity period, revocation status, and PIV or PIV-I content-signing extended key usage. The ACU checks any intermediate CVC's signature and linkage as specified under PIVVCILOADTA. The anchor record contains the validated signer's public key and the IIN derived from that certificate's subjectKeyIdentifier. The ACU loads an anchor only after these checks succeed. An IIN match alone provides no authorization to trust a certificate.

After the load succeeds, the ACU reapplies the PIVMODE configuration. The PD starts a new attempt, establishes SM, and reports Result 0x05, Detail 0x01 (pairing code required). The ACU sends PIVXMITPAIRING and waits for CARDSTATUSR with security status 0x01 (SM and VCI established). The ACU then sends PIVSPE Mode 0x01, with Auto Execute clear, to collect and verify the PIN. After successful verification, the ACU reads the PIV Authentication certificate (Object Identifier 0x5FC105) and uses CRAUTH with key 0x9A and a fresh challenge. Certificate and challenge-response validation remain the ACU's responsibility.

The ACU waits for a successful result at each step before issuing the next command. An error ends this recovery sequence. These are commanded operations after the automatic attempt has ended. The PD returns each command's defined reply; the sequence generates no second PIV Auto result. PIVMODE remains configured for the next credential presentation. An ordinary PIVMODE replacement after pairing or PIN verification clears the state needed for the commanded authentication. Alternatively, after the completed pairing/PIN prerequisite failure, an eligible PIVMODE carrying the explicit Resume TLV can preserve that context and start one new Auto attempt with the exact next unused sequence counter. Subsection 3.12 defines this distinct operation; the illustrative resume bytes appear in the autonomous-template appendix.

If SM or VCI is unsupported, the ACU chooses a supported interface or authentication method explicitly, or rejects the transaction. If a signature or cryptogram check fails, the PD blocks credential operations and retains the diagnostic record. The ACU investigates that failure before requesting a new attempt. Trust recovery is appropriate for missing authorized trust material; failed cryptographic checks are security failures.

This is a control-flow example. The preceding captured SM/VCI packet examples cover pairing and PIN verification. The companion trust-recovery example records the command sequence and validation decisions.

E CRAUTH Worked Examples (Informative)

This example sends a captured P-256 challenge to key 0x9E and returns the captured signature template. The first command fragment contains only Format Marker 0xFF. The second contains Algorithm 0x11, Reference Identifier 0x9E, and the 0x20-byte challenge. The complete request record is 0x0023 bytes. The PD removes the marker and uses the algorithm, key reference, and challenge to construct GENERALAUTHENTICATE.

The credential data comes from the existing contactless card04 SM capture. Credential SM is assumed established as in that capture; the example shows the plaintext at the PD's SM boundary. The packets omit OSDP Secure Channel 3.0 for clarity. The two failure cases are modeled to show terminal error handling. Full exchanges and source references are in the CRAUTH example file. The command and reply codes remain 0xA5 and 0x82.

Table 118: CRAUTH command and reply exchange

Message	Packet bytes (hex)	Notes
ACU CRAUTH	53 01 0F 00 05 A5 23 00 00 00 01 00 FF 6B 63	Total 0x0023; offset 0x0000; fragment 0x0001. Format marker only.
PD ACK	53 81 08 00 05 40 39 35	Fragment accepted.
ACU CRAUTH	53 01 30 00 06 A5 23 00 01 00 22 00 11 9E 9B C1 80 EF 9A 61 6F 9F 05 8B F4 60 44 C8 9A 30 1B BD 90 13 7A DF 12 F9 D9 A9 91 7B 09 4A EF A0 5C FF	Total 0x0023; offset 0x0001; fragment 0x0022.
PD ACK	53 81 08 00 06 40 6A 60	Fragment accepted.
ACU POLL	53 01 08 00 07 60 E9 55	Poll for the result.
PD CRAUTHR	53 81 36 00 07 82 4B 00 00 00 28 00 7C 49 82 47 30 45 02 21 00 83 6F 34 70 E4 CD 87 3C 6C A3 E3 B6 5A D3 6B 17 DF FC E2 7B 56 A8 25 EC 39 A9 DC E3 D4 CB B0 6F 70	Total 0x004B; offset 0x0000; fragment 0x0028.
ACU POLL	53 01 08 00 05 60 8B 33	Poll for the result.
PD CRAUTHR	53 81 31 00 05 82 4B 00 28 00 23 00 95 02 20 61 1E FE 89 BF A8 13 B1 34 BD 11 1D 64 00 01 8B 50 71 1E 6C 9C 5D C8 22 DC 75 24 0B FA 60 D6 EE D9 FA	Total 0x004B; offset 0x0028; fragment 0x0023.

Message	Packet bytes (hex)	Notes
ACU POLL	53 01 08 00 06 60 D8 66	Poll for the result.
PD ACK	53 81 08 00 06 40 6A 60	No completion pending.

Table 119: Modeled failure completions

Failure	Packet bytes (hex)	ACU action
Credential rejection	53 81 0C 00 07 8A 23 10 69 82 2A D5	PIVERROR 0x1023 (security conditions unmet), SW1/SW2 0x6982. End the request.
Failure after a data fragment	53 81 0A 00 05 8A 02 00 0E 51	PIVERROR 0x0002 (smart-card timeout). Discard the partial response.

Draft

F Deprecated `osdp_GENAUTH`

The `osdp_GENAUTH` command is deprecated and expected to be removed in a future revision of this specification. Using a PIV credential to decrypt or process arbitrary data is unsafe and unnecessary for credential verification; implementations **SHOULD** use `osdp_CRAUTH` flows instead. Controllers and PDs **SHOULD** plan migrations away from `osdp_GENAUTH`.

DRAFT